

Biology for Secondary Schools

Student's Book

Form One

THE UNITED REPUBLIC OF TANZANIA
MINISTRY OF EDUCATION,
SCIENCE AND TECHNOLOGY

Certificate of Approval

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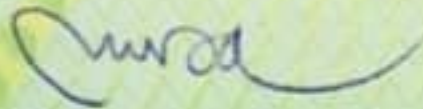
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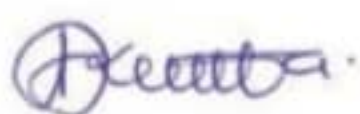
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Dr Aneth A. Komba
Director General

Tanzania Institute of Education



Preface

This textbook, *Biology for Secondary Schools*, is written specifically for Form One students in the United Republic of Tanzania. The textbook is prepared in accordance with the 2023 Biology Syllabus for Ordinary Secondary Education, Forms I-IV, issued by the Ministry of Education, Science and Technology. It is a revised edition of Biology for Secondary Schools Student's Book for Form One that was published in 2021 in accordance with the 2005 syllabus issued by the then Ministry of Education and Vocational Training (MoEVT).

The textbook consists of six chapters, namely Introduction to Biology; Scientific processes in Biology; Cell structure and organisation; Classification of living things; Viruses and major groups of living things; and Nutrition in plants. In addition to the contents, each chapter contains activities, illustrations, and exercises. You are encouraged to do all the activities and attempt all the questions in the exercises. You are also required to prepare a portfolio for keeping activities performed in different lesson. This will enhance your understanding and development of the intended competencies for this level.

Additional learning resources are available at TIE e-Library <https://ol.tie.go.tz> or ol.tie.go.tz.



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Chapter One

Introduction to Biology

Introduction

There are various forms of life on earth ranging from microscopic to macroscopic living things. Such forms of life or living things are studied in Biology. In this chapter, you will learn about the basic concepts and terminologies of Biology, branches of Biology, importance of studying Biology, and the relationship between Biology and other scientific fields. The competencies developed will enable you to use biological principles and processes in everyday life.



Think

Life without Biology

Basic concepts and terminologies in Biology

Task 1.1

Search from library and reliable internet sources information on the basic concepts and terminologies of Biology.

The term 'Biology' originates from two Greek words, **bios** that means 'life' and **logos** that means 'study'. Therefore,

Biology is a branch of science that deals with the study of life. A person who studies Biology is called a **Biologist**.

Living things are also called **organisms**. They include, plants, animals, fungi, and **micro-organisms**. Micro-organisms are very small living things, such as bacteria and some protozoans, which cannot be seen with the naked eyes. The size of organisms varies and ranges from very small such as bacteria to very large such as trees, birds, hippopotamus, and whales, as shown in Figure 1.1.





(a)



(b)



(c)



(d)

Figure 1.1: Living things (a) tree (b) bird (c) hippopotamus (d) whale

Life

Life is a state that distinguishes living things from non-living things. It is characterised by nutrition, respiration, excretion, growth and development, sensitivity or irritability, movement, and reproduction. The basic unit of life is the **cell**. All living things are made up of cells. Cells are so small that they cannot be seen with naked eyes. They can only be seen by using the **microscope**. The microscope is an instrument that is used in scientific studies to enlarge very small things

so that they can be easily seen and examined. The process of enlarging very small things is called magnification.

The branches of Biology include **Botany**, **Zoology** and **Microbiology**. Botany is the study of plants, zoology is the study of animals, and microbiology is the study of microbes. A person who is specialised in the study of plants is called a **Botanist**, a person who deals with the study of animals is called a **Zoologist**, whereas a person who deals with the study of microbes is called a

Microbiologist. However, there are many sub-branches of Biology that deal with different aspects of living things. Examples are given in Table 1.1.

Table: Other branches of Biology

	Branch	Area of study
1	Anatomy	Body structures of organisms
2	Cytology	Structure and functions of cells
3	Ecology	Relationship between organisms and their environment
4	Entomology	Insects
5	Genetics	Heredity and variation
6	Immunology	Immune systems
7	Mycology	Fungi
8	Parasitology	Parasites
9	Taxonomy	Classification of organisms
10	Physiology	Physical and chemical processes taking place within living organisms

Exercise

1. Explain the term Biology.
2. Differentiate Botany from Zoology.
3. Biology is more than a study of animals and plants. Explain.

Importance of studying Biology

Task 1.2

Search from the library and reliable internet sources information on the importance of studying Biology, then write short notes.

The knowledge obtained from studying Biology is important because it:

- (a) Helps to understand and care for the environment properly. It helps to learn about the plants, animals, and micro-organisms that surround us and how they affect us.
- (b) Helps to identify and group living things which makes it easier to learn more about them.
- (c) Helps to appreciate nature and many fascinating things about different organisms and how their functions are learnt in Biology.
- (d) Helps to understand our bodies.
- (e) Helps to acquire research skills

that are useful when carrying out scientific investigations. Such skills include measuring, observing, analysing, and drawing conclusion.

- (f) Helps to improve food production. Scientists can use biological skills to develop higher-yielding and disease-resistant varieties of plants and animals. These improved breeds of plants and animals are called hybrids. Figure 1.2 shows the indigenous and improved breeds of chicken.



Figure 1.2: Chicken (a) indigenous breed (b) improved breed

- (g) Helps to improve the standard of living in human, and care for animals and plants. Skills from Biology can help to determine the causes, symptoms, and methods of transmission of infections and diseases.
- (h) Helps to answer important questions, such as when and where life originate, how plants move, and how animals survive without plants.

- (i) Enables scientists to come up with ways to prevent, treat, and cure diseases. This has greatly improved the quality of people's lives. For instance, the first cure discovered for malaria was quinine. It was extracted from the bark of the Cinchona tree (or quinine tree). Quinine has helped to reduce the number of deaths from malaria, especially in Africa.

- (j) Lays foundation for specialising in careers, such as teaching, nursing, medicine, horticulture, and environmental science.
- (k) Encourages international cooperation through biological research. For instance, biologists from all over the world cooperate to find cure for diseases, such as COVID-19, AIDS, and Ebola.

Relationship between Biology and other scientific fields

Task 1.2

Search for information from the library and reliable internet sources the relationship between Biology and other scientific fields, then write short notes.

Biology is not an isolated field of study. Biology is related to many other fields of study, such as Agriculture, Medicine, Pharmacy, Veterinary Medicine, and Nutrition. Biological researches have a direct impact on the development of these fields.

Activity: Investigating the relationship between Biology and other fields of science

Materials: Newspapers and magazines with articles on Biology, Agriculture, Medicine, Forestry, and other scientific fields, ICT tools, notebook, and pen

Procedure

1. Collect newspapers and magazines with articles on Biology, Agriculture, Medicine, Forestry, Nutrition, and other fields of science.
2. Search from the library or internet sources the articles on Biology, Agriculture, Medicine, Forestry, Nutrition, and other scientific fields.
3. Go through the articles and state which other fields of study, apart from Biology, are mentioned.
4. Relate Biology to other fields of science, such as Agriculture, Medicine, Forestry, and Nutrition.

Question

Explain the fields that you think are related to Biology. Give examples to explain these relationships.

Agriculture

Agriculture is the field of study which deals with growing crops and rearing animals for food, money, or both. Biological research findings on crops and livestock have led to improved agricultural production. Scientists have developed breeds of crops and animals that mature quickly and produce high yields. For instance, indigenous breeds of chicken start laying eggs when

they are about eight months old while the improved breeds known as *layers* can start laying eggs as early as five months. An example of an improved layer breed is shown in Figure 1.3.



Figure 1.3: Improved layer breed

Researchers have also developed breeds of crops and animals that perform well under specific climatic conditions. For example, researchers have been able to develop varieties of maize seeds that are resistant to diseases and that grow well even in harsh conditions.

Medicine and pharmacy

Medicine is the study of the prevention, treatment, and cure of diseases. Pharmacy is the science of preparation and administration of drugs. Anatomy and Immunology form an important part of the study of medicine.

The study of Anatomy helps medical doctors to learn the structure of the body and how it functions. This makes it possible for them to know what is wrong in the body and how it can be treated. In Immunology, scientists study the body's response to disease-causing agents. This knowledge is important in the formulation of vaccines and drugs that are able to prevent and treat diseases, respectively. Some of these drugs and vaccines are made from animals, plants, fungi, or micro-organisms.

Nutrition

The study of Biology is important in understanding the composition and value of different groups of foods. This knowledge is used by dieticians to determine the kind of diets that are suitable for people with different health needs.

Biologists have also developed breeds of livestock and crops that yield products of higher nutritional value. Examples include varieties of sweet potatoes that are rich in vitamin A and a breed of Jersey cattle, as shown in Figure 1.4, which produce milk with low butterfat content. Such breeds are known as improved breeds.



Figure 1.4: Jersey cow

Forestry

Forestry deals with creating, managing, and conserving forests. Forests are sources of fuel (firewood and charcoal), building materials, and medicinal plants. They also serve as water catchments. Biologists have developed varieties of trees that grow well in dry areas. Such trees are used to reclaim dry land for human settlement. Through

Biological research, varieties of fast-growing trees have been developed. The fast-growing trees are planted in deforested areas. This is called reforestation. Reforestation helps to preserve water catchments so as to ensure a continuous supply of water. Forests also prevent desertification. See Figure 1.5 (a) and (b).



(a)



(b)

Figure 1.5: (a) Deforestation due to excessive cutting of trees (b) planted trees

Chapter summary

1. Biology is the study of living things. People who study Biology are called Biologists.
2. The main branches of Biology include Botany, Zoology, and Microbiology.
3. The cell is the basic functional unit of life.

4. The study of Biology is important because:
 - (a) It helps us to gain knowledge about our environment and how to improve it.
 - (b) It is a foundation for specialising in careers, such as nursing, medicine, and teaching.
 - (c) It helps us to acquire research skills, such as measuring, observing, analysis, and conclusion making skills.
 - (d) It enables researchers to produce improved breeds of crops and animals.
 - (e) It enables scientists to come up with the ways of preventing and treating diseases.
 - (f) It encourages international cooperation in biological researches.
5. Biology is related to other fields of study, such as Agriculture, Medicine, Pharmacy, Veterinary Science, Nutrition, and Forestry.

Revision exercise 1

Choose the most correct answer.

1. Which of the following fields of study is NOT related to Biology?
 - (a) Agriculture
 - (b) Astrology
 - (c) Medicine
 - (d) Nutrition
2. Which of the following is a reason for studying Biology?
 - (a) Biology helps us to appreciate and improve nature.
 - (b) Biology deals with non-living objects in our environment.
 - (c) Biology enables us to pollute the environment.
 - (d) Biological research contributes to the development of low-yielding breeds of crops.
3. The branch of Biology that deals with microbes is called:
 - (a) Zoology
 - (b) Microbiology
 - (c) Botany
 - (d) Cytology

4. Which of the following is one of the main branches of Biology?
 - (a) Sociology
 - (b) Agriculture
 - (c) Anthropology
 - (d) Botany
5. Outline four reasons as to why it is important to study Biology.
6. Identify the field and branch of Biology that you will use to solve the following problems:
 - (a) Helping a person with a broken leg.
 - (b) Protecting cattle from tsetse flies.
 - (c) Conserving village river from drying.
7. Differentiate:
 - (a) Unicellular from multicellular organisms
 - (b) Cytology from taxonomy



Chapter Two

Scientific processes in Biology

Introduction

Science is studied through a procedure that follows systematic processes, which require the use of a variety of apparatus and equipment. Among the processes that are conducted to carry out scientific studies include observation, measurement, and experimentation. In this chapter, you will learn about different biological apparatus and equipment, and the use of sense organs to make correct observation. You will also learn how to measure mass, length, and temperature. Likewise, you will learn the steps for carrying out scientific studies, commonly known as the scientific methods. The competencies developed will enable you to apply scientific methods, appropriately use biological apparatus, make accurate observations using sense organs, and perform accurate measurements in various daily life situations.



Think

Studying Biology without conducting experiments

Common Biology laboratory apparatus, equipment and other resources

Task 2.1

Search from reliable internet sources the video or images showing biology apparatus and equipment and their uses. Note down what you have observed.

Biology laboratory is a special room or building designed for carrying out biological experiments. A **laboratory** should have good qualities and facilities which include large door which open outwards, adequate space for carrying out experiments, proper lighting, good ventilation, a source of water, a source of heating, and adequate space for storing apparatus, chemicals, models, and specimens.

A wide variety of apparatus (singular is apparatus) and equipment are needed in order to conduct Biology practical. The following are some common Biology laboratory apparatus, equipment and resources:

A hand lens

This is mainly used to magnify specimens that are under observation. A hand lens enables a scientist to magnify small organisms or their parts in order to observe them clearly. An example of a hand lens is shown in Figure 2.1.



Figure 2.1: Hand lens

Scientists usually make drawings of objects which they observe using a hand lens. The size of drawing that they make is usually larger or smaller than the actual object.

Magnification is calculated as follows:

$$\text{Magnification of a drawing} = \frac{\text{Size of drawing}}{\text{Size of an actual object}}$$

For example, if the length of the drawing is 8 cm and the actual length of the object is 2 cm, then:

$$\text{Magnification of the drawing} = \frac{8}{2} = 4$$

Sweep nets

A sweep net is used for catching small flying organisms, such as butterflies, grasshoppers and other small animals. An example of a sweep net is shown in Figure 2.2.



Figure 2.2: Sweep net

Scoop nets

These are used for catching fish and other aquatic animals. An example of a scoop net is shown in Figure 2.3.



Figure 2.3: Scoop net

Pooter

This is a device that scientists use to pick up small organisms, such as insects without hurting them. The pooter has two tubes. The scientist sucks through one tube and the organism is drawn into the pooter through the other tube. An example of a pooter is shown in Figure 2.4.



Figure 2.4: Pooter

Quadrat

A quadrat is a wooden square or metallic grid which is used to estimate

the number of organisms in an area. An example of a quadrat is shown in Figure 2.5.

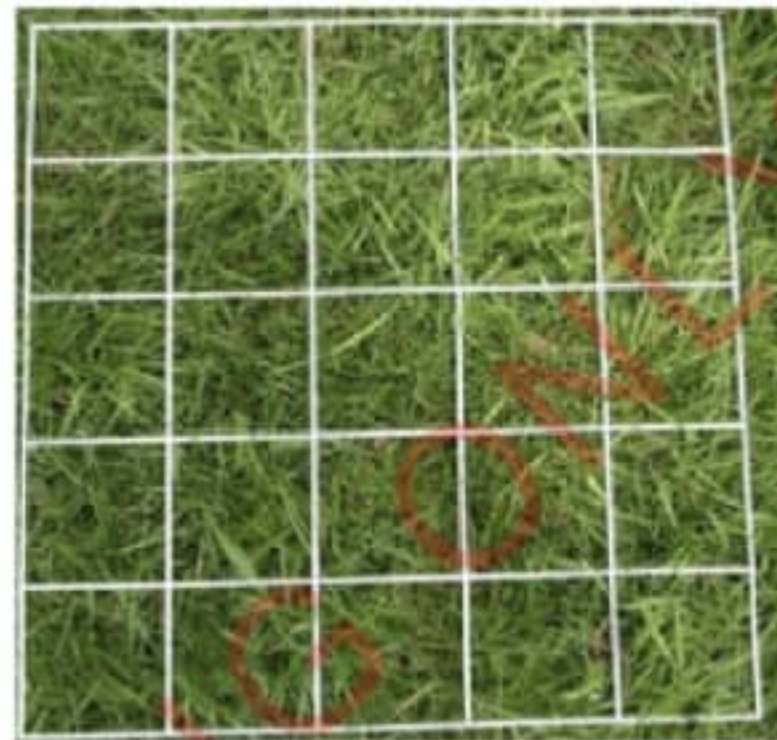


Figure 2.5: Quadrat

Specimen bottle

A specimen bottle is a glass or plastic container in which specimens are kept for storage. An example of a specimen bottle is shown in Figure 2.6.



Figure 2.6: Specimen bottle

A dissecting kit

A dissecting kit contains tools for dissection of specimens. An example of a dissecting kit is shown in Figure 2.7. Some of the items included in the dissecting kit are:

- (a) Forceps for holding specimens.
- (b) A pair of scissors for cutting specimens or objects.
- (c) Scalpels for cutting.
- (d) A needle for loosening internal parts.
- (e) Pins for holding specimens in place.
- (f) Hand lens for magnifying small specimens or their parts.



Figure 2.7: Dissecting kit

Thermometer

A thermometer is an instrument used to measure temperature. An example of a thermometer is shown in Figure 2.8.



Figure 2.8: Thermometer

Heat sources

Heat sources are used for heating various things in the laboratory. Examples of heat sources in the laboratory are Bunsen burner, kerosene stove, and spirit burner, as shown in Figure 2.9.



(a)



(b)



(c)

Figure 2.9: Heat sources (a) Bunsen burner (b) kerosene stove (c) spirit burner

Mounting needle

A mounting needle is used to lift small delicate specimens. An example of a mounting needle is shown in Figure 2.10.



Figure 2.10: Mounting needle

Microscope slide and coverslip

The microscope slide is a small flat piece of glass that is used to hold specimen under observation. A specimen to be observed under a microscope is placed on a slide with a drop of a liquid and covered with a coverslip. An example of a microscope slide and coverslip is shown in Figure 2.11.

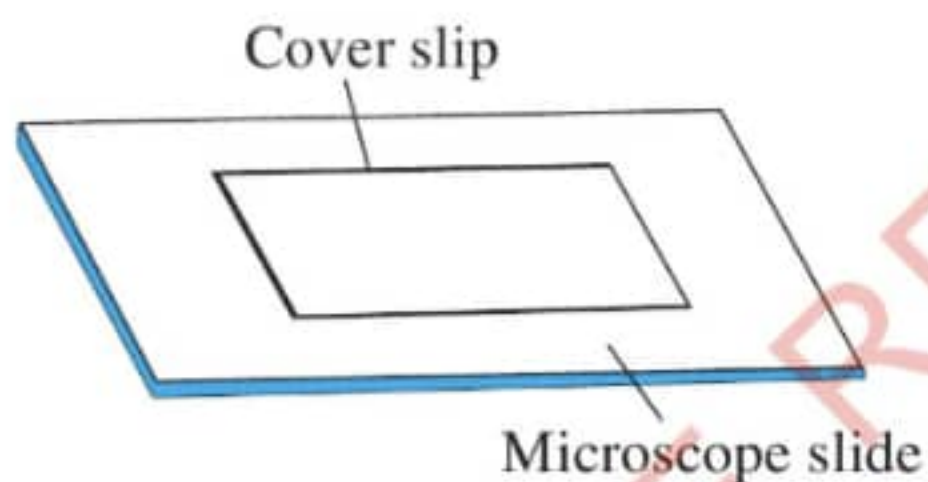


Figure 2.11: Microscope slide and coverslip

Microscope

A microscope is an instrument that is used during scientific studies to magnify very small specimens so that their details can be seen. The smallest structures of cells, tissues, and organs

of organisms can be studied well if magnified by the microscope. There are two types of microscopes: Electron microscope and Light microscope.

The electron microscope

The electron microscope uses beams of electrons to magnify specimens. The maximum useful magnification of most electron microscopes is about a million times (1 000 000 $\times$). These microscopes are very expensive. They are mostly found in research centres or research institutions.

The light microscope

The light microscope is the type of microscope that depends on light to illuminate and magnify tiny specimens. For this matter, it differs from the electron microscope by having lower magnification. The maximum useful magnification of most light microscopes is about one thousand times (1 000 $\times$). This type of microscope is commonly used in schools and health facilities. An example of the light microscope is shown in Figure 2.12.

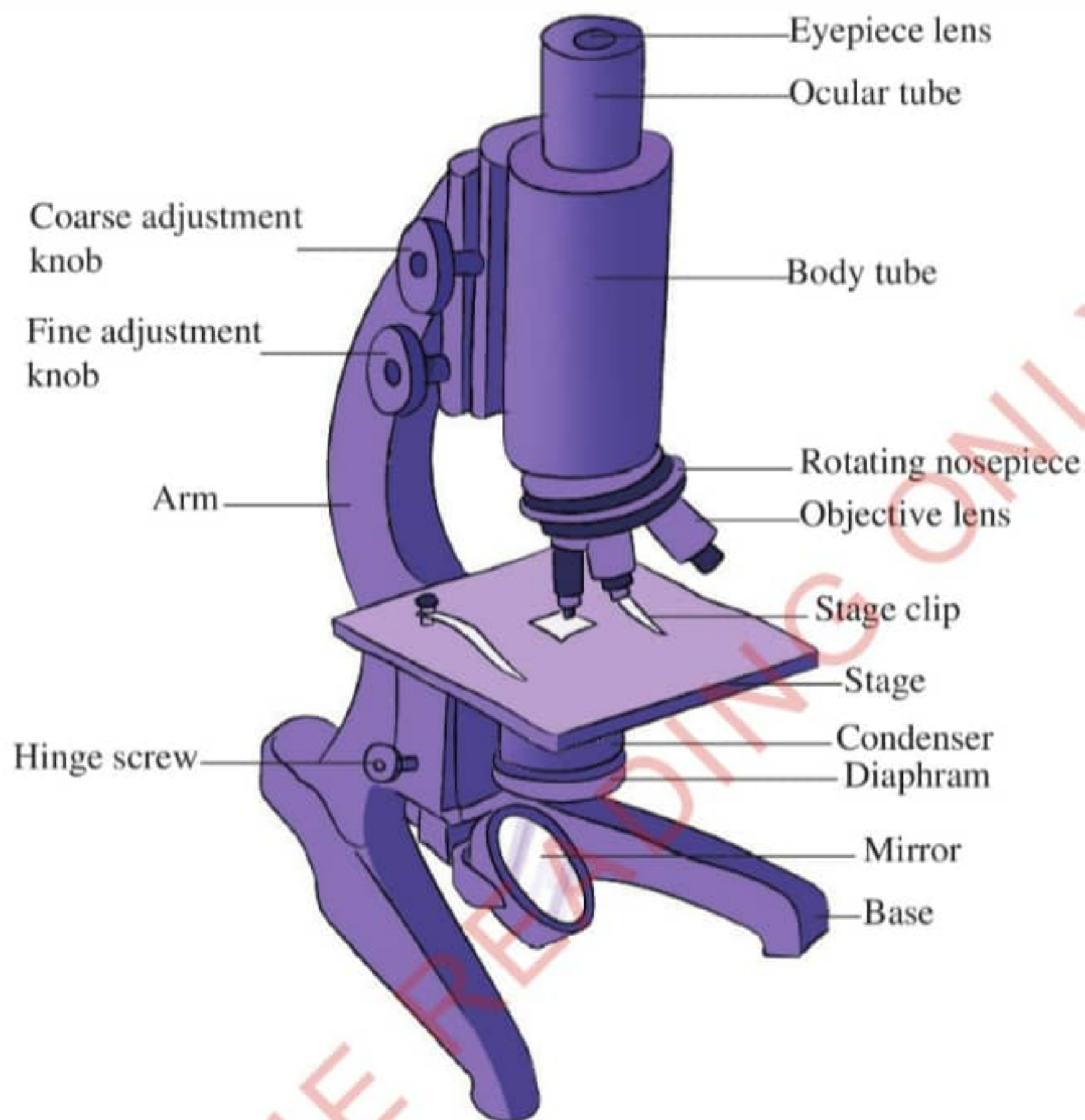


Figure 2.12: Parts of the light microscope

The parts of the light microscope are described in Table 2.1.

Table 2.1: Parts of the light microscope and their functions

Part	Function
Eyepiece	Has a lens that magnifies the specimen five times ($5\times$), ten times ($10\times$), or fifteen times ($15\times$)
Body tube	Supports the eyepiece and the rotating nosepiece
Rotating nosepiece	Supports the objective lenses. Rotates to allow changes from one lens to another
Objective lenses	Magnify the object under observation. Light microscope usually has three objective lenses: low power, medium power, and high power. A high-power lens reveals more details of the specimen
Coarse adjustment knob	Rises or lowers the body tube in order to bring the image into focus

Part	Function
Fine adjustment knob	Rises or lowers body tube in order to bring the image into sharp focus
Arm	Supports the body tube, knobs, stage, diaphragm and mirror. It is one of the parts that is held when moving the microscope
Ocular tube	Allows light to pass straight from the objective lens to the eyepiece lens
Stage	Surface on which the specimen is placed
Stage clip	Holds the slide containing the specimen in place
Mirror	Reflects and directs light to the specimen under observation
Condenser	Concentrates light onto the specimen that is placed on the stage
Diaphragm	Regulates the amount of light passing from the mirror to the condenser
Hinge screw	Rises and lowers the stage to keep the specimen in the correct position for observation
Base	Provides firm support for the microscope

How to use the light microscope

- Place the microscope on the laboratory bench or table. Make sure it is not too close to the edge. Position the microscope such that the arm faces you and the stage faces away from you.
- Mount the specimen on the microscope slide. Cover it with a coverslip.
- Make sure that the low objective lens is in line with the eyepiece lens.
- Place the slide with the specimen on the stage. Hold it in place with the stage clip.

- While looking through the eyepiece, use your hand to adjust the mirror so that the light is directed to the specimen on the stage. Always keep both eyes open when looking through the eyepiece.
- Adjust the coarse adjustment knob to bring the specimen into focus.
- Adjust the fine adjustment knob to bring the specimen into a sharp focus.

Total magnification = eyepiece magnification × objective lens magnification

For example, if the eyepiece magnification is $5\times$ and the objective lens magnification is $40\times$, then the total magnification is $5 \times 40 = 200\times$

- (h) Rotate the nosepiece to a higher power objective lens, if you want to observe more details of the specimen. Figure 2.13 shows a student using the light microscope.

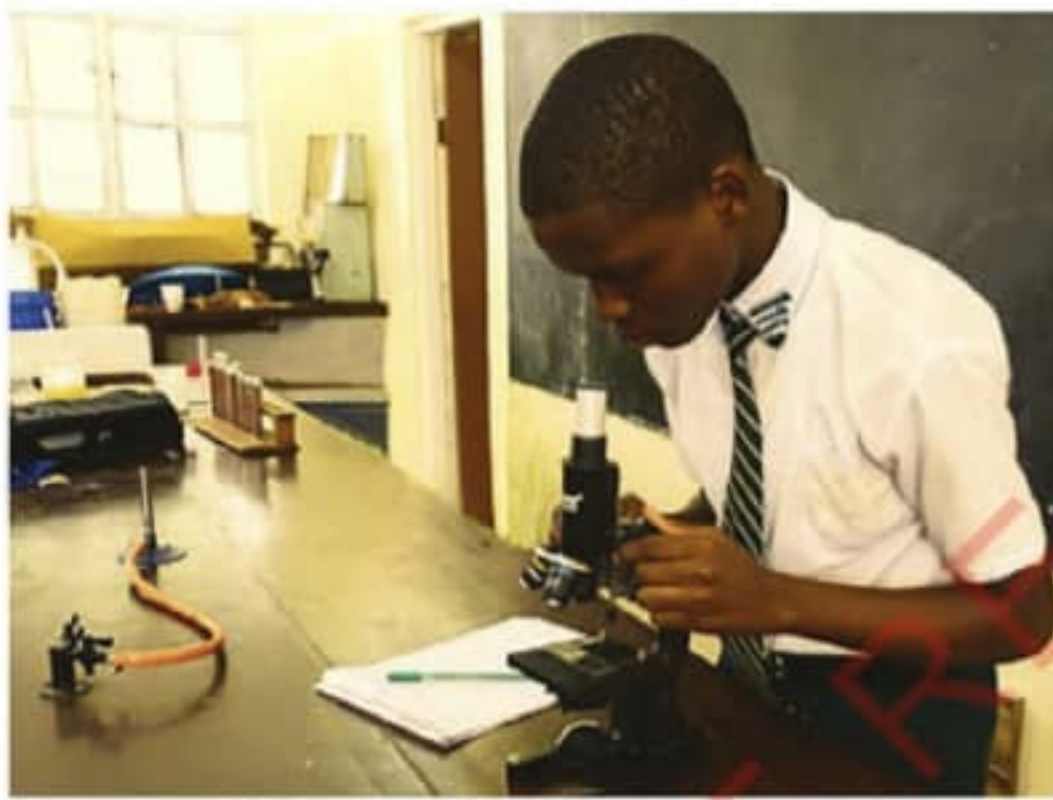


Figure 2.13: Student using the light microscope

How to care for the light microscope

Turn the adjustment knob slowly. Always start with the coarse adjustment knob.

- (a) Lift the microscope using both hands, one hand holding the arm and the other holding the base.
- (b) When not in use, cover the microscope with clean cloth and store it in a dry, dust-free place.
- (c) Clean dirty lenses using lens tissue or a soft cloth.
- (d) Do not place the microscope at the edge of the table or bench as it could be knocked over.

- (e) When the microscope is not in use for a long time, remove the lenses and put them in a desiccator.
- (f) Do not touch the microscope with wet hands.
- (g) Do not touch the surface of the mirror or the lenses with your fingers.
- (h) Lubricate the moving parts regularly.
- (i) Always use the coverslip in order to protect the lenses of the microscope from getting into contact with the specimen.
- (j) Remove the slides from the stage immediately after use.
- (k) Move the objective lenses up when storing the microscope.
- (l) When storing the microscope, rotate the rotating nosepiece so that the low magnification lens is in line with the eyepiece lens.

Activity 2.1: Using the light microscope

Materials: Light microscope, microscope slide and coverslip, razor blade, knife or scalpel, onion bulb, forceps, mounting needle or glass rod, iodine solution, and blotting paper

Procedure

1. Use the knife to cut the onion bulb vertically or lengthwise into four pieces.

Safety precaution

Take precautions when using sharp objects, such as knife and razor blade.

2. Obtain a fresh leaf from one of the quarters.
3. Use a pair of forceps to peel a thin layer from a leaf, as shown in Figure 2.14.



Figure 2.14: Peeling a thin layer from an onion leaf

4. Trim the thin layer to approximately 5 mm x 5 mm using the scalpel or knife.
5. Put a drop of water on the microscope slide. Place the trimmed thin layer of onion leaf in the drop of water using a glass rod or a mounting needle, as shown in Figure 2.15.



Figure 2.15: Placing the thin layer of onion leaf on the slide

6. Gently cover the specimen with a coverslip making sure that no air bubbles are trapped inside, as shown in Figure 2.16.



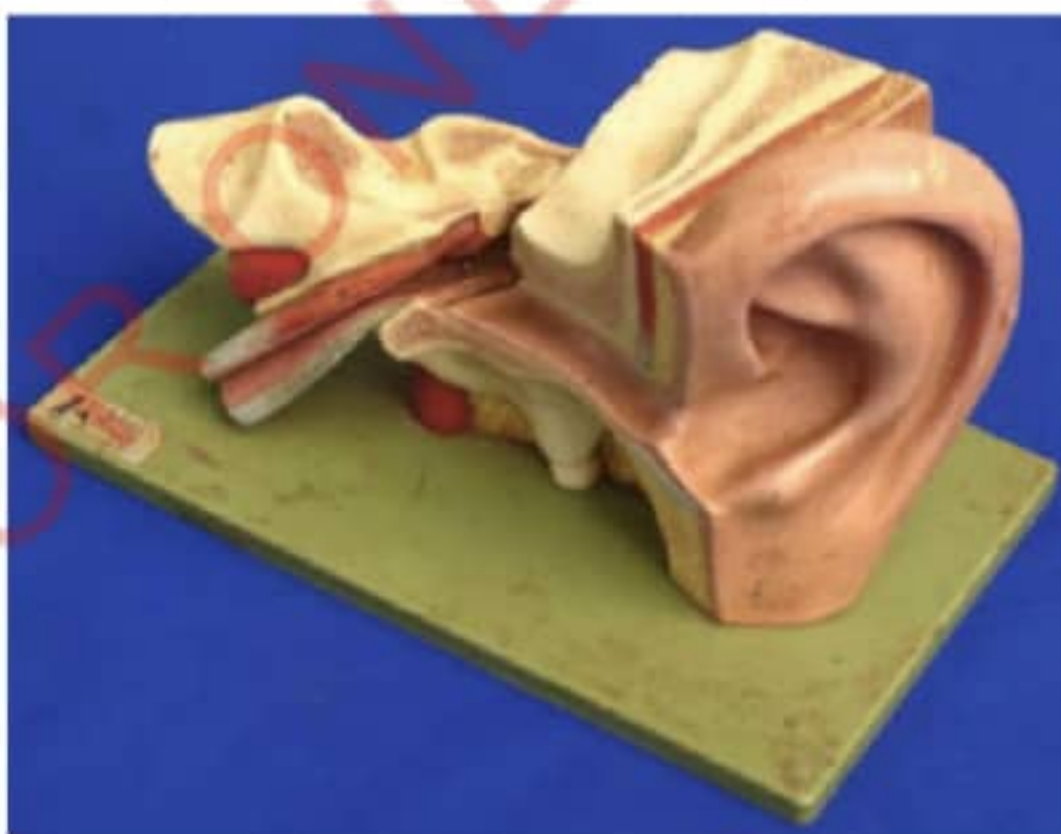
Figure 2.16: Covering the specimen using the coverslip

7. Dry any excess water from the sides of the coverslip using a blotting paper.
8. Mount the slide on the stage of the light microscope. Hold it in place with the stage clips.
9. Observe the specimen under the low power objective lens, then observe under the medium power objective lens.
10. Remove the specimen from the microscope stage. Remove the specimen from the slide and put a new specimen. Put a drop of iodine on the specimen. Repeat the procedures. Iodine makes the specimen clear. Use blotting paper to eliminate any excess iodine.

11. Observe the specimen again under low power and medium power objective lenses.
12. What did you observe?

Biological models

Biological models are sculptures that represent a particular organ or system of an organism. They are usually made in three dimensions, hence representing the real part of an organism. Models are used during teaching and learning process to reinforce concepts. Examples of models that are commonly used in teaching and learning Biology, include models of the human eye, lungs, heart, ear, and skeleton. Most of the models used in Biology are made up of plastic materials. Examples of biological models are shown in Figures 2.17 (a) and (b).



(a)



(b)

Figure 2.17: Models of (a) the human ear (b) the human skeleton

Preserved specimens

Preserved specimens are collected organisms or parts of organisms that are preserved for learning purposes. Organisms or parts of organisms that are usually preserved include plant leaves, roots, fruits as well as insects, and other small animals. Figure 2.18 shows examples of preserved specimens.



Figure 2.18: Preserved specimens

Cages

A cage is an enclosure often made up of wire mesh or bars or wood, in which birds, mice, rabbits, or other small animals are kept, as shown in Figure 2.19.



Figure 2.19: Rabbit in a cage

Aquarium

Aquarium is a special vessel that is used in the laboratory for keeping live aquatic organisms, such as fish, as shown in Figure 2.20. Aquaria can be of different shapes and sizes.



Figure 2.20: Aquarium

Basic skills in scientific studies

Task 2.2

Search from library and reliable internet sources the information on basic skills in scientific studies. Write short notes.

There are basic skills that are essential in scientific studies. The skills are needed in studying Biology. They include observation, measurement, and experimentation.

Observation

Observation is one of the methods of studying Biology. Through observation one can study living things in many ways. Observation is done by using **sense organs**. An example of using sense organs in making observations is shown in Figure 2.21.

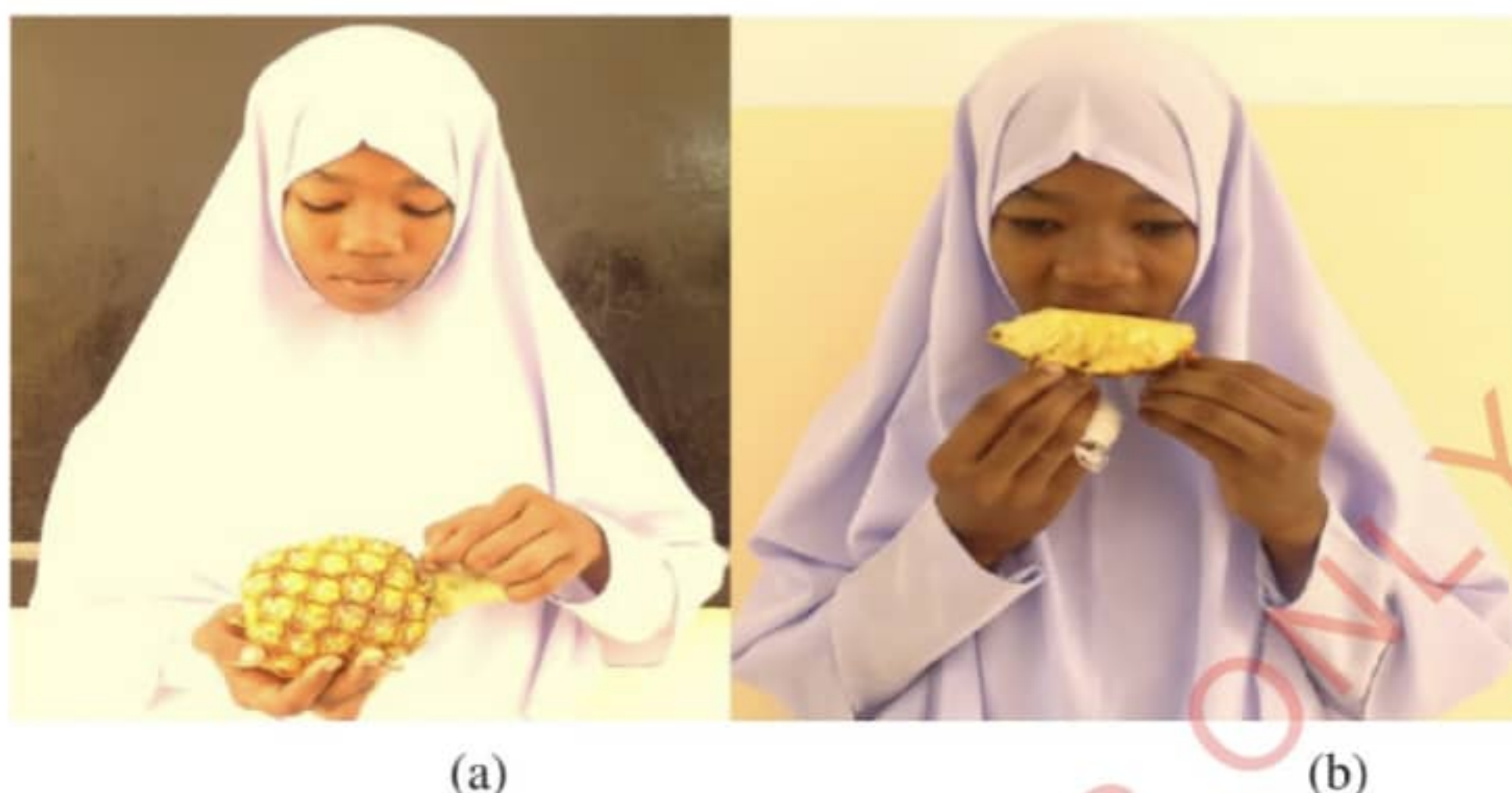


Figure 2.21: Using sense organs to make observation (a) seeing and touching
(b) tasting

We have five sense organs. These are eyes, ears, skin, nose, and tongue. The observation process with the help of sense organs enables you to study living organisms. For instance, if you have a coconut, you can use your sense organs to determine that:

- (a) It is round by seeing,
- (b) It contains a liquid by hearing the sound when shaken,
- (c) It has a rough husk by touching,
- (d) It has a good smell by smelling (after breaking the coconut), and
- (e) It has a sweet taste by tasting (after breaking the coconut).

Activity 2.2: Making observations using sense organs

Materials: Colourful pictures, insects, preserved specimens, flowers, perfume, whistle, bell, stones, wooden objects, pieces of cloth, ice, hot water, and lukewarm water

Procedure

1. Carefully observe the specimens provided using your sense organs.

Safety precautions

Do not taste anything in the laboratory. They may be contaminated with poisonous materials or they may be poisonous themselves.

2. Note down the features of each specimen you have observed.
3. Draw a table similar to Table 2.2 in your exercise book. Record the observations by putting a tick (✓) on each object.

Table 2.2: Observation of objects using sense organs

Object	Sense			
	Sight	Hearing	Smell	Touch
1. Flower				
2. Perfume				
3. Whistle				
4. Cloth				

The use of sense organs to make observations has many advantages in everyday life. For example, when you see dark clouds, you know that it is likely to rain. When you hear the school bell ringing in the morning, you know it is time to begin the school routine. You like to wear a perfume because it smells nice. When you get near fire, you feel the heat and move away. When cooking you taste food to ensure it is delicious.

However, one of the disadvantages of observations made using the senses is that they are often subjective. This means that they are based on a person's interests, opinions, or situation rather than facts. For instance, one may feel a bowl of water lukewarm, while another may feel it warm and yet

another person may feel it hot. For this reason, scientists use standard units of measurement when carrying out scientific investigations. These units remain constant everywhere.

Measurements

Scientists use specific instruments and units of measurement in their investigations. The standard system of measurement used by scientists all over the world is called the *International System of Units (SI units)*. This system ensures precision in the presentation of scientific results and the uniformity of data presentation internationally. It also enables scientists to confirm results from other parts of the world. Table 2.3 shows some common measurements in Biology and their units.

Table 2.3: Fundamental quantities

Measurement	SI unit	Symbol	Other common units
Mass	Kilogram	kg	Grams (g)
Length	Metre	m	Millimetre (mm), Centimetre (cm) Kilometre (km)
Temperature	kelvin	K	Degrees Celsius ($^{\circ}\text{C}$), Degrees Fahrenheit ($^{\circ}\text{F}$)
Time	Second	s	Minutes (min), Hour (hr)

Mass

Mass is the quantity of matter of an object. Everything in the world is made up of matter. Mass is measured using a weighing balance or a weighing scale. It is often expressed in grams (g) or kilograms (kg), whereby $1000\text{ g} = 1\text{ kg}$. The weighing balance can be manual or digital. Examples of weighing scales are shown in Figure 2.22 and 2.23.



Figure 2.22: Weighing scales (a) beam balance (b) digital scale



Figure 2.23: Using a beam balance to measure mass

Activity 2.3: Measuring the mass of objects

Materials: Weighing balance, weights (0.5 kg, 1 kg, 2 kg, and 5 kg), stones, flour, sugar, notebook, coins, textbook, and pencil or pen

Procedure

1. Put a weighing scale on a flat surface, such as a table.
2. Put half a kilogram weight on the platform.
3. Add a small amount of flour until the weighing scale balances.
4. What is the mass of the flour you have measured?
5. Record your results in your notebook.
6. Repeat procedures 2, 3, and 4 above using different weights to measure the different materials you have collected.
7. Prepare a short report of your results.

Length

Length is a measurement of the distance or dimension from one point to another. For example, you can measure the height of a human being, and plant, as well as the length of a leaf and fish. Figure 2.24 shows examples of instruments for measuring length and Figure 2.25 shows how to use one of the length measuring instruments.

Length can be measured using a ruler or tape measure. Common units of expressing length are millimetres (mm), centimetres (cm), metres (m), and kilometres (km). These units of length can be converted into other units, as shown in Table 2.4.

Table 2.4: Conversion of some measurements

10 mm = 1 cm	1000 m = 1 km
100 cm = 1 m	100 000 cm = 1 km



(a)



(b)



Figure 2.24: Instruments for measuring length (a) ruler (b) tape measures



Figure 2.25: Measuring the height of a student

Activity 2.4: Measuring length of objects

Materials: Ruler, paper, a leaf or a small dried fish, notebook, and a pen or pencil

Procedure

1. Take a leaf and put it on a piece of paper. Trace the outline of the leaf.
2. Put two marks at the extreme points of the leaf outline.
3. Join the two points by a straight line.
4. Put the zero mark of the ruler at first mark and read the length at the second point.
5. Record the measurement in your notebook.
6. Compare your findings with those of your classmates. Did you get similar measurements?

Temperature

Temperature is the degree of hotness or coldness of a substance. It is a physical property that explains the common notions of hot and cold. The common units for measuring temperature are kelvin (K), degrees Celsius ($^{\circ}\text{C}$), and degrees Fahrenheit ($^{\circ}\text{F}$). These units relate as follows:

$$K = ^{\circ}\text{C} + 273.15$$

$$^{\circ}\text{F} = \left[\left(\frac{9}{5} \right) \times ^{\circ}\text{C} \right] + 32$$

$$^{\circ}\text{C} = \frac{5}{9} (F - 32)$$

The normal human body temperature is between 36°C and 37°C .

Thermometer is used to measure temperature. It can measure human body temperature and temperature of liquids, such as water. The boiling

point of water is 100°C and its freezing point is 0°C . Figure 2.26 shows an example of using a thermometer to measure temperature.



Figure 2.26: Using thermometer to measure temperature of water

Activity 2.5: Measuring the temperature of different things

Materials: Thermometer, beakers, hot water, cold water, and ice cubes

Procedure

1. Carefully take a thermometer from its case and hold it at the top. Do not touch the bulb.
2. Shake it for a few seconds, then read the number that corresponds to the mark.
3. Put the thermometer bulb under your armpit and wait for two minutes.
4. Read the number that corresponds to the mark.

5. What is your body temperature?
6. Use the thermometer to measure the temperature of hot water, cold water, and the ice cubes by dipping the bulb into a container with the substance you want to measure. Observe the movement of the mark inside the thermometer until it stops moving. Record the number that corresponds to the mark.
7. Record the temperature in degrees Celsius in your notebook.
8. Compare your results with those of your classmates. Are they similar?

Scientific methods

Task 2.3

Search from library and reliable internet sources the information on scientific methods.

Science is the knowledge and systematic study of the structure and behaviour of the natural and physical world. It is based on facts that can be proven through observation and experiments. In order to study situations and come up with solutions to the problems, scientists follow a set of steps called the scientific method. The scientific method is a way of studying things by testing facts systematically. It follows seven steps which are: identifying the problem, formulating a hypothesis, experimentation, observation and recording data, data analysis and interpretation, conclusion, and reporting results.

Identifying the problem

This is where a scientist makes a puzzling observation. The observation can be made in the environment or from the work of other scientists. An example of such an observation is: 'Earthworms are mostly found in cool and well-shaded soil'. After making an observation, the scientist asks questions that can be answered by gathering evidence. For example: 'What conditions make earthworms prefer cool shaded areas to open well-lit areas?'

Formulating a hypothesis

A hypothesis (plural is hypotheses) is a suggestion of the answer to the question asked. It is an intelligent guess that tries to explain an observation. For example: 'Earthworms prefer cool well-shaded areas because these areas have higher moisture than open well-lit areas.'

Experimentation

An experiment is a scientific procedure carried out under controlled conditions to determine whether a hypothesis is correct or not. In an experiment, a scientist uses **variables** to test a hypothesis. Variables are conditions or factors that can change or can be changed. Examples of variables include temperature, speed, and light. They can be classified as dependent, independent, and controlled variables.

Dependent variable: This is the condition or factor that is measured or tested to obtain the results of the experiment.

Independent or manipulated variable: This is a condition or a factor that the researcher changes systematically to obtain different results.

Controlled variable: This is the condition or factor that is kept constant during the experiment. For example,

Figure 2.27 shows an experimental set up that can be used to test if earthworms prefer moist areas.

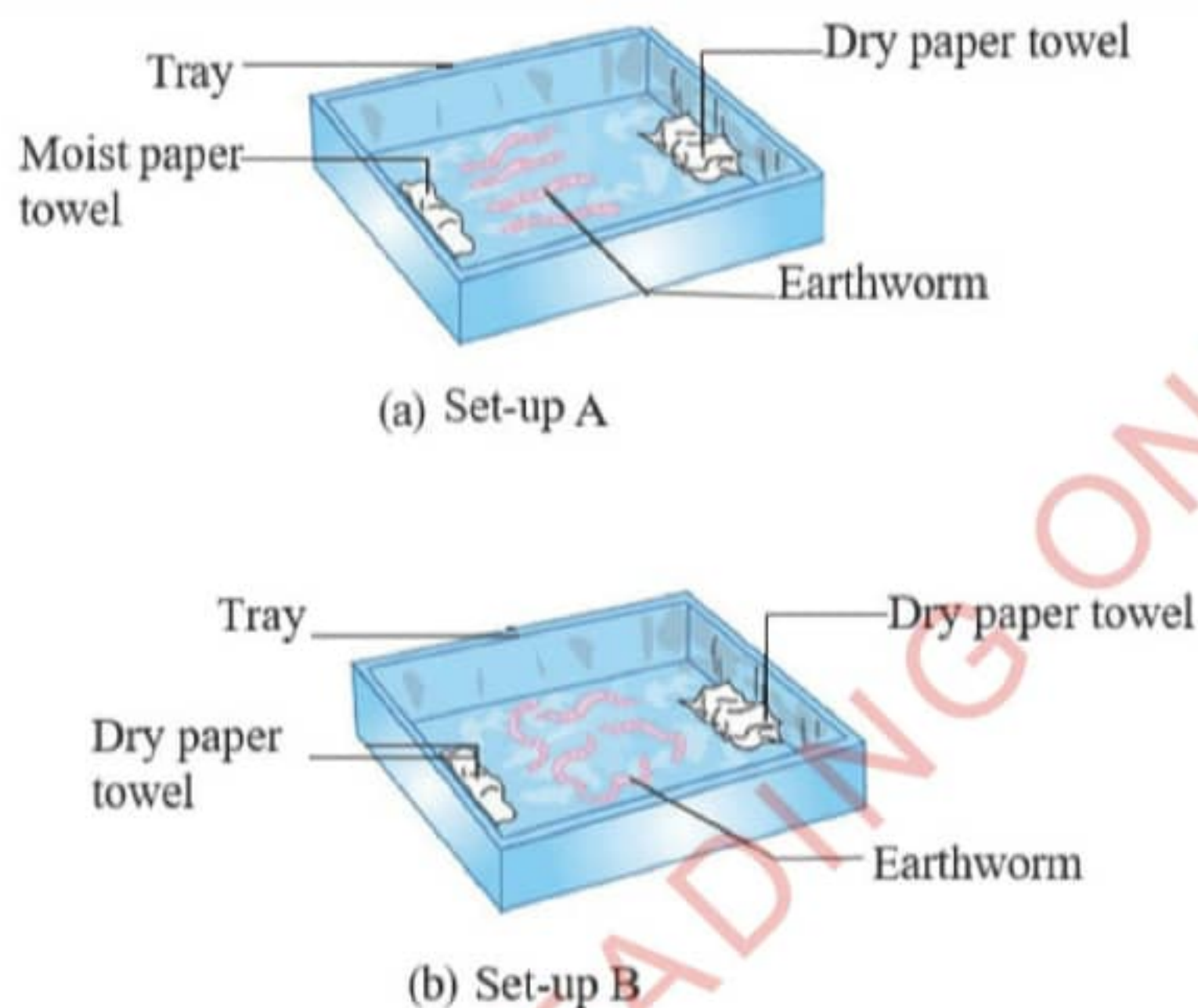


Figure 2.27: Experiment to investigate the conditions preferred by earthworms

In set-up A:

- The direction in which the earthworms move is the factor under investigation. This is the dependent variable.
- The moisture content of one of the paper towels is varied. This is the independent variable.
- The amount of light allowed is kept constant. This is the controlled variable.

Set-up B: is the **control experiment**. In this set-up, the paper towels are not moistened. The independent variable is not changed. The control experiment proves that the independent variable

determines the direction of movement of earthworms.

Observation and data recording

The scientist observes what happens from the time the experiment is set up to the time it ends. It is important to note all the changes made in the independent variable and the resulting changes in the dependent variable. For example, the results of the experiment in Figure 2.27 can be recorded in a table, as shown in Table 2.5 below. The results in the first row are given as an example.

Table 2.5: Results

Drops of water on paper towel	Movement of earthworms
0	Random
2	
4	
6	
8	
10	
12	

Data analysis and interpretation

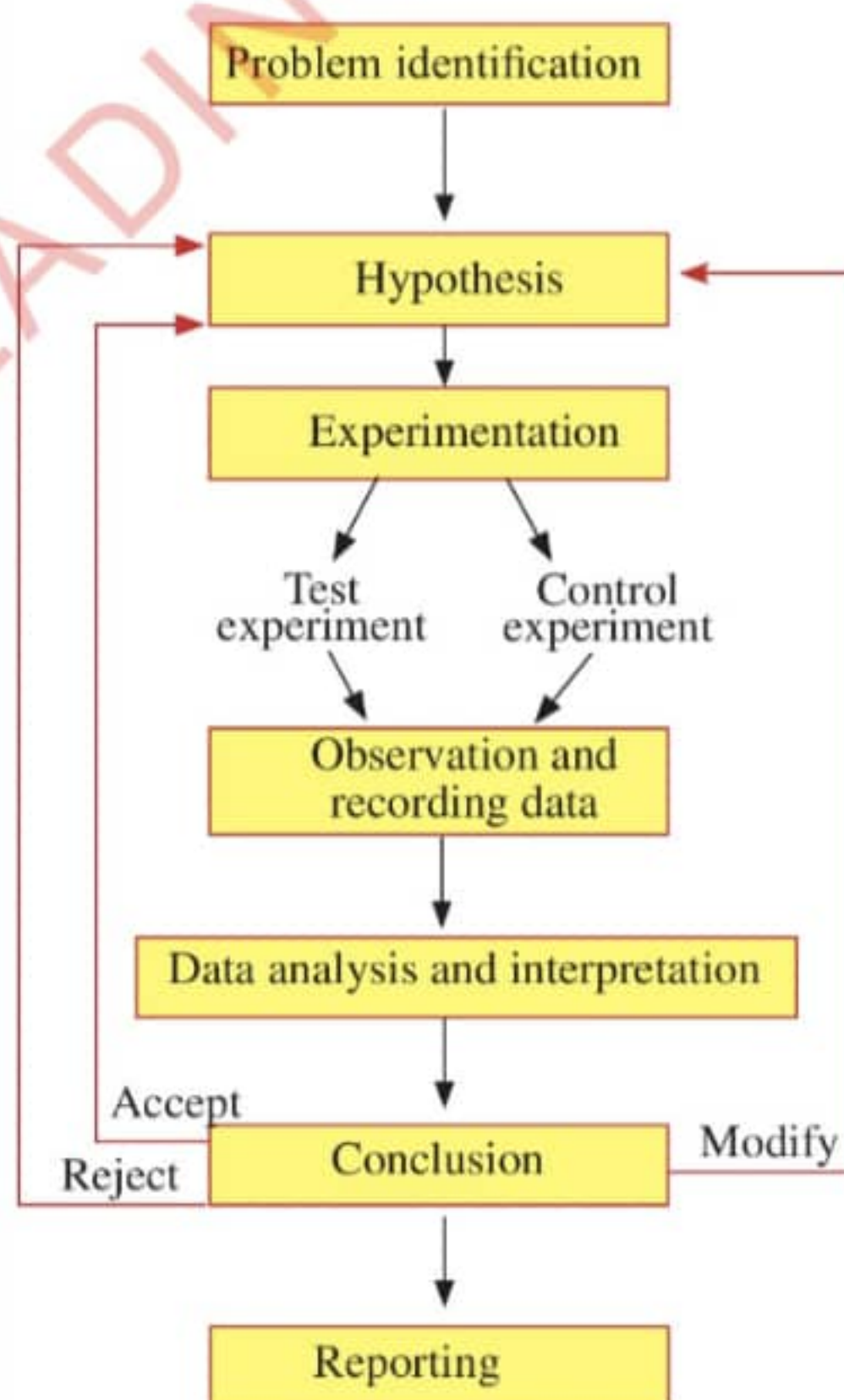
At the end of the experiment, the scientist analyses the observations and data recorded. The scientist may also look for patterns or trends in the data. For example, in this experiment it may happen that: 'As more drops of water were added to one paper towel, more earthworms moved to that side of the tray.'

Conclusion

A conclusion is a statement that summarizes what a scientist has learnt from an experiment. When scientists reach a conclusion, they state whether the data or information collected supports their hypothesis or not. Conclusions often lead scientists to pose new questions and plan new experiments to answer them.

It is important to remember that results of an experiment may prove

or disprove the hypothesis. From this experiment, it can be concluded that: 'Earthworms prefer moist conditions to dry conditions.' This confirms the hypothesis. Based on the conclusion, the hypothesis can be accepted, modified, or rejected. Another hypothesis is then formulated which suggests why the first hypothesis was correct or wrong. Hence, scientific research is a never-ending process, as shown in Figure 2.28.

**Figure 2.28:** Steps in the scientific methods**Reporting results**

Scientists communicate their results to others in a final report. They can present

the results to the scientific community by publishing their final report in a journal or at seminars, conferences, workshops, and meetings.

Note: If your results do not support the hypothesis:

- (a) Do not stop to repeat the experiment.
- (b) Give possible reasons for the difference between your hypothesis and the experimental results.
- (c) Give ideas for further investigations to find answers to the problem.

Simple biological experiments

Biological experiments can be done to study different features of living things. As you learn Biology, you will often carry out experiments to test hypotheses or to learn more about organisms.

Activity 2.6: Observing different types of plant leaves

Materials: Different types of plant leaves, hand lens, notebook, and a pen or pencil

Procedure

1. Collect different types of leaves from plants in the school or home environment.

Caution

Avoid collecting hairy or thorny plants and those known to be poisonous.

2. Examine each leaf closely as follows:
 - (a) Look at the size, shape, and colour.
 - (b) Feel the texture of the leaf.
3. Using a hand lens, observe the patterns on each leaf.
4. Note down what you have observed.

Questions

1. How are the leaves similar?
2. How are the leaves different?

Activity 2.7: Collecting and observing the characteristics of animals

Materials: Preserved specimens of insects and other small animals, sweep nets, a pen or pencil, notebook, bottles, and hand lenses

Procedure

1. Use sweep nets to catch insects, such as grasshoppers, termites, cockroaches, and butterflies around school compound, as shown in Figure 2.29. Sweep nets can also be used to catch small animals, such as millipedes and frogs.



Figure 2.29: Using sweep nets to collect insects and other small animals

- Put the collected insects and other small animals in specimen bottles.

Safety precaution

Care should be taken when collecting dangerous organisms such as wasps, scorpions, centipedes, and bees.

- Carefully observe the external features of the organisms you

have caught then use hand lens to observe the details of the same features. The features include number of legs, eyes, and the way they move, as shown in Figure 2.30.



Figure 2.30: Using hand lenses to observe insects and other small animals

- Record your observations in a notebook.
- Release or preserve the organisms you have used in your observation

Chapter summary

- Biological apparatus are instruments and equipment needed for effective studying of Biology.
- Some of the apparatus used in studying Biology are hand lens, sweep nets, scoop nets, pooters, specimen bottles, quadrats, dissecting kits, mounting needles, slides, coverslips, and microscopes.
- A microscope is an instrument that is used during scientific studies to magnify very small specimens so that their details can be seen clearly.
- There are two types of microscopes: light microscope and electron microscope.

5. The total magnification in a microscope is calculated using the following formula.

$$\text{Eyepiece magnification} \times \text{Objective lens magnification}$$

6. Observation, measuring, and experimentation are skills needed when studying Biology.
7. Sense organs used to make observations are:
- Eyes for seeing.
 - Nose for smelling.
 - Ears for hearing.
 - Tongue for tasting.
 - Skin for feeling.
8. Scientific measurements are taken using specific instruments and are expressed in specific units. Some basic biological measures are:

Measure	Instrument	SI unit
Mass	Weighing balance	Kilogram (kg)
Length	Ruler, tape measure	Metre (m)
Temperature	Thermometer	kelvin (K)

9. The scientific method is a set of steps that scientists use to study things systematically. It involves:
- Identifying the problem.
 - Formulating a hypothesis.
 - Experimentation.
 - Observation and recording data.
 - Data interpretation.
 - Conclusion.
 - Reporting results.
10. Simple biological experiments are performed to study different features of living things.

Revision exercise 2

1. Identify the apparatus or equipment you can use to perform each of the following biological activities:
 - (a) Catch a frog from a pond in order to study its external features.
 - (b) Collect insects and small animals from the school compound.
 - (c) Estimate the number of small organisms per square metre.
2. As a biology student, how would you help your friend to label and state the functions of the parts of the microscope labelled A-E in Figure 2.31?

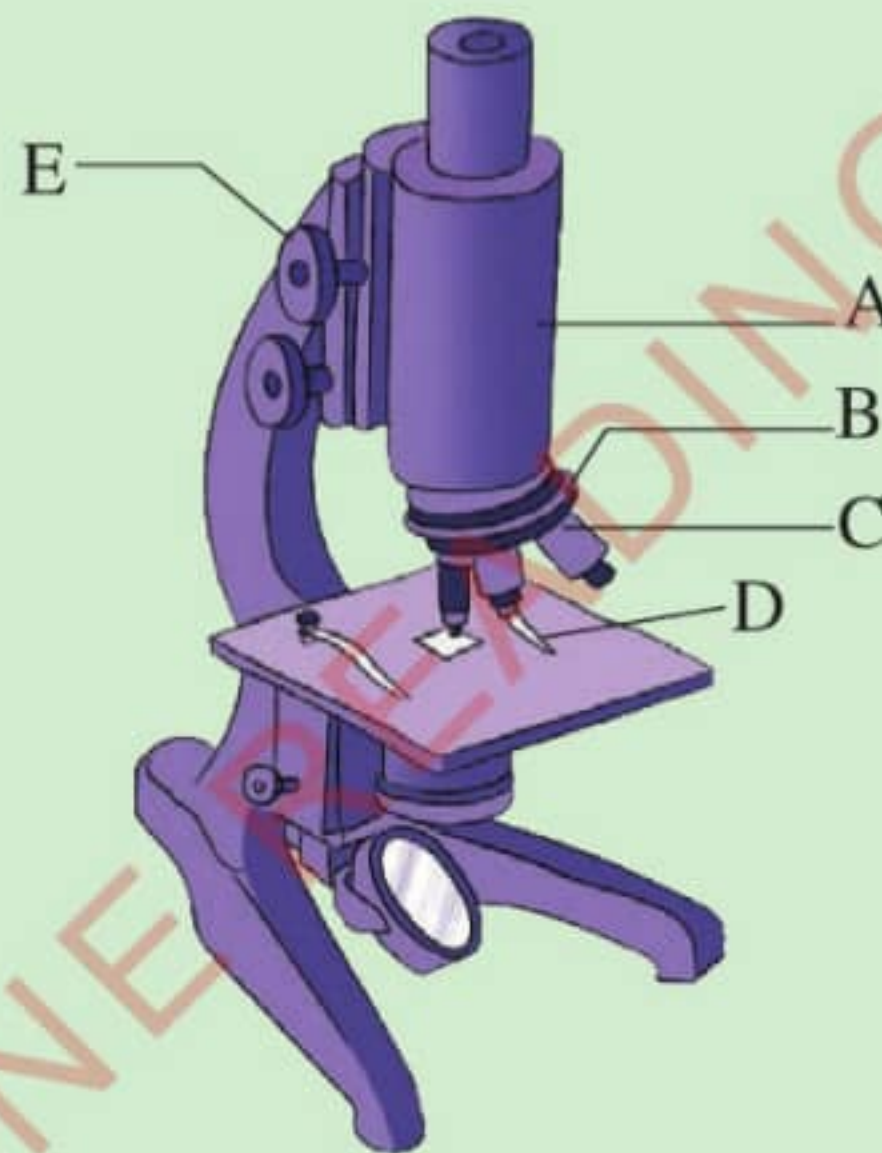


Figure 2.31

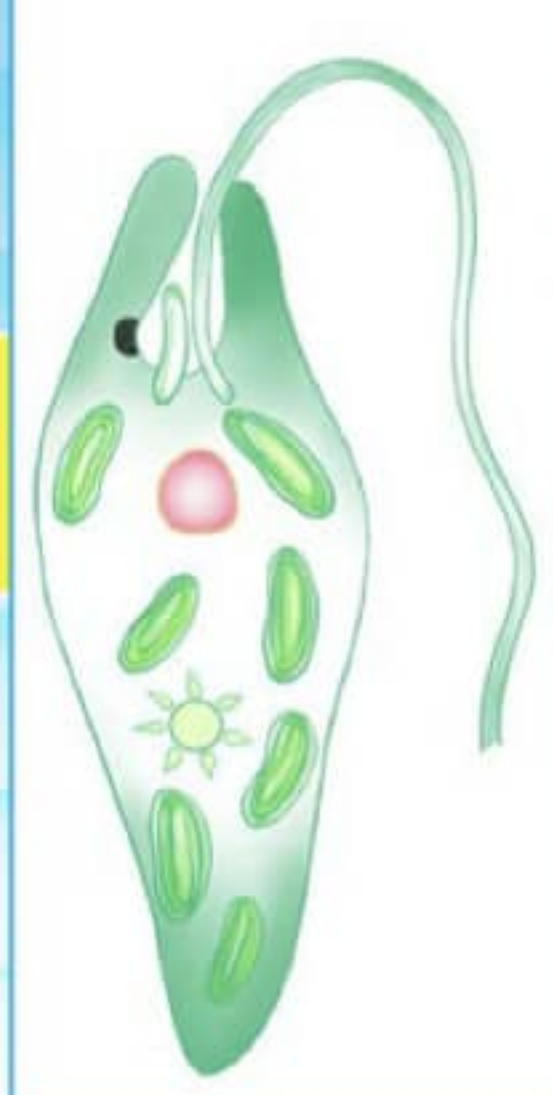
3. Amina was performing an experiment. Some of her specimens were very delicate and small. Which apparatus was she supposed to use in order to lift these specimens from the specimen bottle?
4. John attempted to observe plant cells using a prepared slide under a light microscope, but he was unable to obtain a clear image due to the following challenges:
 - (a) The slide containing specimen did not settle on the stage for observation.
 - (b) He was unable to achieve sharp focus.
 - (c) The image was too bright, making it difficult to see the detail of specimen.

Educate your friend on how to address the challenges faced in order to obtain a clear image of the plant cells.

5. Biology students were assigned to observe the external features of a variety of preserved specimens.
 - (a) Mention the sense organ that the students would primarily use, and explain its function.
 - (b) Give one disadvantage of using sense organs to make observations.
6. Why is it important for scientists to use a standard system of measurement?
7. Name the instrument you would use to measure the following and state their corresponding SI units.
 - (a) Time
 - (b) Mass
 - (c) Temperature
 - (d) Length
8.
 - (a) Convert the following temperatures into degrees Fahrenheit
 - (i) 36°C (ii) 40°C
 - (b) Convert the following temperatures into kelvin
 - (i) 100°C (ii) 0°C
 - (c) Convert the following temperatures into degrees Celsius
 - (i) 57 K (ii) 273 K (iii) 142°F (iv) 43°F
 - (d) How many millimetres are in
 - (i) 1 cm ? (ii) 1 m ?
 - (e) Convert the following into metres
 - (i) 300 mm (ii) 5500 cm
9. Imagine you are a student. Your Biology teacher has tasked you to measure the length and width of a leaf.
 - (a) What instruments would you use to ensure accurate measurements?
 - (b) What steps would you follow to obtain reliable data?
10. As a biologist, you have observed that children in your village are getting frequent fever accompanied with vomiting and headaches. This is not observed in a neighbouring village where there is wide use of mosquito

nets. You are supposed to address this problem using scientific method. Explain the steps that you would follow to address this problem.

11. During an experiment, a scientist noticed that some of the factors can be manipulated. What is the general term or name used to refer to such factors?
12. What is the importance of control experiment when testing a given hypothesis?
13. Identify the step of the scientific method in each of the following situations:
 - (a) John is not feeling well, he goes to see a medical doctor at a nearby hospital.
 - (b) The doctor asks John several questions about how he feels.
 - (c) The doctor orders John's body temperature, blood, and urine samples be measured and diagnosed in the laboratory.
 - (d) The laboratory technician diagnoses malaria parasites in John's blood.
 - (e) The doctor confirms that John has malaria and prescribes medicine for him.
14. A form one student was presenting his research finding to fellow students. He made a concluding statement that "Scientific research is a never-ending process". Give possible reasons for his statement.



Chapter Three

Cell structure and organization

Introduction

All living things are made up of cells. The cell is the basic functional unit of life. In this chapter, you will learn about the meaning of the cell, characteristics of the cell, and the functions of different parts of animal and plant cells. You will also learn the similarities and differences between plant and animal cells. In addition, you will learn the concept of cell differentiation and its importance to the formation of tissues, organs, and organ systems. The competencies developed will enable you to learn better other biological processes.



Think

Life without cells

The cell

Task 3.1

Search from library and reliable internet sources the information about cell.

In 1665, an English scientist Robert Hooke revolutionized our understanding of life by designing a

microscope and examining cork from a tree's bark. He discovered that the cork consisted of numerous box-like structures that are closely packed together, resembling a honeycomb. These structures were named 'cells'. Since then, it has been revealed that living things are made up of one or more cells, which serve as fundamental units of life. All essential life processes occur within the cell.

The remarkable observations and conclusions made by Hooke, along

with other scientists, paved the way for the development of the cell theory. This widely accepted theory indicates the relationship between cells and living things. The following are components of the cell theory.

- (a) All living things are composed of one or more cells.
- (b) Cells are the basic unit of structure and functions of living things.
- (c) All cells are produced from other cells.
- (d) Cells contain inheritable information which controls their activities.
- (e) All cells are basically similar in chemical composition.
- (f) All life processes take place in the cells.

The cell theory applies to all living things, no matter how big or small they are. Since cells are common to all living things, they can provide information about life. Scientists can use the concept of cell theory to learn about nutrition, growth, reproduction, respiration, movement, sensitivity, and excretion.

Types of cells

Task 3.2

Search from reliable internet sources the simulations/videos/images that show prokaryotic and eukaryotic cells. Note down what you have observed.

Activity 3.1: Exploring prokaryotic and eukaryotic cells under the microscope

Materials: Microscope, prepared slides of prokaryotic cells and eukaryotic cells or charts showing prokaryotic cells and eukaryotic cells, notebook, and pencil or pen

Procedure

1. Prepare the microscope with the appropriate magnification power for the slides.
2. Mount the slide of prokaryotic cell on the light microscope's stage. Hold it in place with the stage clips.
3. Observe the structure of the prokaryotic cell under the microscope.
4. Replace the slide with a prepared slide of eukaryotic cell and focus on different areas of the slide. Observe the structure of the eukaryotic cell.
5. Record the observations.
6. Describe your observations.

Question

Based on your observations, explain the similarities and differences between prokaryotic cell and eukaryotic cell.

Prokaryotic cells

The prokaryotic cell is a type of cell in which the nuclear materials are not bound with a membrane. The cell is found in organisms called prokaryotes. Prokaryotes are single-celled organisms, such as bacteria and blue-green bacteria.

The following are characteristics of prokaryotic cells:

- (a) They do not have true nucleus, as they lack membrane bound nucleus.
- (b) They do not have membrane-bound organelles, such as mitochondria.
- (c) They reproduce by binary fission and sometimes by conjugation.
- (d) Mostly they are smaller in size than eukaryotic cells.
- (e) Some prokaryotes have whip-like structure called flagella for locomotion or hair-like structure called pili for attachment.
- (f) Have circular Deoxyribonucleic acid (DNA).
- (g) Can be rod-shaped, spherical-shaped, spiral-shaped, or comma-shaped. Figure 3.1 shows various shapes of prokaryotes.



Figure 3.1: Shapes of prokaryotes (a) spiral-shaped (b) spherical-shaped (c) rod-shaped (d) comma-shaped (e) spirochaete-shaped

Eukaryotic cell

These are types of cells whose organelles are enclosed in a membrane. The cells are found in organisms called eukaryotes.

The following are the characteristics of eukaryotic cells.

- (a) They have a membrane-bound nucleus.
- (b) They are generally larger and more complex than prokaryotic cells.
- (c) They contain specialised membrane-bound organelles, such as mitochondria and chloroplasts.
- (d) They have linear DNA.

Animal and plant cells

Both animals and plants are made up of eukaryotic cells. However, these cells differ in their structure and functions.

The animal cell

Task 3.3

Search from reliable internet sources the simulations showing animal cells. Note down what you have observed.

An animal cell is a type of eukaryotic cell that lacks cell wall and has true membrane bound nucleus. Figure 3.2 shows the structure of an animal cell, as seen under the light microscope.

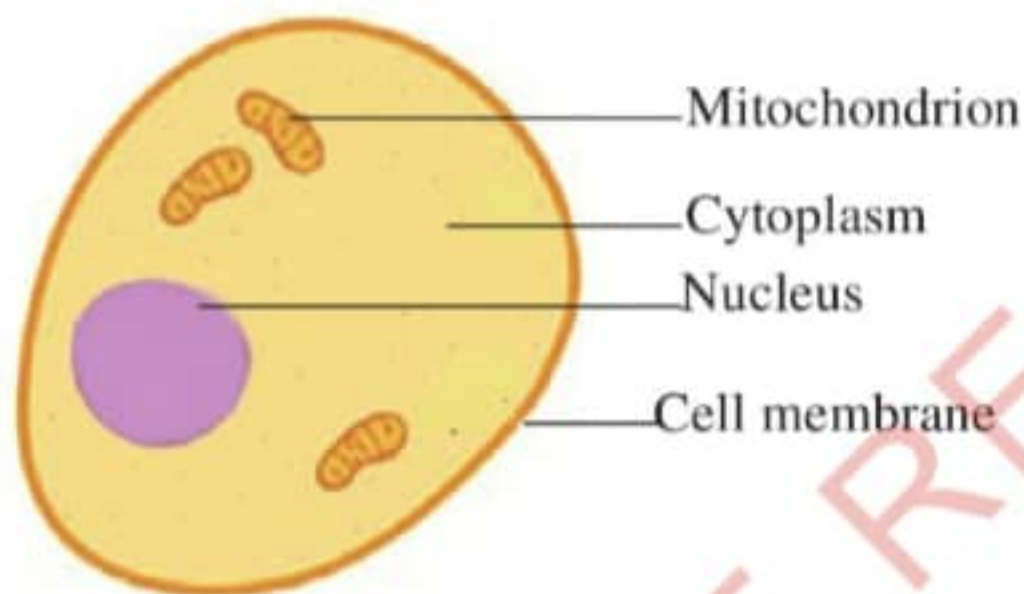


Figure 3.2: A generalised structure of an animal cell

Activity 3.2: Observing animal cells

Materials: Prepared slides of a cheek cell, a microscope, ICT tools, notebook, and a pen or pencil

Procedure

1. Prepare the microscope with the appropriate magnification power.
2. Mount the prepared slide on the stage of the light microscope.
3. Use the microscope to focus on different areas of the slide.

4. Use a low-power objective lens to observe the prepared slide.
5. Draw what you have observed.
6. Search from the library, internet sources and other biology books the structures of human cheek cells.
7. Compare what you have observed and what you have searched from different sources with Figure 3.3.

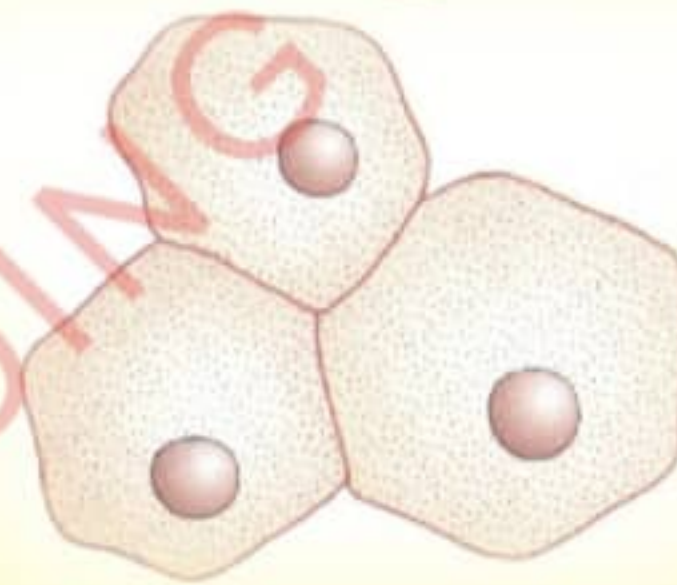


Figure 3.3: Human cheek cells

Question

Describe the structure of human cheek cells.

Functions of different parts of an animal cell

Task 3.4

Search information from biology books and reliable internet sources on the parts of an animal cell and their functions.

Cell membrane

The cell membrane is a thin layer that encloses the whole cell. The cell membrane is made up of two layers

of lipids and protein molecules. This membrane is flexible and semi-permeable. Semi-permeable means that it allows certain substances to pass in or out of the cell. Flexibility of animal cell membrane and the lack of cell wall gives it an irregular shape.

Cytoplasm

This is a jelly-like substance made up of water and dissolved chemical substances. The cytoplasm is the site for many chemical reactions in the cell. Cell organelles, such as the vacuoles, nucleus, and mitochondria are suspended in the cytoplasm.

Nucleus

This is a spherical organelle suspended in the cytoplasm. The nucleus consists of nucleolus and a fluid called nucleoplasm. It is surrounded by a membrane called the nuclear membrane. The following are functions of the nucleus.

- (a) It determines the chemical processes that take place in the cell.
- (b) It controls the functions of the cell.
- (c) It determines the cell's size, shape, and functions.
- (d) It determines the hereditary characteristics of a cell.

Cell vacuoles

Cell vacuoles are fluid-filled spaces bound by a membrane. Animal cells have small and temporary vacuoles, mainly used to secrete and excrete wastes from the cell.

Mitochondria

Mitochondria (singular is mitochondrion) are oval-shaped organelles that have two membranes, as shown in Figure 3.4. The outer membrane is smooth. The inner membrane has folds called cristae (singular is crista). The role of mitochondria is to produce energy for the cell. That is why they are sometimes referred to as 'power houses of cells'.

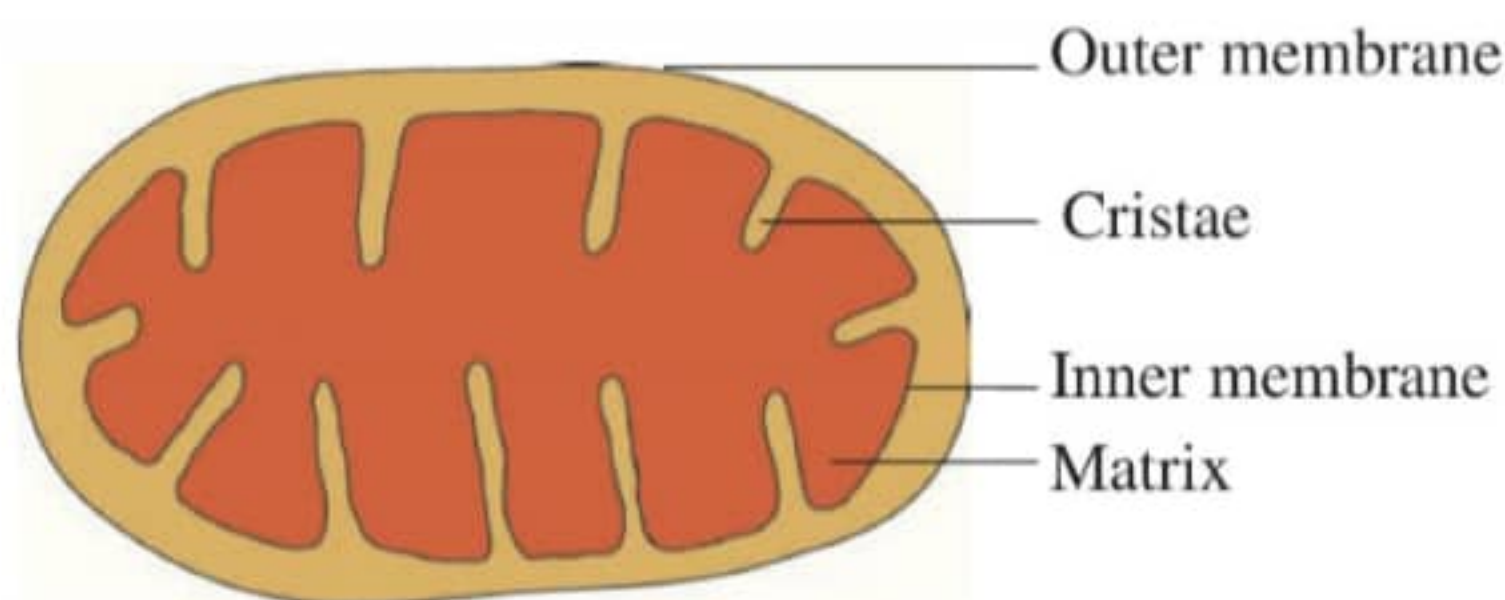


Fig 3.4: The structure of a mitochondrion

The plant cell

Task 3.5

Search from reliable internet sources the simulations/videos that show plant cells. Note down what you have observed.

Figure 3.5 shows a generalised structure of a plant cell as seen under the light microscope.

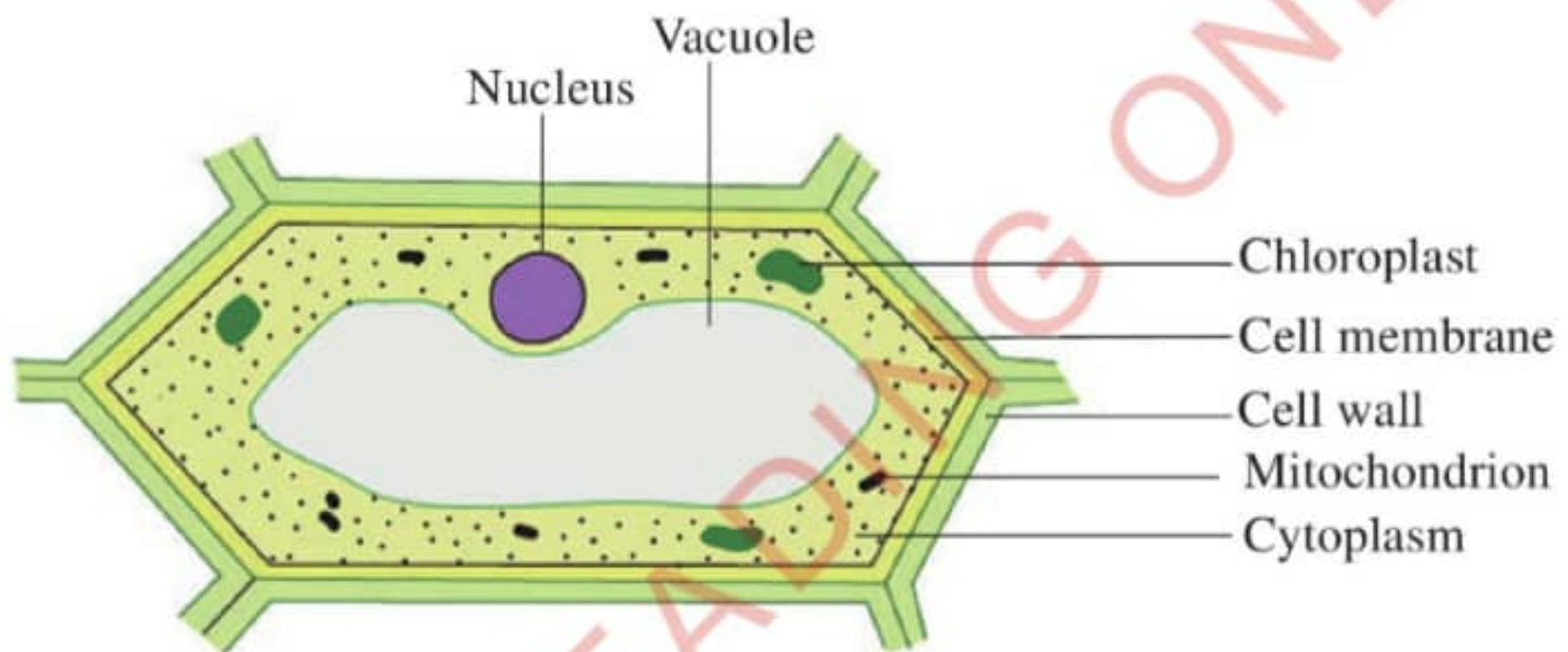


Figure 3.5: A generalised structure of a plant cell

Similar to the animal cell, the plant cell has a cell membrane, cytoplasm, nucleus, and mitochondria. These organelles function in the same way as those of an animal cell. However, plant cells have additional structures which serve specific roles in plants. These structures are:

The cell wall

This is an outer cover made of cellulose that surrounds the cell membrane. The cell wall is fully permeable. It allows the passage of water and minerals. The cell wall protects and supports the cell. The cell wall gives the plant cell a definite shape.

Chloroplasts

Chloroplasts are oval organelles that contain green pigments in plants. This pigment is called **chlorophyll**. It is important in **photosynthesis**, the process by which green plants make their own food. Chlorophyll absorbs light energy needed for photosynthesis.

Cell vacuole

Plant cells have a large and permanent vacuole that usually occupies the central part of the cell. This vacuole contains sap and is surrounded by a membrane called **tonoplast**. The cell sap causes the cytoplasm to be pushed outwards against the cell wall. This makes the cell firm, hence helps plant cells to maintain their shape.

Activity 3.3: Observing the plant cell

Materials: Onion, knife, forceps, mounting needle, microscope, microscope slide, coverslip, notebook, and a pen or pencil

Procedure

1. Separate a fleshy leaf from an onion bulb.
2. Cut a small square of the leaf.

Safety precaution

Take precaution when using sharp objects such as knife, forceps, and mounting needle.

3. Using forceps, peel off the epidermis from the inner surface of the square.
4. Put it on a microscope slide, and add a drop of water.
5. Using forceps, a mounting needle, or sharp pencil point, carefully lower a coverslip over the epidermis specimen.

6. Use a low-power objective lens to examine the specimen under the microscope.
7. Draw what you see under the microscope. Compare your diagram with Figure 3.6

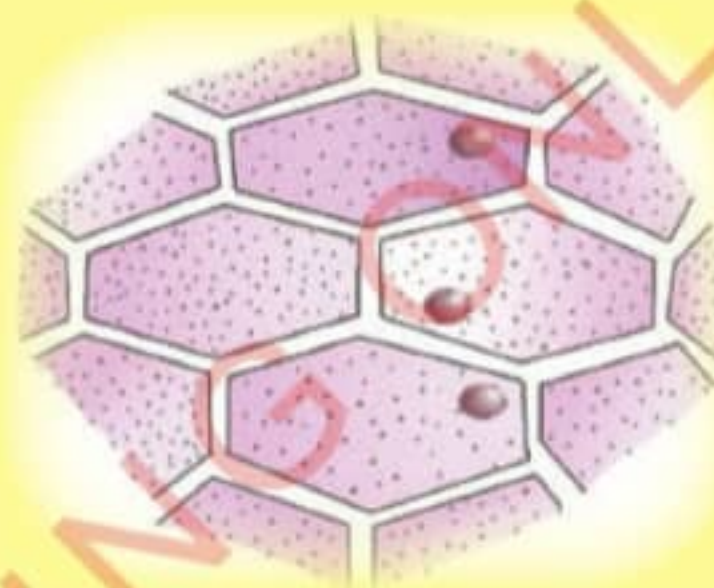


Figure 3.6: Structure of onion epidermal cells

Question

1. Describe the structure of a plant cell.
2. Based on your observations on Activities 3.2 and 3.3, explain the similarities and differences between plant and animal cells.

Similarities and differences between animal and plant cells**Similarities**

Animal and plant cells are similar in that both have a cell membrane, cytoplasm, a nucleus, cell vacuoles and mitochondria.

Differences

The differences between plant and animal cells are shown in Table 3.1.

Table: Differences between plant and animal cells

Plant cell	Animal cell
1. Has a cell wall	1. Has no cell wall
2. Has chloroplast	2. Lacks chloroplast
3. Has a large and permanent centred vacuole	3. Has small and temporary vacuoles
4. It is regular in shape	4. It is irregular in shape
5. Nucleus is located at the periphery	5. Nucleus is centrally positioned
6. Stores food in the form of starch	6. Stores food in the form of glycogen

Exercise

1. What are the differences between prokaryotic and eukaryotic cells?
2. Mention the structures that are found in plant but not in animal cells.
3. Draw a diagram of an animal cell and label the parts which perform the following functions.
 - (a) Energy production
 - (b) Control all the functions of the cell
 - (c) Suspended organelles
 - (d) Allow passage of some substances

Cell differentiation**Task 3.6**

Search from the library and reliable internet sources the information about cell differentiation, then write short notes.

An organism that is made up of one cell is called a **unicellular** organism. Examples of such organisms are

Amoeba sp., **Paramecium** sp., and bacteria. In such organisms, only one cell carries out all the life processes, such as respiration, reproduction, and excretion.

Organisms made up of more than one cell are called **multicellular** organisms. A multicellular organism consists of a few to millions of cells. These cells have different functions and have features that make them better suited to carry out these functions. This is called **cell differentiation**. Cell differentiation is the process in which a cell changes from one form to another and becomes more specialised to perform specific functions.

A group of cells that perform the same function form a **tissue**. Examples of animal tissues are bone, muscle, and blood. Examples of plant tissues are xylem and phloem. Figure 3.7 shows some examples of animal tissues.

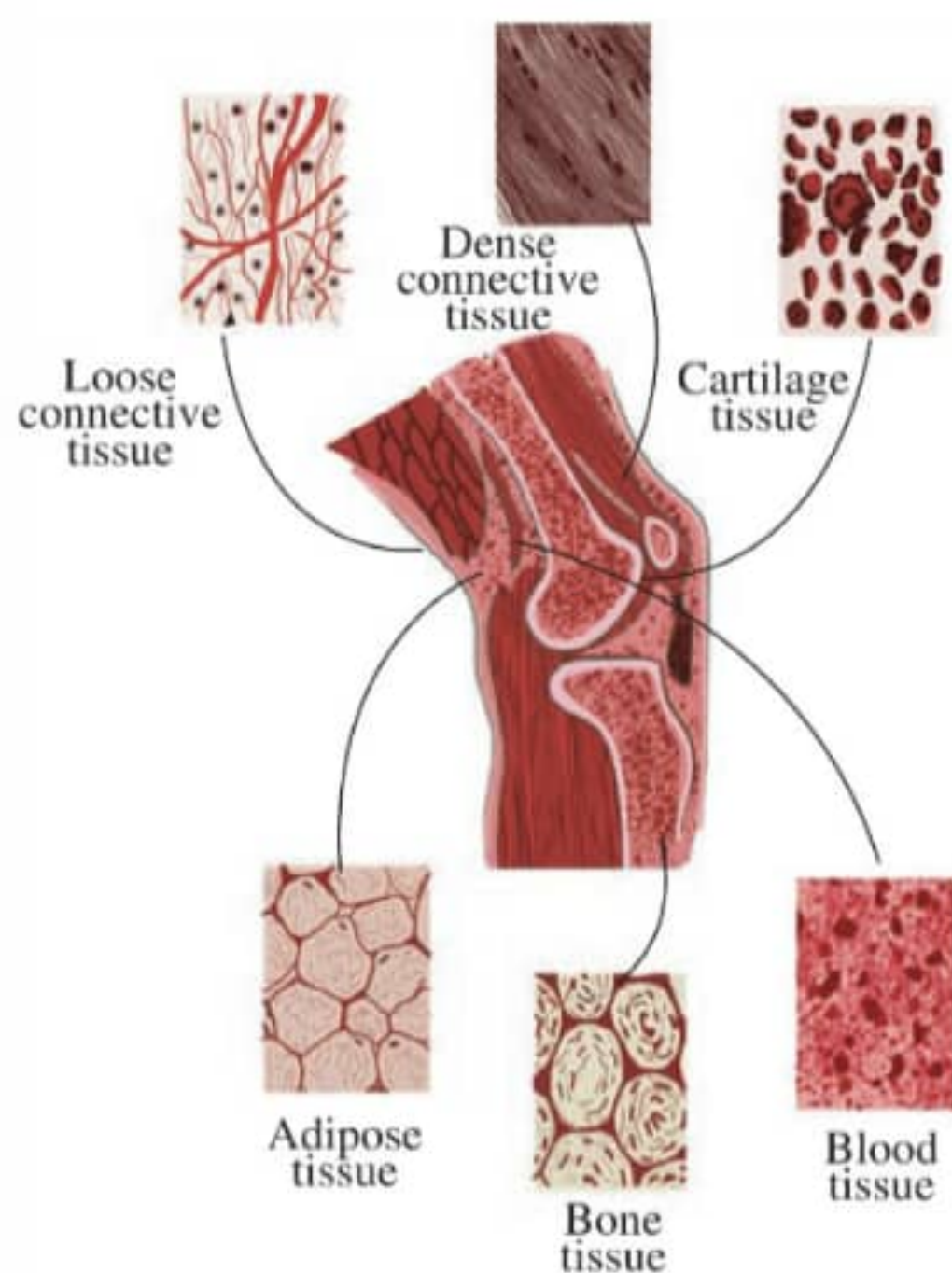


Figure 3.7: Examples of animal tissues

An **organ** consists of different tissues that work together to perform a certain function. Animal organs include the heart, liver, stomach, and brain. Plant organs include the stem, flowers, fruits, leaves, and roots. Figure 3.8 shows an example of a plant organ.



Figure 3.8: A flower

An **organ system** is made up of organs that work together to perform a certain function. Examples of systems are respiratory system, digestive system, reproductive system, hormonal system, skeletal system, and blood circulatory system. For example, the blood circulatory system transports blood to all parts of the body. It consists of the heart, blood vessels, and blood itself.

Most multicellular organisms are made up of different organ systems working together. Therefore, there is special organisation from the cell to tissue, tissue to organ, organ to organ system to organism. Figure 3.9 shows an example of organ system in the human body.

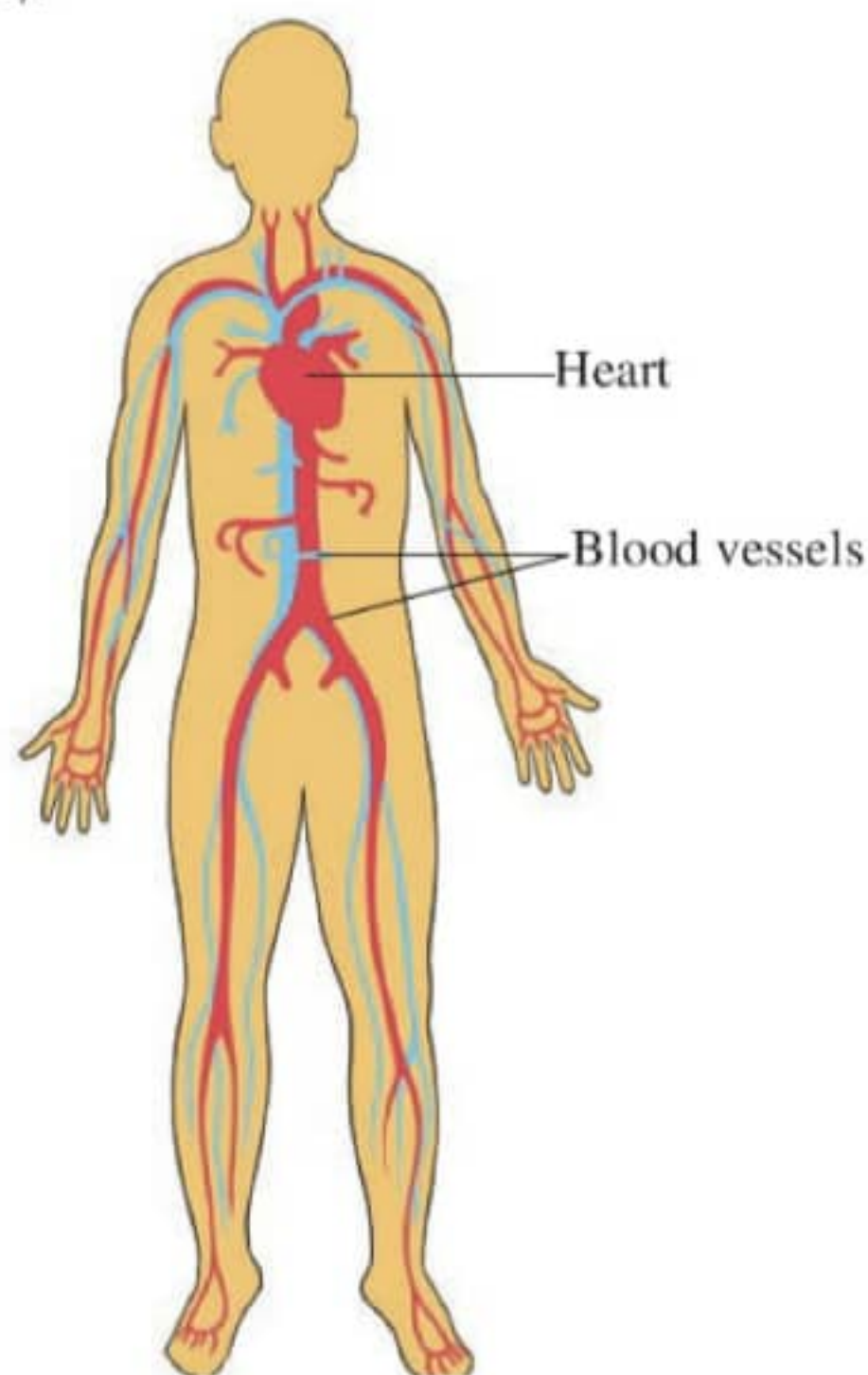


Figure 3.9: The human blood circulatory system

Importance of cell differentiation

Cell differentiation leads to division of labour. Division of labour among the cells means specific cells performing specific functions. This helps the body to carry out all life processes at the same time and more efficiently.

Specialised animal cells

The following are examples of specialised animal cells.

White blood cells

White blood cells are also called leucocytes. They protect the body against illness and diseases. The cells can change their shapes so as to engulf and destroy harmful microorganisms. Some cells contain digestive enzymes which destroy the microorganisms. Figure 3.10 shows the structure of a white blood cell.

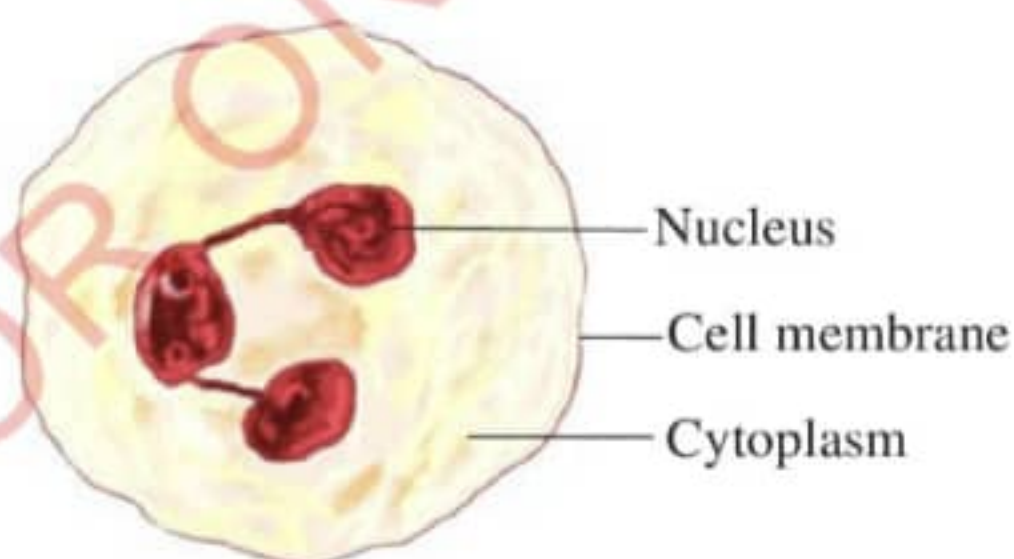


Figure 3.10: Structure of a white blood cell

Red blood cells

Red blood cells are also called erythrocytes. They lack nuclei and are bi-concave in shape. This provides a large surface area for transportation of oxygen from the lungs to various parts of the body. They also contain haemoglobin which carries oxygen to different parts of the body. Figure 3.11 shows the structure of a red blood cell.



Figure 3.11: Structure of a red blood cell

Sperm cells

The sperm cell fertilizes the female egg during reproduction. The sperm cell has a head and a tail. The tail enables the sperm cell to swim to the egg. Figure 3.12 shows the structure of a sperm.

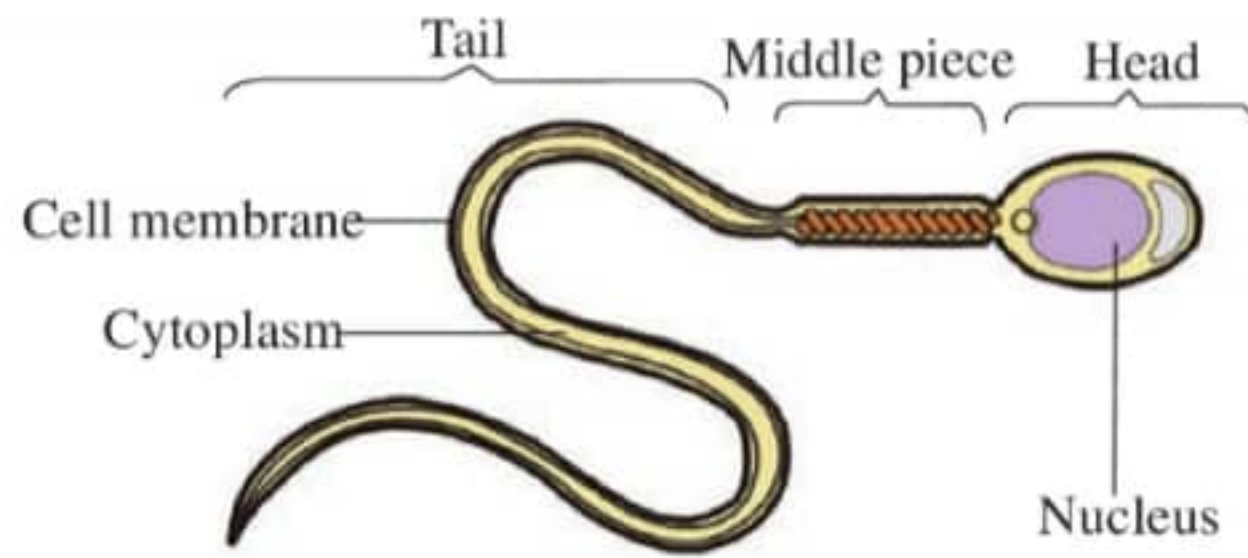


Figure 3.12: Structure of a sperm cell

Specialised plant cells

The following are examples of specialised plant cells:

Root hair cells

Root hair cells absorb water and mineral salts. They help to increase the surface area for absorption. Figure 3.13 shows the structure of a root hair cell.

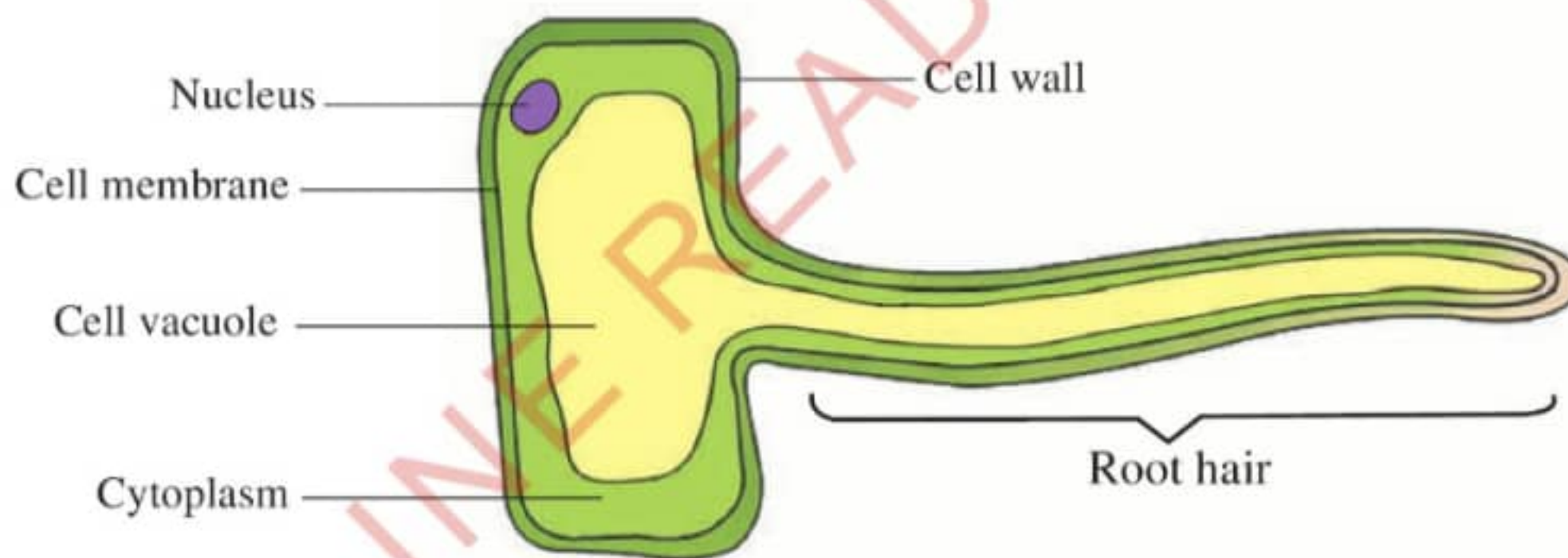


Figure 3.13: Structure of root hair cell

Activity 3.4: Observing root hair structure in plants

Materials: Fresh root sample, microscope, microscope slide, coverslip, water, dropper or pipette, and stain (optional), notebook, and pen or pencil

Procedure:

1. Obtain a fresh root sample from a young plant. Make sure the root

- is intact and has visible root hairs.
2. Gently rinse the root under running water to remove any soil particles.
3. Take a microscope slide and place a small drop of water on it.
4. Carefully, place the root sample on the water droplet, ensuring the root hairs are facing down.
5. When using a stain, add a drop or two of the stain to the water on

the slide. This step is optional but can enhance visibility.

6. Gently, cover the specimen with a coverslip. Gently, press down on the coverslip to remove any air bubbles and ensure the sample is adequately flattened.
7. Place the prepared slide on the microscope stage. Observe the slide starting with the lowest magnification power and then adjust the focus knobs to bring the root hair into focus.
8. Carefully, observe the root hairs under the microscope, and take note of what you have observed.

Question

Based on your observation, describe the structure of the root hair.

Guard cells

Guard cells surround the stomata. They control the opening and closing of **stomata** (singular is **stoma**) of the leaf. The inner walls of guard cells are thicker than the outer walls. This makes them expand irregularly. When the guard cells expand, the stoma opens. When they contract, the stoma closes. Stomata are tiny pores used for gaseous exchange and loss of excess water. Figure 3.14 shows the structure of a guard cell.

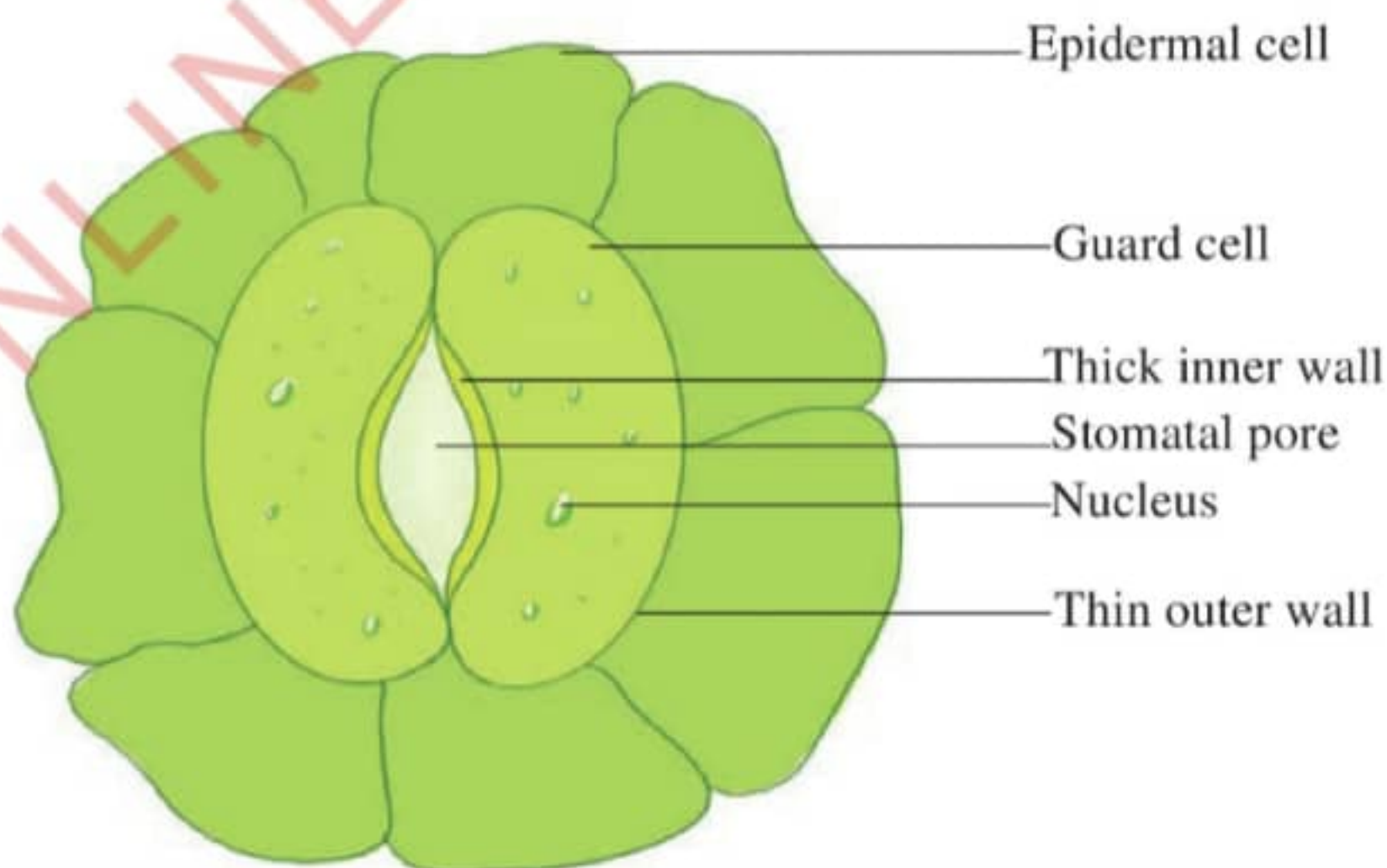


Figure 3.14: Structure of a guard cell showing stoma

Xylem vessels

Xylem vessels are made up of hollow dead cells with walls made up of lignin. Lignin is a tough rigid material that makes up the wall of xylem vessels. The cells are connected to form xylem vessels. These vessels transport water and minerals from the root to the leaves. They also provide support to the plant. Figure 3.15 shows the structure of xylem vessels.

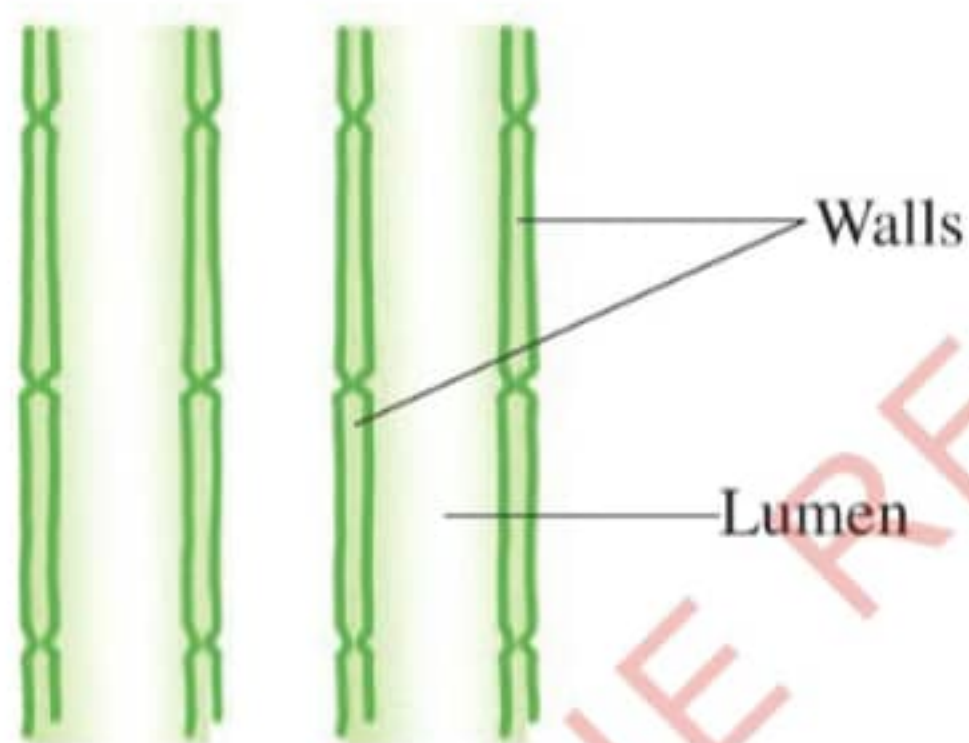


Figure 3.15: Structure of xylem vessels

Activity 3.5: Observing cell structure and differentiation in plants

Materials: ICT tools, notebook, and pen or pencil

Procedure:

1. Visit reliable online sources and select the appropriate simulations/video that describe cell structure and differentiation in plants.
2. Use the selected simulations/video to learn the concept of cell structure and differentiation in plants.

Question

What have you learned from the simulations?

Activity 3.6: Observing xylem in plant stems

Materials: Fresh herbaceous plant stem, microscope, microscope slide, coverslip, water, dropper or pipette, sharp knife, notebook, and a pen or pencil.

Procedure

1. Select a fresh herbaceous plant with a thin stem. Herbaceous plants are non-woody plants with soft stems that are easier to work on.
2. Carefully, cut a thin cross-section of the stem using a sharp knife. The cross-section should be about 1-2 mm thick.
3. Place the cross-section of the stem on a clean glass slide.
4. Add a few drops of water to the cross-section to keep it moist and prevent it from drying out.
5. Carefully, place a coverslip over the cross-section, ensuring it covers the entire area of the specimen.
6. Gently, press down on the coverslip to remove any air bubbles and ensure the slide is well-prepared for observation.
7. Place the prepared slide on the stage of the microscope.
8. Start with the lowest magnification power and adjust the focus knobs

to bring the cross-section into focus.

9. Gradually, increase the magnification power to higher levels to see the finer details of the xylem tissue.
10. Observe the cross-section under the microscope. Look for

elongated cells with thick walls, which represent the xylem tissue. You may see them arranged in bundles.

Question

Describe what you have observed in number 10.

Palisade cells

Palisade cells found in plant leaves, contain large amounts of chloroplasts. Chloroplasts are the sites for photosynthesis. They contain chlorophyll which traps sunlight energy during photosynthesis. Figure 3.16 shows the structure of a palisade cell.

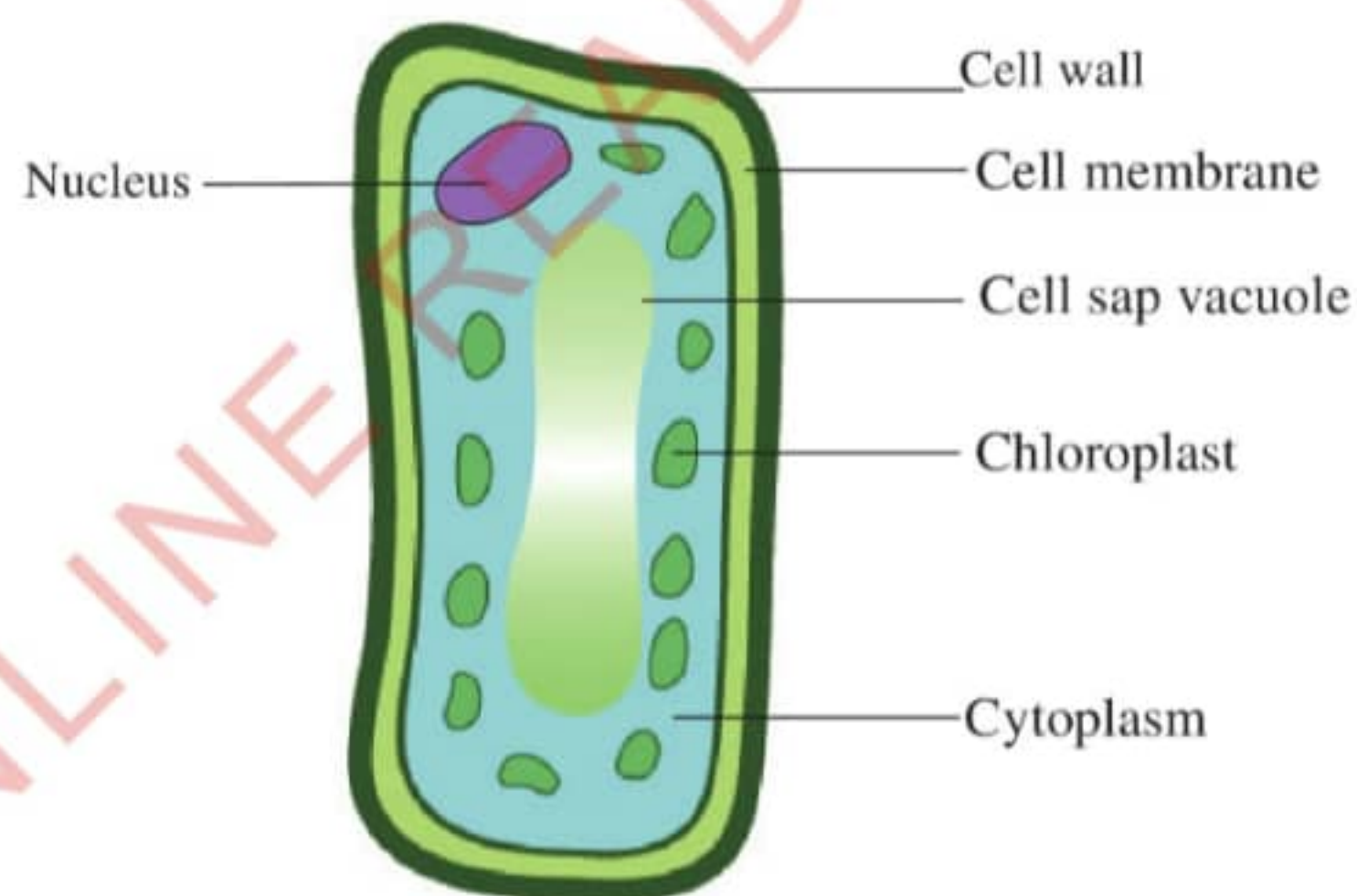


Figure 3.16: Structure of a palisade cell

Chapter summary

1. The cell is the basic functional unit of life.
2. There are two main types of cells: prokaryotic and eukaryotic.
3. Animal cells and plant cells are eukaryotic cells.
4. The structures that are found in plant cells but not in animal cells are the cell wall, chloroplasts, and cell sap vacuoles.
5. Cell differentiation refers to the way cells are specialized to perform a specific function.

6. A tissue is a group of cells that perform the same function.
7. An organ is made up of various tissues that work together to perform a specific function.
8. An organ system consists of organs that work together to perform a specific function.
9. Examples of specialized animal cells are white blood cells, red blood cells, and sperm cells.
10. Examples of specialized plant cells are guard cells, root hair cells, and palisade cells.

Revision exercise 3

1. One of the following is NOT a component of the cell theory:
 - (a) All living things are composed of cells.
 - (b) All cells are produced from other cells.
 - (c) All cells are basically similar in chemical composition.
 - (d) All life processes take place outside the cells.
2. One of the following is NOT a characteristic of prokaryotic cells.
 - (a) They have no nuclear membrane.
 - (b) They reproduce by binary fission.
 - (c) They have circular DNA.
 - (d) They have mitochondria.
3. Which of the following is the functional unit of life?
 - (a) Tissue
 - (b) Cell
 - (c) Organ
 - (d) System
4. Which of the following organism has differentiated cells?
 - (a) Bacteria
 - (b) Amoeba
 - (c) Plants
 - (d) Paramecium

5. Write **TRUE** for correct statements and **FALSE** for incorrect statements in the spaces provided.
- (a) A tissue comprises different organs that perform the same function.

- (b) Chloroplasts and mitochondria perform similar functions. _____
- (c) Palisade cell is one of the specialised cells in plants. _____
- (d) A cell membrane is similar in function to a cell wall. _____
6. The following terms are arranged in alphabetical order. Rearrange them into an appropriate biological sequence, starting with the lowest level.
- cell → organ → organism → system → tissue
7. Name the parts of the cell described in each of the following statements.
- (a) The semi-permeable membrane that encloses the cytoplasm of a cell.
- (b) A large cellular organelle that contains hereditary information.
- (c) The site for respiration in the cell.
- (d) A jelly-like mixture consisting mostly of water, located between the cell membrane and the nucleus.
- (e) A part that protects, supports, and gives shape to plant cells.
8. Differentiate:
- (a) Prokaryotic from eukaryotic cells
- (b) Organ from tissue
9. Compare the structures of a plant and an animal cell as seen under the light microscope.
10. Explain what will happen if the following are removed from the plant or animal cell.
- (a) Nucleus
- (b) Cell wall
- (c) Chloroplast
- (d) Mitochondria
11. Using examples of plant and animal cells, explain the meaning of cell differentiation.
12. Describe the functions of any three specialised cells in plants.



Chapter Four

Classification of living things

Introduction

There are different kinds of organisms in the world. Some are too small, they can not be seen with naked eyes. Such organisms are called micro-organisms. Others can be seen with the naked eyes and are called macro-organisms. Due to the existing diversity of organisms, it is difficult to correctly identify and determine the exact position of organisms in classification. In this chapter, you will learn about the concept of classification, classification systems, ranks of classification, and binomial nomenclature. The competencies developed will enable you to identify and classify newly discovered organisms correctly by comparing their characteristics with the already classified organisms.



Think

Living things with no assigned names

Concept of classification

Task 4.1

From reliable internet sources, library and other biology books, read about the meaning and importance of classification.

The process of grouping organisms according to their similarities and differences is called classification. Similar organisms are placed in the same group. The

similarities can be in terms of their evolutionary relationship or the way they carry out life processes, such as feeding and reproduction. The field of biology that deals with classifying organisms is called taxonomy. Organisms, such as snails, orange trees, crocodiles, birds, lions, and baobab trees belong to different groups. See Figure 4.1.



(a)



(b)



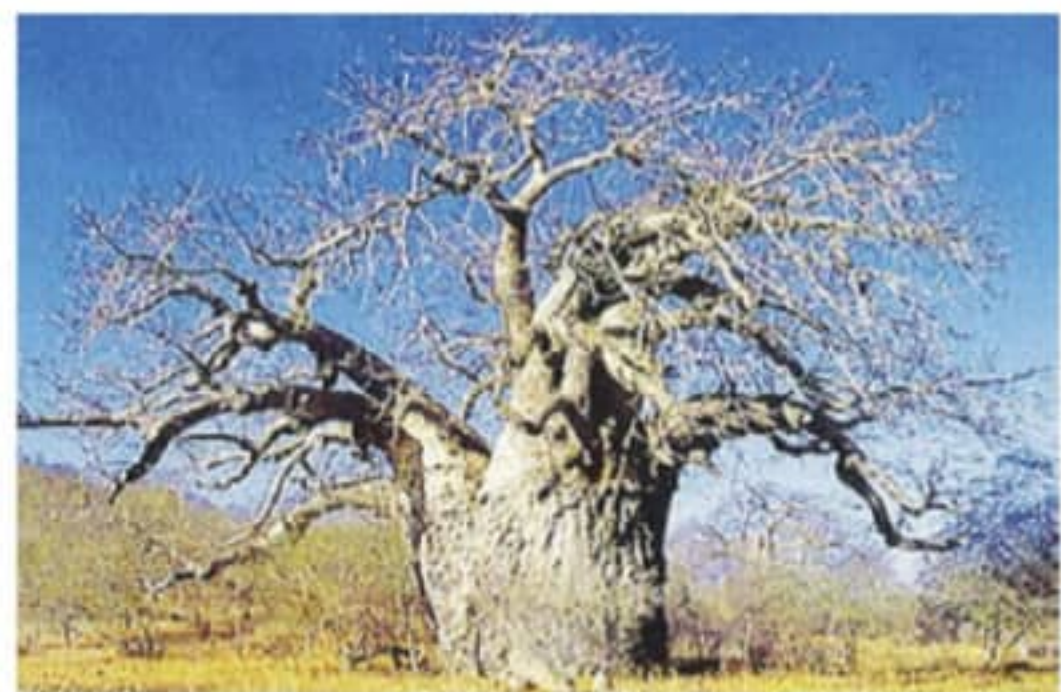
(c)



(d)



(e)



(f)

Figure 4.1: Variety of living things (a) snail (b) orange tree (c) crocodile (d) birds (e) lion (f) baobab tree

Activity 4.1: Grouping living things according to their similarities and differences

Materials: A variety of small plants, insects, and other animals, notebook, and a pen or pencil

Procedure

1. Walk around the school compound and collect a variety of plants, insects, and other small animals.
2. Observe and group the collected organisms according to their similarities and differences.
3. Write the similarities and differences among the grouped organisms.

Caution

- (a) Avoid collecting dangerous organisms, such as wasps, scorpions, and snake; and poisonous plants, such as milkweed and oleander.
- (b) Avoid collecting weak or injured specimens as they will lead to incorrect observations.

Questions

1. What did you learn from the activity?
2. Why do you think it is important to classify organisms?

The importance of classification

Classification is important in the study of living things because of the following reasons.

- (a) There are millions of species of living things in the world. Grouping them makes it easier to study and identify them, because the members of the group have many characteristics in common.
- (b) It makes communication among biologists in the world easier.
- (c) It enables scientists to predict characteristics of an organism based on the characteristics of a group it belongs to. When we know the characteristics of a group then we can predict the characteristics of an organism in that group. For example, both an eagle and a chicken are birds. If we know the structure of the heart of a chicken, we can predict the structure of the heart of an eagle even if we have not seen it.
- (d) It helps to understand how organisms are related in terms of their evolutionary history.

Classification systems

There are two main types of biological classification systems, namely artificial and natural classification systems.

Task 4.2

Read from reliable online sources, library and other biology books the information about artificial and natural classification systems.

Artificial system of classification

Artificial classification system groups organisms according to a few observable features, for example, the presence of legs or wings. Based on this, bees, birds, and bats would be grouped together because they have wings. Snakes, earthworms, and snails would also be grouped together because they do not have legs. Artificial classification system also involves the use of some other characteristics, such as size, how organisms move, where they live, or what they eat.

In the third century BC, Aristotle (a Greek philosopher) used artificial classification system to classify plants. He grouped them according to their shape, size, and their usefulness or unusefulness to humans.

Activity 4.2: Classifying living things using artificial system of classification

Materials: A variety of living things or charts that show pictures of a bird, bat, bee, snake, snail, an earthworm, notebook and pen or a pencil

Procedure

1. Observe the organisms in Figure 4.2 and note as many observable features as you can.

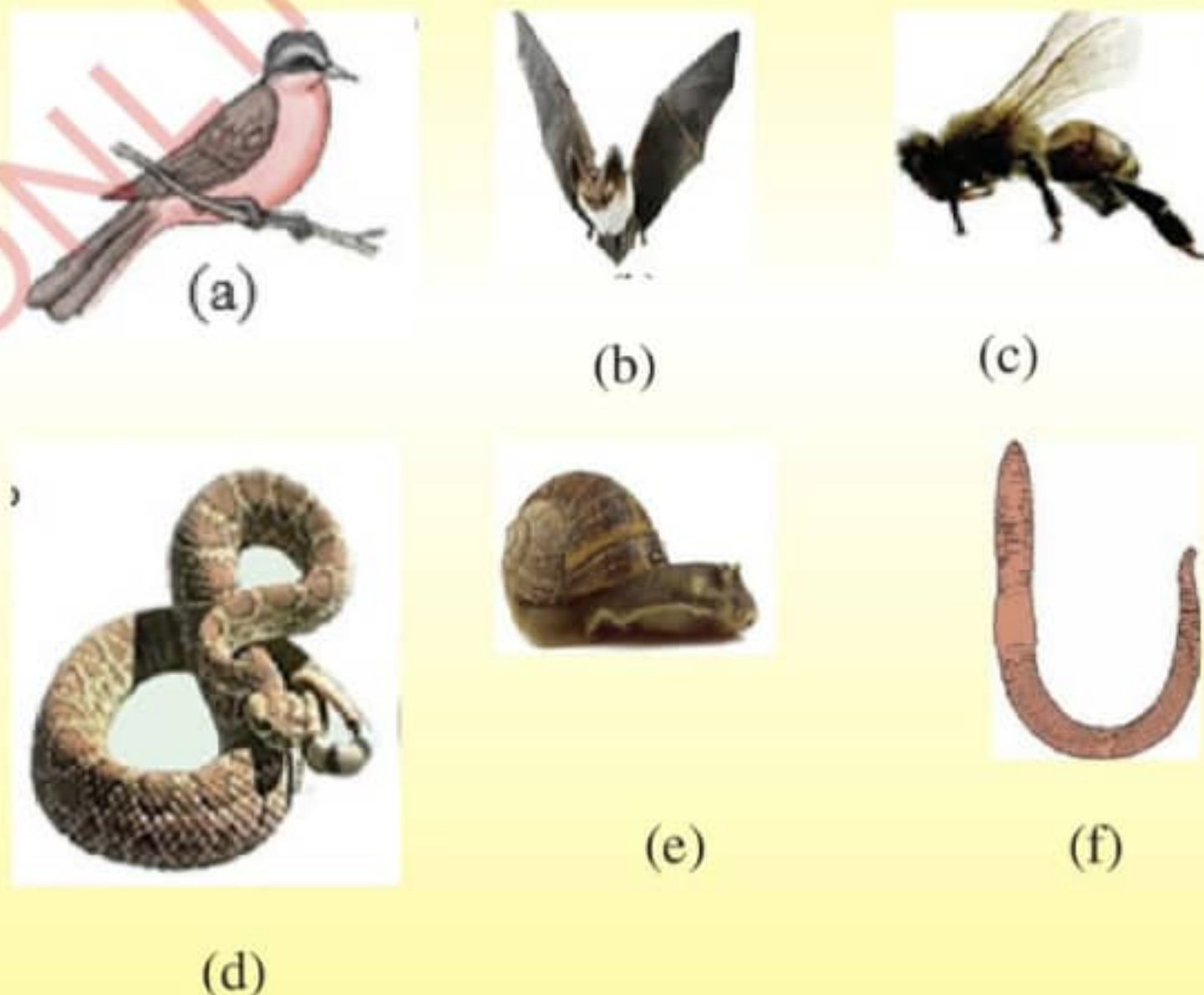


Figure 4.2: Variety of organisms (a) bird (b) bat (c) bee (d) snake (e) snail (f) earthworm

2. Classify the organisms in Figure 4.2 based on the following features:

- (a) How big or small their size is
- (b) How they move
- (c) Where they live

3. How many organisms did you get in each of the categories in procedure 2?

Advantages of artificial system of classification

- (a) It is based on simple and easily observable features. Artificial classification is therefore straightforward and easy to use.
- (b) It is simple to identify and classify organisms since newly discovered organisms with just a few known features can be easily fitted in.
- (c) It is stable because it does not change with time or new discoveries.
- (d) It is less expensive because it uses few observable features to group the organisms.
- (e) It does not require special classification techniques or highly skilled personnel.

Disadvantages of artificial system of classification

- (a) Some organisms that are unrelated in their internal structure such as bats

and birds can be grouped together because of the resemblances of their few external features.

- (b) Similar organisms may be placed in different groups because of the use of only few observable features, for example, bats and rats can be grouped in different groups.
- (c) It provides only limited information about each member. For example, organisms, such as bacteria and some fungi are grouped as micro-organisms because they are microscopic. This classification ignores other features, such as modes of feeding, reproduction, and cell structure.
- (d) It does not allow the prediction of information; hence limits more advancement in taxonomy.
- (e) It is less accurate because it uses only few observable characteristics, such as mode of feeding, habitat, locomotion, or presence of wings.

Natural system of classification

In the natural system of classification, organisms are grouped based on many features in common, particularly those related to evolutionary relationships. In this system, characters that show *homology* or similarity of origin must be distinguished from those showing *analogy* or similarity of use. For example, the arms of man, forelimbs

of whales, and wings of birds and bats are homologous structures. In contrast, the wings of birds and wings of insects perform similar functions but they are different origins hence, they are **analogous**. Natural classification system uses homologous characters.

Advantages of natural system of classification

- (a) It allows organisms which are genetically and evolutionally related to be grouped together.
- (b) It is more accurate, since it involves scientific research to gather enough information before the actual placement of an organism into a particular group is decided.
- (c) It enables placing of newly

discovered organisms into correct groups they belong.

- (d) It allows addition of new features when they are discovered.

Disadvantages of natural system of classification

- (a) It is very expensive since it involves experiments, researches, and high level of classification skills.
- (b) It is tedious and time consuming since it uses many features in grouping organisms.
- (c) It is not stable because it can change any time due to discovery of new features or information. Table 4.1 shows the differences between artificial and natural systems of classification.

Table 4.1: Differences between artificial and natural system of classification

Artificial system of classification	Natural system of classification
Considers external features only	Considers both internal and external features
Requires simple skills	Requires advanced scientific skills
Based on an individual's interest	Based on international standards
It is usually fast and less expensive	Takes more time and is expensive
Less accurate	More accurate
Has limited predictive value	Has high predictive value

Activity 4.3: Classifying organisms based on artificial and natural system of classification

Materials: A variety of living things or charts showing pictures of organisms

Procedures

1. Observe the external features of organisms and note as many features as you can.
2. (a) Group the organisms based on artificial system of classification.
(b) Group the organisms based on natural system of classification.

Questions

What are the main features used to place organisms into procedures number 2 (a) and (b)?

Major groups of living things

Task 4.3

Search from library and reliable internet sources the information on the major groups of living things and ranks of classification.

The following are the five major groups of living things that constitute the five kingdoms.

- (a) Kingdom Monera, example, bacteria.
- (b) Kingdom Protocista, examples, amoeba, *Plasmodium* sp., and *Euglena* sp.

- (c) Kingdom Fungi, examples, mushroom and yeast.
- (d) Kingdom Plantae, examples, maize, bean, and banana plants.
- (e) Kingdom Animalia, examples, human beings, dogs, and rats.

However, viruses have not been included in these five kingdoms because they possess features of both living and non-living things. Examples of viruses include Corona virus, Ebola virus, and HIV.

Ranks of classification

In classification of living things, there are seven main ranks or taxonomic units which are arranged hierarchically in a descending order from the highest to the lowest rank. Each unit is called a taxon (plural is taxa). The main taxa are Kingdom, Phylum or Division, Class, Order, Family, Genus, and Species. The highest rank is Kingdom and the lowest is Species. Hierarchical arrangement of the taxonomic units or ranks is referred to taxonomic hierarchy.

Kingdom Animalia is subdivided into phyla (singular is phylum) while kingdom Plantae is subdivided into divisions. The phyla or divisions are further subdivided into classes. Each class is subdivided into orders. An order is subdivided into families and families are subdivided into genera (singular is genus). A genus is subdivided into

species. A species is a group of closely related organisms that can interbreed freely under natural conditions and produce fertile offspring.

The flow chart in Figure 4.3 shows how kingdom Plantae and kingdom Animalia are subdivided.

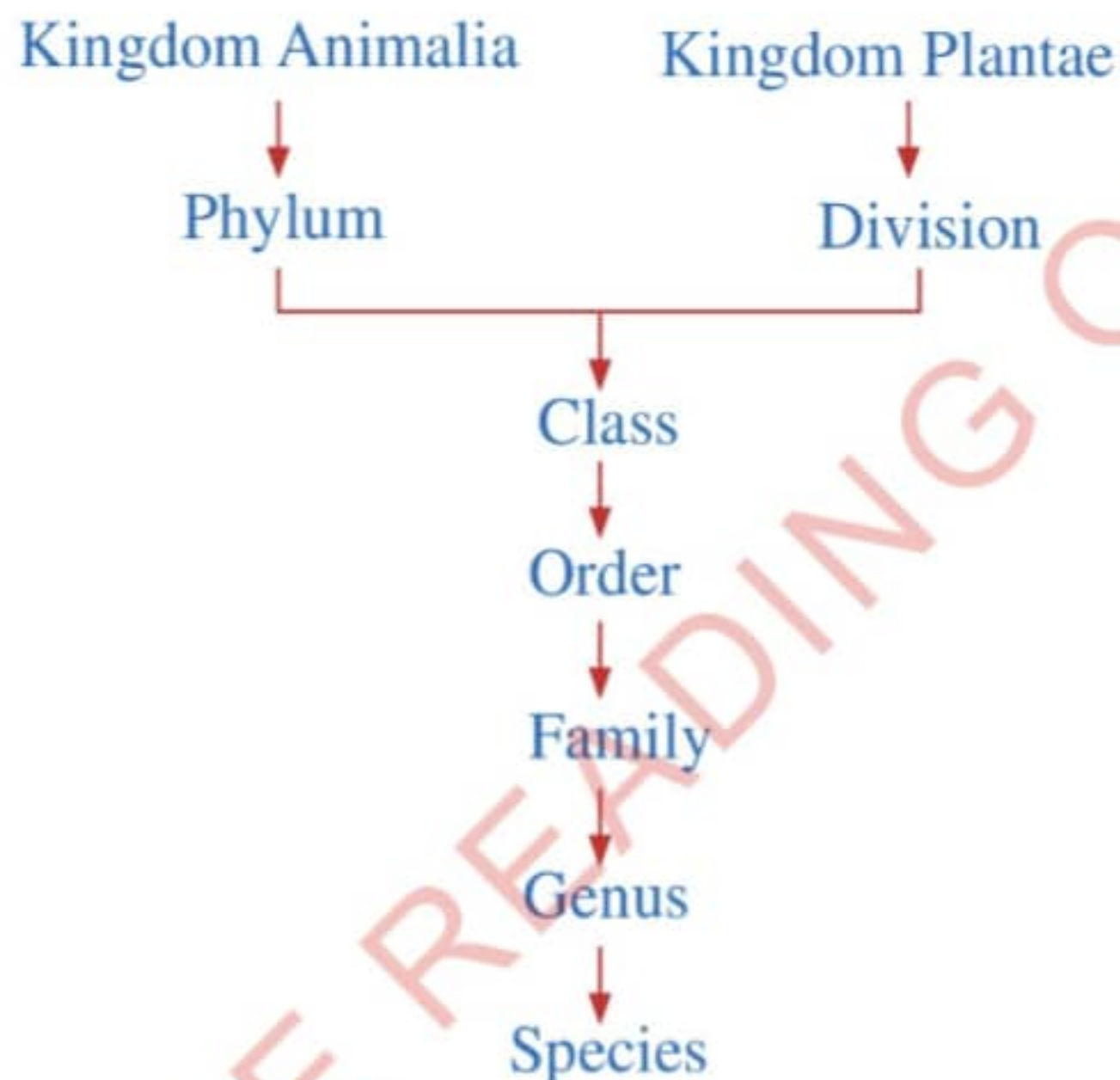


Figure 4.3 Taxonomic ranks of classification

Kingdoms contain different varieties of organisms. At kingdom level, organisms share few features in common. As you move down the ranks, each subsequent unit has fewer organisms and the members of each unit have many features in common.

At the species level, organisms share many features. Members of the same species can interbreed to produce fertile offspring, but organisms from different species do not interbreed. If they interbreed, they do not produce

fertile offspring. This is because of differences in genetic make-up, behaviour, and morphological features, such as structure of sexual organs. However, there are a few exceptions to this rule. For example, a horse and a donkey can interbreed to produce a mule. Mules are usually but not always infertile. Dogs and wolves can interbreed to produce fertile offspring. In Table 4.2, a few common organisms have been classified from kingdom to species level.

Table 4.2: Classification of some common organisms

<i>Taxonomic unit</i>	<i>Human being</i>	<i>Cat</i>	<i>Maize plant</i>
Kingdom	Animalia	Animalia	Plantae
Phylum/Division	Chordata	Chordata	Angiospermophyta
Class	Mammalia	Mammalia	Monocotyledoneae
Order	Primate	Carnivora	Graminales
Family	Hominidae	Felidae	Poaceae
Genus	Homo	Felis	Zea
Species	Homo sapiens	Felis catus	Zea mays

Note that the names of higher ranks (above species) always start with a capital letter. For example, the human being is in kingdom Animalia, phylum Chordata, class Mammalia, order Primate, family Hominidae, genus Homo, and species Homo sapiens.

Binomial nomenclature

Task 4.4

Search from the library and reliable internet sources the information about naming of organisms.

Binomial nomenclature is a scientific system of naming organisms using a two-part name. The first part of the name represents the *genus*. It is also called the generic name. The second part of the name represents the **specific epithet**. The two parts form the name of a species or scientific name. This system was first introduced by Carl Linnaeus, who is also referred to as the father of classification.

As an example, the scientific name of a human being is Homo sapiens. 'Homo' is the generic name and 'sapiens' is the specific epithet. Scientific names of some other common organisms are shown in Table 4.3.

Rules of binomial nomenclature

The following rules are observed when writing scientific names.

- All scientific names must be written in Latin language or if is in a different language, the name must be Latinised.
- The name should have two parts; the first part represents the genus and the second part represents the specific epithet.
- The generic name is written before the specific epithet.
- The generic name must start with a capital letter and specific epithet begins with a small letter.
- In published documents, such as books, scientific names should

be written in italics, for example *Zea mays* and *Homo sapiens*. When handwritten, the names must be underlined separately for example, Zea mays and Homo sapiens. Examples of how the scientific names of some common organisms are written in documents is shown in Table 4.3.

Table 4.3: Scientific names of common organisms

Common name	Scientific name
<i>Common frog</i>	<i>Rana temporaria</i>
<i>Domestic cat</i>	<i>Felis catus</i>
<i>Coconut palm</i>	<i>Cocos nucifera</i>
<i>Mango tree</i>	<i>Mangifera indica</i>
<i>Garlic</i>	<i>Allium sativum</i>
<i>Onion</i>	<i>Allium cepa</i>
<i>Pea plant</i>	<i>Pisum sativum</i>
<i>Housefly</i>	<i>Musca domestica</i>
<i>Lion</i>	<i>Panthera leo</i>
<i>Mimosa plant</i>	<i>Mimosa pudica</i>

Chapter summary

1. Classification is the grouping of organisms based on their similarities and differences.
2. Classification is important because:
 - (a) It makes it easier to identify and study the millions of organisms in the world.
 - (b) Each organism (species) is referred to by the same name all over the world.
 - (c) It is easier to study organisms when they are in groups.
 - (d) It enables scientists to make predictions of information.
3. There are two main system of classification systems:
 - (a) Artificial system of classification is based on few observable features of organisms.
 - (b) Natural system of classification is based on many features in common particularly those related to evolutionary relationship.

4. In natural system of classification, organisms are divided into kingdoms, phyla or divisions, classes, orders, families, genera, and species. These units are called taxa (singular is taxon).
5. The members of a species are the most closely related. They can interbreed freely under natural conditions to produce fertile offspring.
6. Currently, five kingdoms are recognised, namely Monera, Protocista, Fungi, Plantae, and Animalia.
7. Binomial nomenclature is the system of giving scientific names to organisms. A scientific name has two parts:
 - (a) The first part is the generic name.
 - (b) The second part is the specific epithet.

Revision exercise 4

1. Match each item from Column A with its corresponding item in Column B.

List A	List B
(i) Kingdoms	(a) Natural and artificial systems
(ii) Classification ranks	(b) Division, genus, species
(iii) Types of classification systems	(c) <i>Zea mays</i>
(iv) Scientific naming	(d) Monera, Animalia
(v) Human being	(e) Binomial nomenclature
(vi) Maize plant	(f) <i>Pisum sativum</i>
(vii) The father of classification	(g) <i>Felis catus</i>
	(h) <i>Homo sapiens</i>
	(i) Aristotle
	(j) Carl Linnaeus

2. Write **TRUE** for a correct statement and **FALSE** for an incorrect statement in the space provided.
 - (a) Artificial system of classification is based on few observable features.

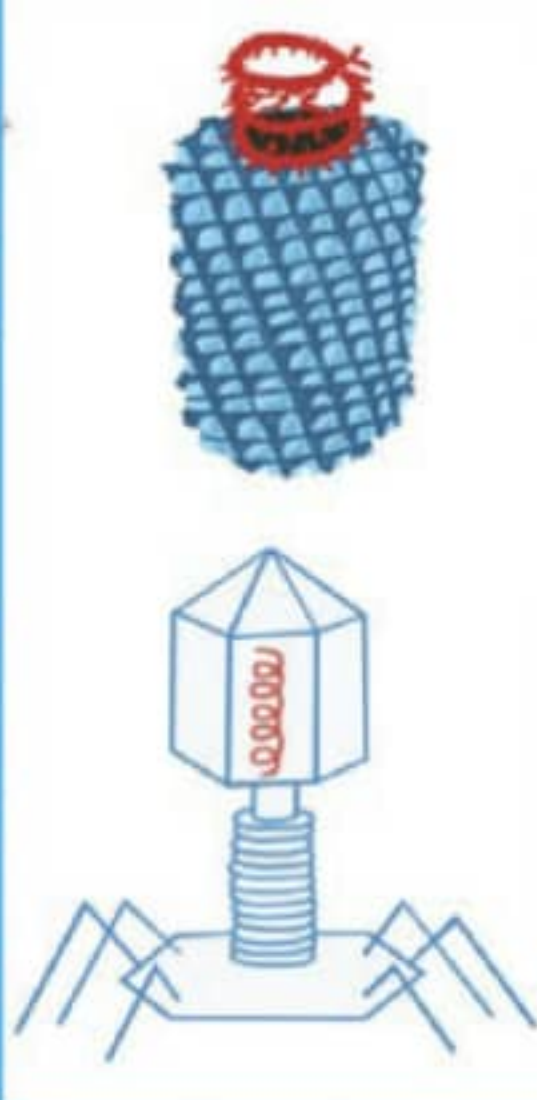
 - (b) The process of sorting living things into groups is called classification.

 - (c) Organisms of the same specie exhibit a low degree of similarity. _____

- (d) The lowest rank in classification is the genus. _____
- (e) Classification is based on similarities and differences among organisms.
- _____

3. *Differentiate:*

- (a) Classification from taxonomy.
- (b) Natural from artificial systems of classification.
4. *Explain the concept of the biological species.*
5. How would you apply the concept of classification in a real-world context?
6. Starting from the highest rank, outline the hierarchical ranks of classification in biology.
7. *Explain why a chicken and a duck cannot interbreed even if they are kept in the same coop.*
8. In a biology class, your fellow students classified pigeon and butterfly in the same group based on the presence of wings. Explain the shortcomings of their classification.
9. Diana wrote the scientific name of human beings as homosapiens. Explain why she was wrong.



Chapter Five

Viruses and major groups of living things

Introduction

There are many species of living organisms in the world. These organisms are placed in groups based on their common or shared characteristics. Each group has its own features that make it distinct from others. In this chapter, you will learn about viruses and five kingdoms of organisms, namely, Monera, Protocista, Fungi, Plantae and Animalia. The competencies developed will enable you to distinguish members of these groups, understand their interrelationships and explore their importance to human beings and other organisms.



Think

A world without the knowledge of classification

Viruses

Task 5.1

From the library and reliable internet sources, search for information about the characteristics, advantages and disadvantages of viruses.

A virus is an extremely small fragment of nucleic acid (DNA or RNA) surrounded by a protein coat. It is smaller than a living

cell. A virus is not a cell. Examples of viruses are shown in Figure 5.1

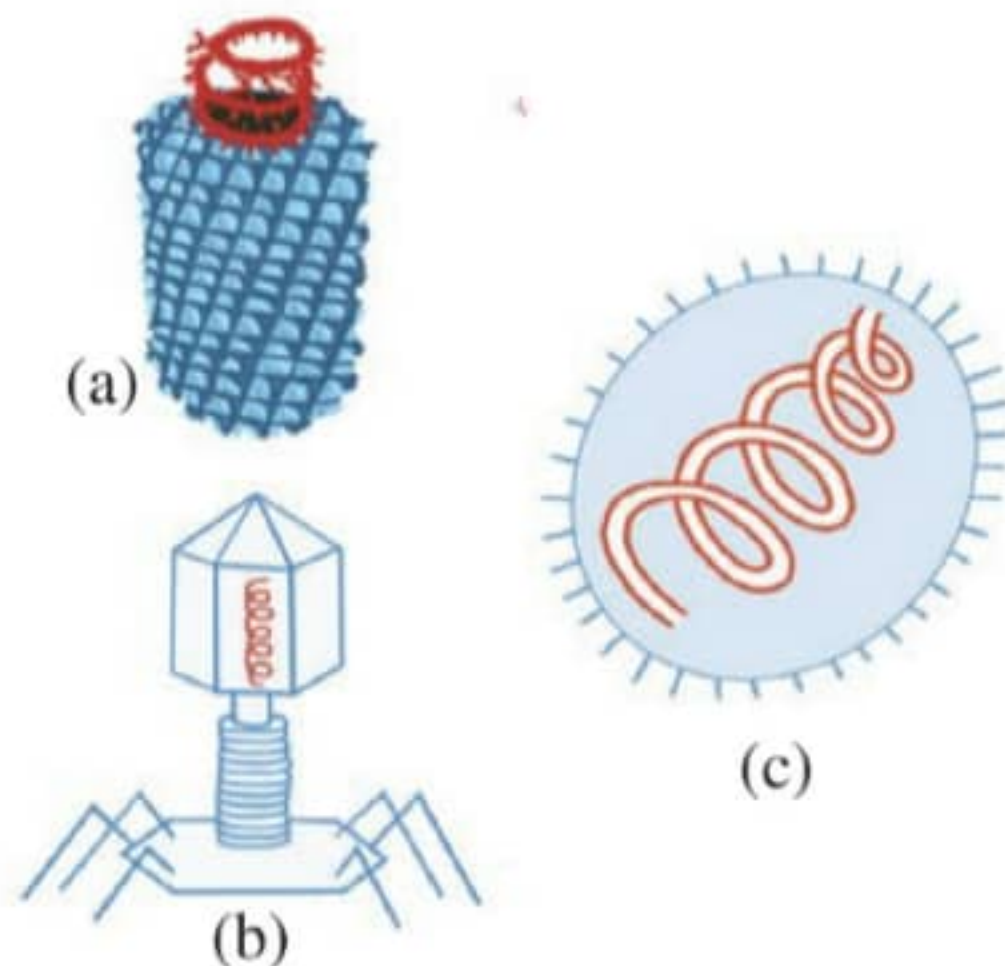
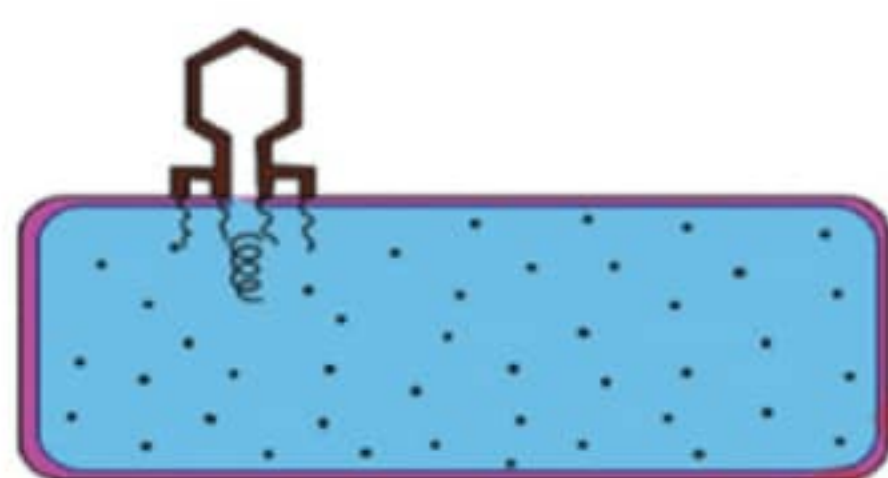


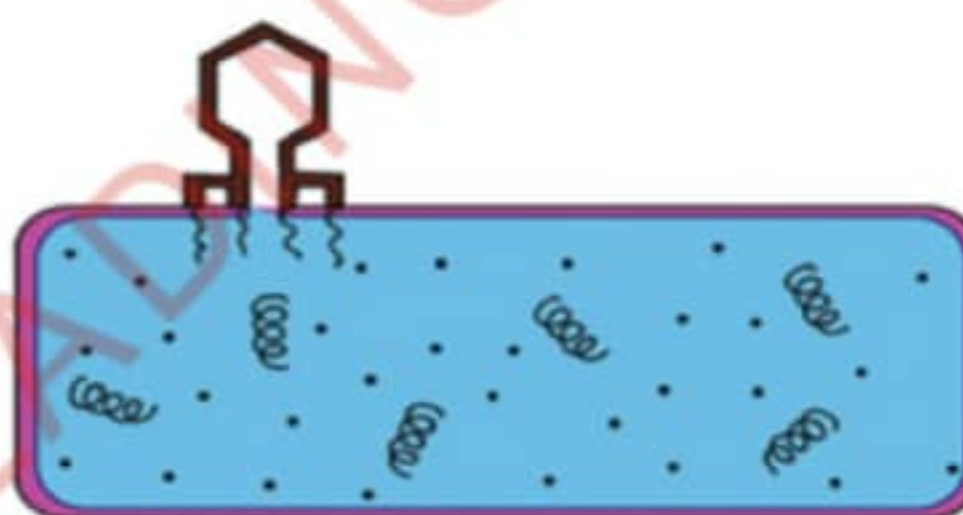
Figure 5.1: Different types of viruses (a) Tobacco mosaic virus (b) Bacteriophage virus (c) Influenza virus

Characteristics of viruses

- (a) They do not have a nucleus, cytoplasm, or cell organelle.
- (b) They have a simple structure consisting of a small piece of nucleic acid (DNA or RNA) surrounded by a protein coat called a capsid. Some viruses have viral envelopes. These are membranes enclosing the capsids. The envelopes are made up of proteins from the host cell.
- (c) They cannot reproduce on their own. They must attack a host cell and use the materials in that cell to reproduce. This is called obligate parasitism. The ability of a virus to reproduce inside the cell and crystallize in the absence of a living host places them between living and non-living things. Figure 5.2 shows how some viruses attacking and reproducing inside the host cell.



(a) Virus attach itself to a host cell and inject DNA into the cell



(b) The virus DNA undergoes replication inside the host cell and forms viral components



(c) The viral components are assembled to form new viruses which are complete with a protein coat



(d) The host bursts to release new viruses

Figure 5.2: Viral reproduction

- (d) When outside a host cell, they show no symptoms of life (do not grow, feed, excrete, or respire). They exist in a dormant state.
- (e) They are host-specific. This means that a certain type of virus attacks only a specific host. For example, the rabies virus affects only mammals and HIV attacks only certain types of white blood cells in human beings.

- (f) Most viruses are infectious: This means they can cause diseases to their hosts.

The structure of viruses

Viruses are composed of strands of genetic material (DNA or RNA) which

forms the core. The core is enclosed by a protein coat called capsid as in bacteriophage. DNA or RNA are of various shapes and sizes according to the type of the virus. Examples of the structures of viruses are shown in Figure 5.3.

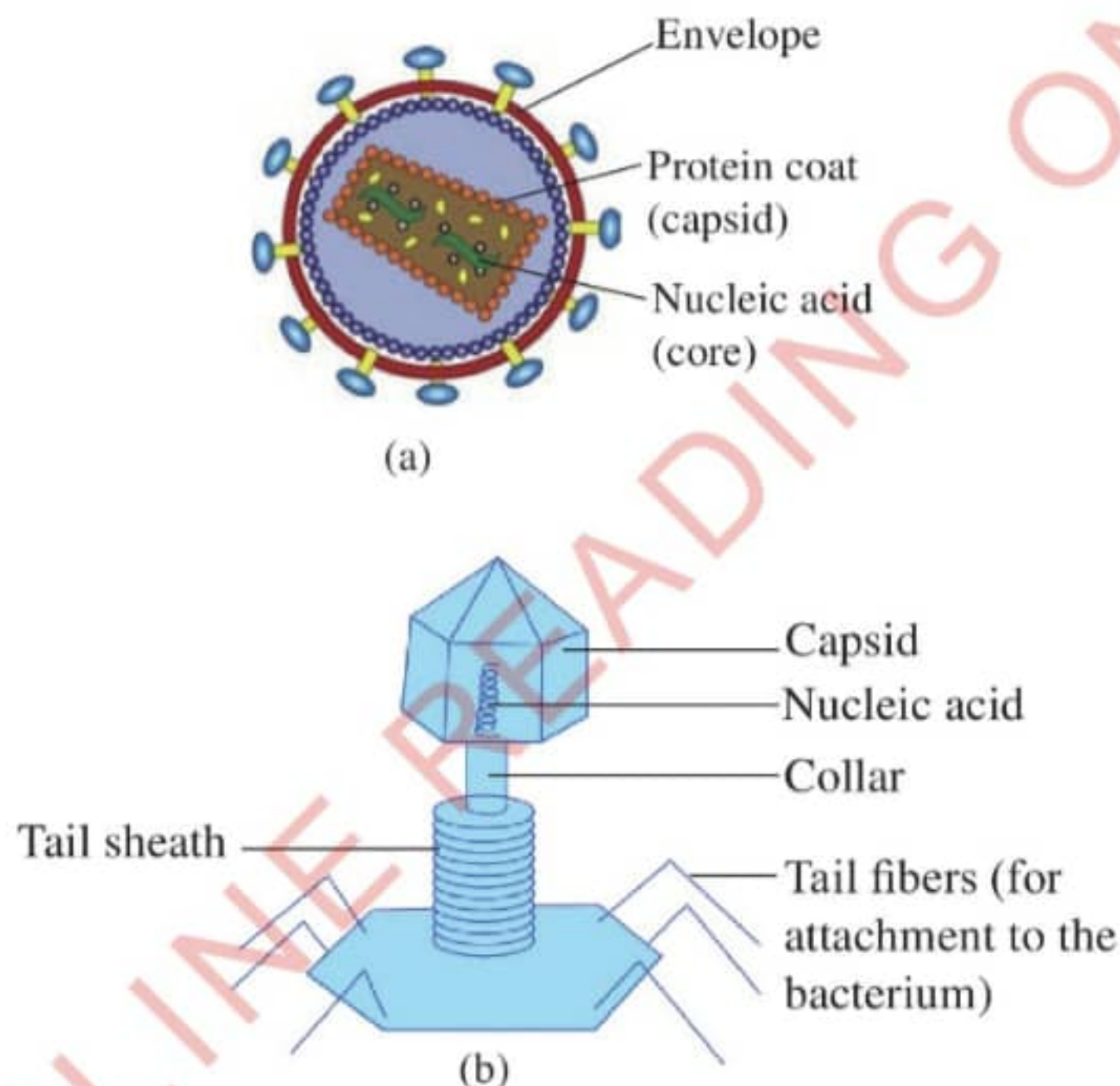


Figure 5.3: Structure of (a) Human Immunodeficiency Virus (b) Bacteriophage

Advantages of viruses

- (a) Viruses are important in the study of cell and molecular Biology. They are used by scientists to manipulate and investigate the functions of cells.
- (b) Some viruses are used to make vaccines. For example, the first vaccine against smallpox was a small dose of the virus that causes cowpox, which is a mild infection.

On recovering from cowpox, the body developed antibodies that could resist both cowpox and smallpox.

- (c) Bacteriophages are viruses that attack bacteria. They help in controlling bacterial infections and diseases.
- (d) Some viruses are used in biological control to eradicate pests, such as insects. This is due

to their characteristics of being host specific and infectious.

Disadvantages of viruses

- (a) Most viruses are pathogenic. They cause infections and diseases, such as tobacco mosaic disease, tomato spotted wilt disease, cassava mosaic disease, rabies, chickenpox, COVID-19, polio, and AIDS. Pathogenic viruses can reproduce very fast, leading to large-scale epidemics.
- (b) Viruses can alter themselves often, and thus become difficult to cure viral infections. For example, there are many different types of viruses that can cause common cold and influenza.

Exercise 5.1

1. Why are viruses regarded as non-living things?
2. Explain why some viruses are used in agriculture to fight against pests and disease vectors.
3. Viruses can be regarded as a threat to human health. Explain.

Kingdom Monera

Task 5.2

From different reliable internet sources and biology books, read about the characteristics of bacteria, their advantages and disadvantages.

Kingdom Monera consists of bacteria and blue-green algae. The scientific study of bacteria is called Bacteriology.

Characteristics of bacteria

- (a) They are prokaryotic as their cells lack a well-defined nucleus. Their nucleus has no nuclear membrane. They also lack other membrane bound organelles.
- (b) They are small unicellular organisms. Some bacteria stick together to form chains or clusters called colonies.
- (c) Some are free-living while others are parasites or saprophytes. Free-living bacteria means that they exist on their own. Parasitic bacteria depend on other organisms known as hosts for their needs. Examples of hosts are human beings and other animals. Saprophytic bacteria get their food from dead organic matter by external digestion.
- (d) Free-living bacteria and some parasitic bacteria have flagella for movement.
- (e) Bacteria have a slimy outer layer. This layer helps to protect them from parasites like viruses and predators, such as protozoa.
- (f) They reproduce asexually by binary fission or through spores.
- (g) They occur in various shapes. There are five known shapes of bacteria. These are spherical-

shaped bacteria or cocci (singular is coccus), rod-shaped or bacilli (singular is bacillus), spiral-shaped bacteria or spirilla (singular is spirillum), and comma-shaped bacteria or vibrio (singular is vibrio). The following Table shows various shapes of bacteria.

Table: Groups of bacteria based on their shapes

Shape	Group name	Examples
 Spherical-shaped bacteria	Cocci	<i>Streptococcus pneumoniae</i> which causes pneumonia and <i>Neisseria gonorrhoeae</i> which cause gonorrhoea
 Rod shaped bacteria	Bacilli	<i>Mycobacterium tuberculosis</i> that causes tuberculosis <i>Escherichia coli</i> that is found in the guts of human beings can cause diarrhoea and urinary tract infections
 Spiral-shaped bacteria	Spirilla	<i>Treponema pallidum</i> that causes syphilis
 Comma-shaped bacteria	Vibrio	<i>Vibrio cholerae</i> that causes cholera
 Corkscrew bacteria	Spirochaetes	<i>Borrelia</i> sp. which cause Lyme disease and relapsing fever

Activity 5.1: *Observing features of bacteria*

Materials: Microscope, prepared microscope slides of bacteria, charts showing structure of bacteria, ICT tools, notebook, and pen or pencil

Procedure

1. Observe the charts that show structure of bacteria or prepared slides of bacteria using the microscope.

2. Search from reliable internet sources the videos that show structures and shapes of bacteria.
3. Describe the shapes and structural features of the bacteria you observed.
4. Based on your observations and comparisons with diagrams in Figure 5.4, discuss any similarities and differences you noted between the observed bacteria and the diagrams in Figure 5.4.

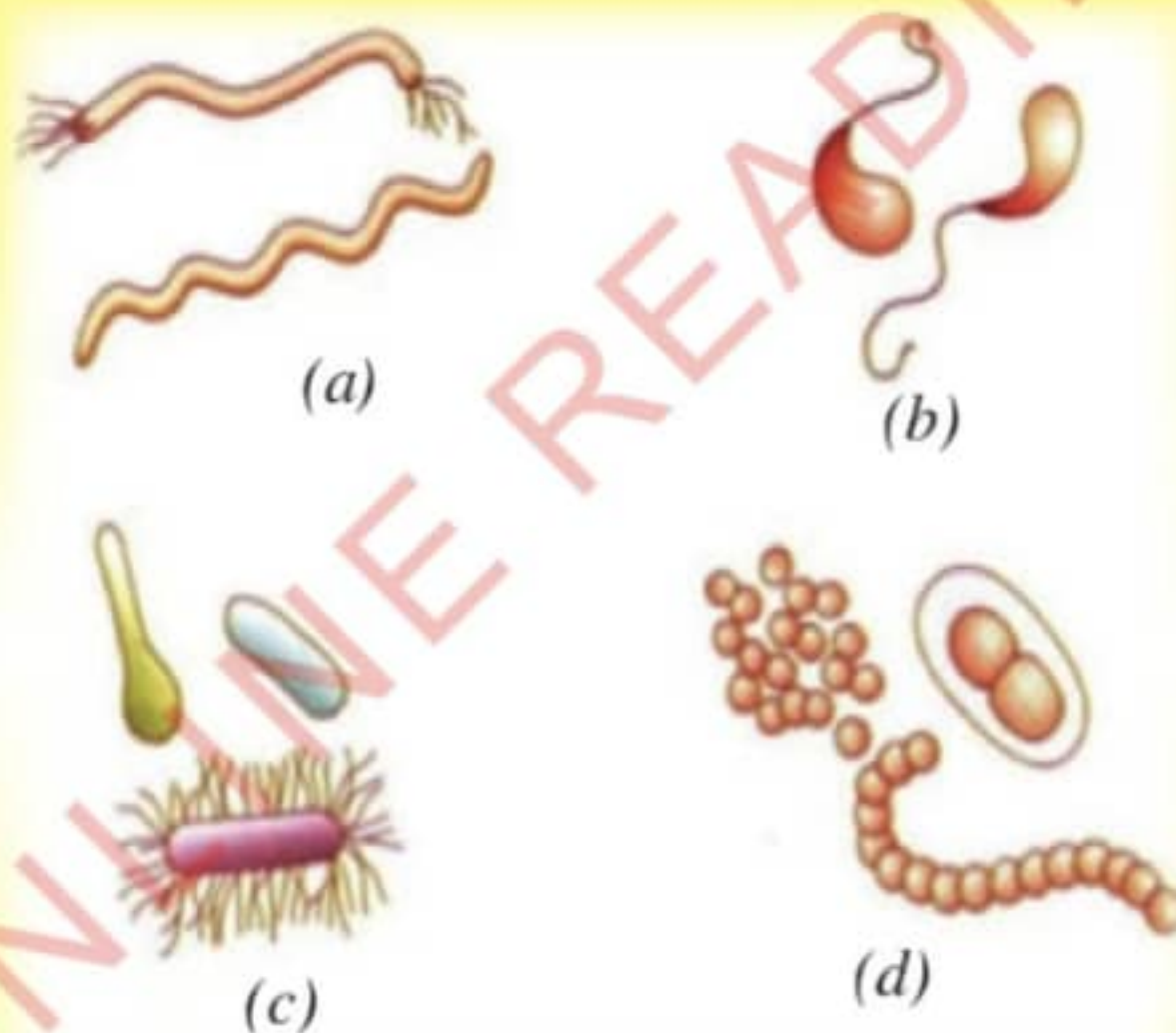


Figure 5.4: Structures and shapes of bacteria (a) spirilla (b) vibrio (c) bacilli (d) cocci

Structure of bacteria

Bacteria have circular DNA suspended in the cytoplasm without being enclosed by the nuclear membrane. The cell wall encloses the cell membrane. Figure 5.5 shows a generalised structure of a bacterium.

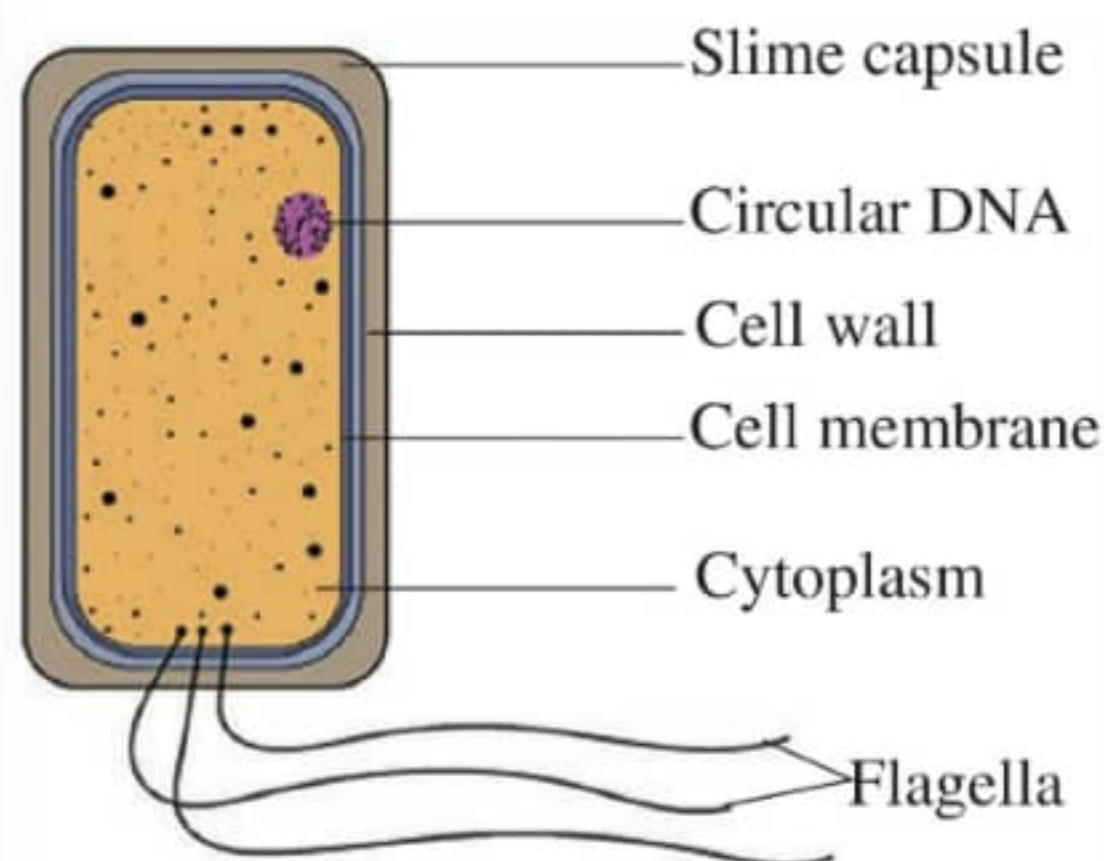


Figure 5.5: Generalised structure of a bacterium

Bacteria can be pathogenic or non-pathogenic.

Task 5.3

Search for information on pathogenic and non-pathogenic bacteria from the library and reliable internet sources.

Pathogenic bacteria

These are bacteria that can cause infections and diseases. Some of the diseases caused by bacteria in plants are fire blight and crown gall. In animals, bacteria cause diseases, such as tuberculosis, tetanus, cholera, and gonorrhoea. Pathogenic bacteria have features that facilitate the spread of diseases and infections which include fimbriae or pili and flagella.

Fimbriae or pili

Fimbriae are hair-like structures on the surface of some bacteria. For example, *Escherichia coli* have

fimbriae that attach themselves to the lining of the urinary tract or the intestines. Example of the structure of bacterium with fimbriae is shown in Figure 5.6.

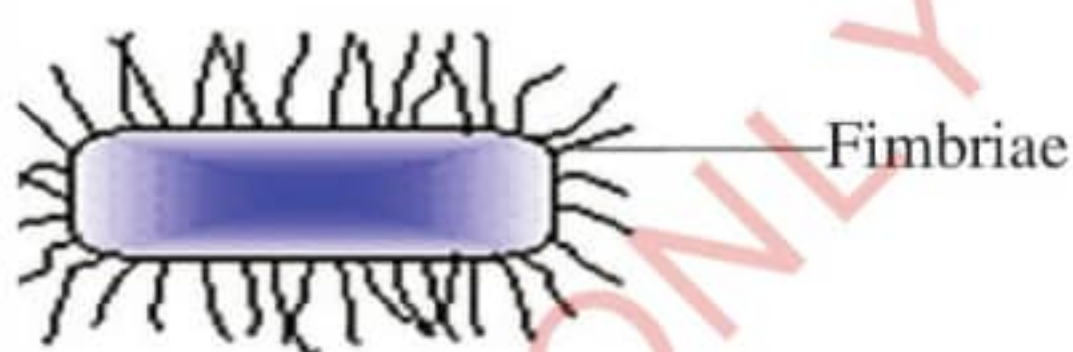


Figure 5.6: Bacterium with fimbriae

Flagella

These are long tail-like structures that help some pathogenic bacteria to move. An example of bacteria with flagella is shown in Figure 5.5.

Toxins

Some bacteria produce toxic compounds that harm their hosts. Human body reacts to the produced toxins by inducing vomiting and diarrhoea in order to remove those toxins. For instance, toxins released by *Clostridium* sp. can lead to severe diarrhoea due to body reaction. Bacteria also perform invasion and colonization; transformation; and resistance. Invasion and colonization happen when bacteria attack cells and tissues and reproduce rapidly. Bacteria can transform themselves and become resistant to modes of treatment. For example, the bacterium that causes tuberculosis has become resistant to many antibiotics by transforming itself.

Non-pathogenic bacteria

These are bacteria that are harmless even when they are in a plants or animal's body. For example, **Rhizobium** sp. that are found in the root nodules of leguminous plants, help to convert atmospheric nitrogen into nitrates in the soil. Bacteria in the human gut synthesize vitamins K and B12.

Advantages of bacteria

- (a) Some non-pathogenic bacteria feed on substances that are harmful to the environment, and in the process neutralize them. For example, they neutralize petroleum waste from petroleum industries, dyes and pesticides, thereby making the environment clean.
- (b) Bacteria in the stomach of ruminant animals, such as goats and cattle secrete cellulase enzyme. Cellulase enzyme helps in the digestion of cellulose which forms a large part of the diet of ruminants.
- (c) Nitrogen-fixing bacteria live in the root nodules of leguminous plants. They convert free nitrogen in the air into nitrates in soil which can be used by plants to make proteins.
- (d) Many bacteria are decomposers. They break down the dead bodies of plants and animals to release important elements, such as nitrogen, carbon, and phosphorus which can be recycled.

- (e) Some bacteria produce lactic acid as a result of fermentation of carbohydrates. This is important in the production of cheese, yoghurt, vinegar, and alcohol.
- (f) Some bacteria are used to produce antibiotics, which are used to treat bacterial infections.
- (g) Bacteria are used in genetic engineering to produce hormones, such as insulin and human growth hormone.
- (h) Bacteria such as **Escherichia coli** in the human gut synthesize vitamins K and B12.

Disadvantages of bacteria

- (a) Many bacteria cause infections and diseases in animals. Some of the common bacterial diseases of animals are typhoid fever caused by **Salmonella typhi**, Tuberculosis (TB) caused by **Mycobacterium tuberculosis** and gonorrhoea caused by **Neisseria gonorrhoeae**.
- (b) Some bacteria cause crops diseases. Examples of these diseases are fire blight on pea and apple, and leaf spot in cotton.
- (c) Some bacteria cause food spoilage by decaying stored food and making it unsuitable for human consumption.
- (d) Sulphur bacteria produce sulphuric acid, which causes damage to buried metal pipes.

- (e) Denitrifying bacteria convert soil nitrates into atmospheric nitrogen; therefore, they reduce soil fertility.

Exercise 5.2

1. Your friend has disposed a fruit to a dump site. Explain what may happen to a fruit overtime and provide reasons why the process is important for environmental conservation.
2. Students are advised to wash their hands after handling samples of bacteria. What is the reason for this advice?
3. Describe a generalised structure of a bacterium.

Kingdom Protoctista

Members of this kingdom include *Amoeba* sp., *Euglena* sp., *Plasmodium* sp., *Paramecium* sp., and *Trypanosoma* sp.

Task 5.4

From the library and reliable internet sources, search information about protoctists (characteristics, advantages, and disadvantages).

Characteristics of protoctists

- (a) They are eukaryotic as they have nucleus and other membrane bound organelles.
- (b) Most of them are unicellular organisms.

- (c) Most of them live in or near water, or in moist places.
- (d) Some protoctists are autotrophs (manufacture their own food) while others are heterotrophs (obtain nutrients from other organisms).
- (e) They include mobile and sessile species.
- (f) Some protoctists reproduce sexually and others asexually while others reproduce both sexually and asexually.
- (g) Many protoctists have locomotory structures, such as cilia, flagella or pseudopodia.

Phyla of kingdom Protoctista

Protoctista is subdivided into several phyla. Only five phyla of this kingdom will be described in this section. These phyla are Rhizopoda, Apicomplexa, Euglenophyta, Ciliophora, and Zoomastigina.

Phylum Rhizopoda

Organisms included in phylum Rhizopoda move and feed using structures known as pseudopodia (singular is pseudopodium). An example of organisms found in this phylum is *Amoeba* sp. Parasitic amoeba includes *Entamoeba histolytica*, which feeds on cells of the human colon and cause amoebic dysentery. Free living amoeba are found in sea water, freshwater, and in the soil. They can also be found in muddy ponds and slow

flowing streams containing decaying organic matter. A generalised structure of amoeba is shown in Figure 5.7.

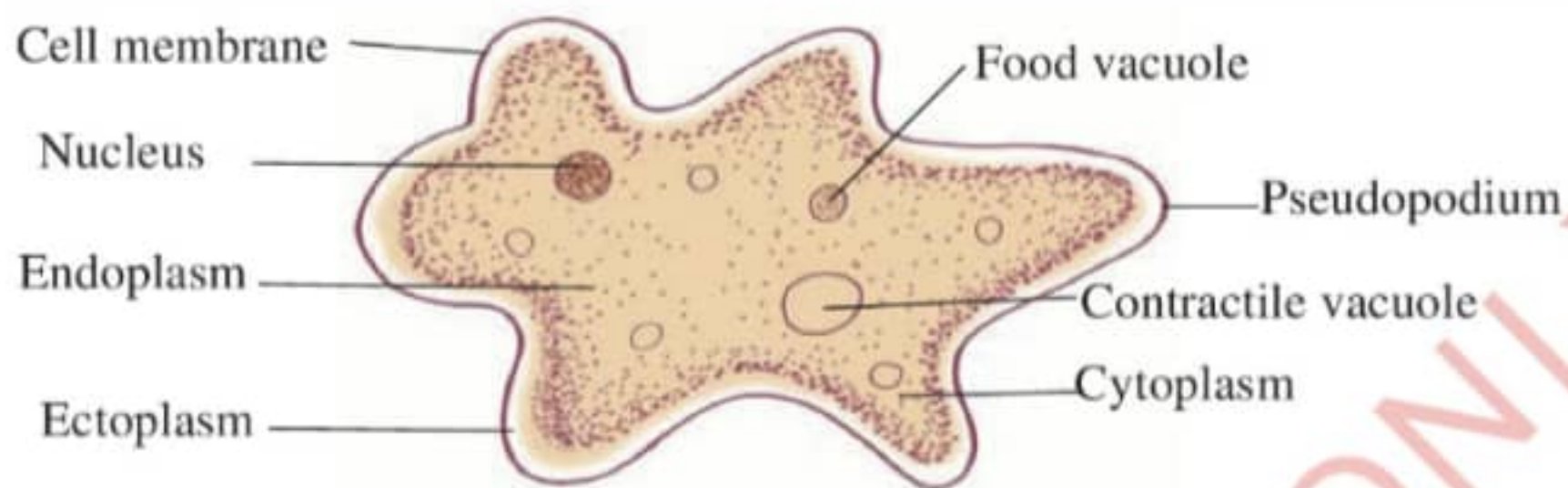


Figure 5.7: Generalised structure of amoeba

Characteristics of amoeba

- Amoeba have two layers of cytoplasm; a viscous outer layer called ectoplasm and a more fluid internal layer called endoplasm.
- Most of them are free living, feeding on plant and animal matter. Others are parasitic in animals, including human beings.
- They have pseudopodia which are used for both locomotion and feeding, as shown in Figure 5.8.

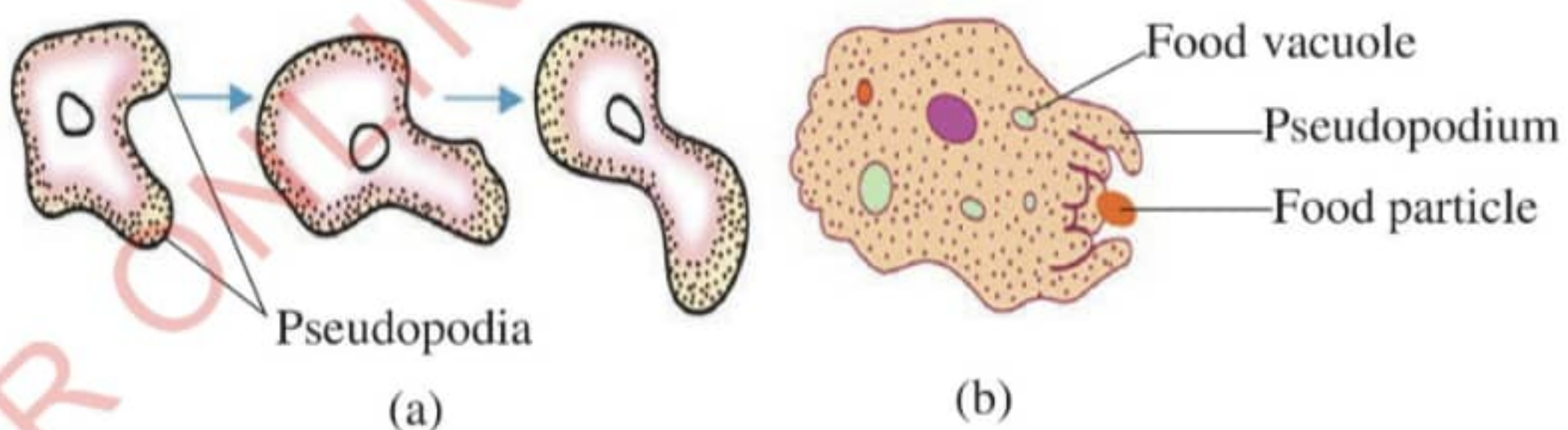


Figure 5.8: Amoeba using pseudopodia to (a) move (b) feed

- They have contractile vacuoles which regulate the amount of water in the freshwater amoeba.
- They form temporary food vacuoles to hold and digest food particles.
- In amoeba, waste products, such as urea and ammonia are excreted by simple diffusion. Diffusion is the movement of substances from a region

of high concentration to a region of low concentration. When there is a high concentration of urea or ammonia in the amoeba, these substances flow out of the amoeba into the surrounding water.

- (g) In amoeba, oxygen and carbon dioxide gases are exchanged by diffusion.
- (h) They are unicellular organisms.
- (i) They reproduce asexually by binary fission, as shown in Figure 5.9.

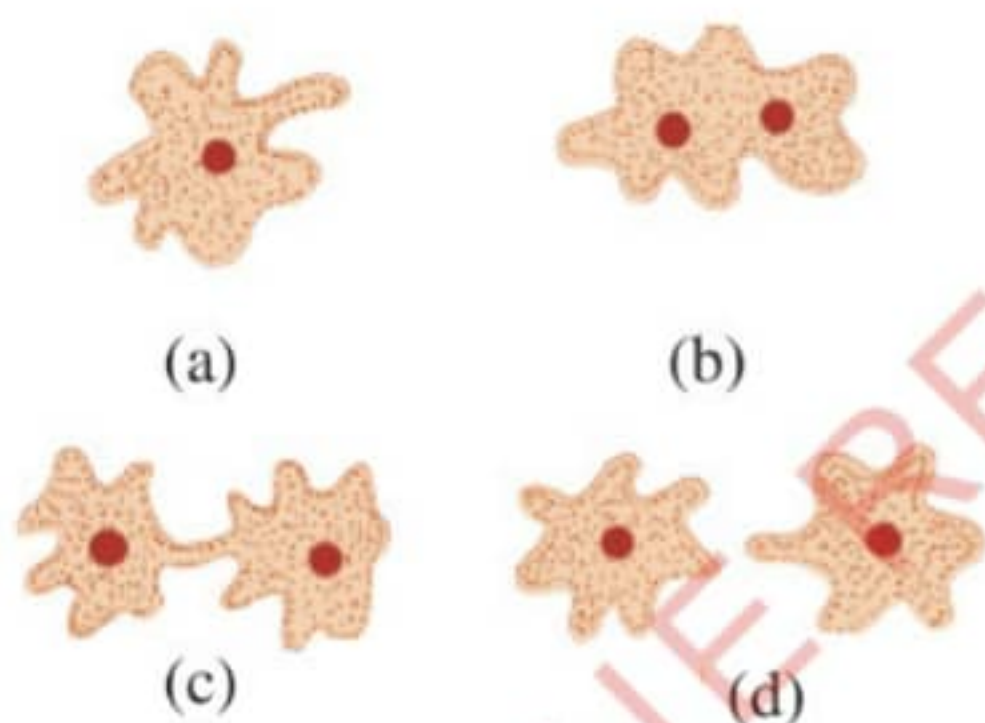


Figure 5.9: Binary fission in amoeba
 (a) mother cell (b) mother cell nucleus divides (c) mother cell cytoplasm divides (d) two new daughter cells

Task 5.5

Use reliable internet sources to view simulations and videos showing:

- (a) Movement in amoeba
- (b) Feeding in amoeba
- (c) Asexual reproduction (binary fission) in amoeba

Advantage of amoeba

- (a) Amoeba are commonly used in laboratories to study cell structure and function.
- (b) Amoeba are important decomposers in aquatic environments, feeding on organic particles in the water.

Disadvantage of amoeba

Some species of Amoeba cause diseases. For example, *Entamoeba histolytica* causes amoebic dysentery in humans and *Entamoeba gingivalis* cause teeth and gum diseases.

Exercise 5.3

- Imagine you are a scientist studying micro-organisms in a pond ecosystem. While examining a water sample under a microscope, you come across a single-celled organism with unique appearance. Describe the characteristics you would look for to determine whether this organism is a protoctists.
- You are a medical researcher investigating an outbreak of amoebic dysentery in a community. You have collected samples from patients and identified the causative organism using a microscope. Describe the key characteristics of this organism that would help you confirm that the organism is the one responsible for causing amoebic dysentery.
- Explain the advantages of amoeba.

Phylum Euglenophyta

This phylum consists of unicellular aquatic organisms most of which live in freshwater and possess flagella. *Euglena* sp. is an example of organisms found in this phylum. Figure 5.10 shows the structure of *Euglena* sp.

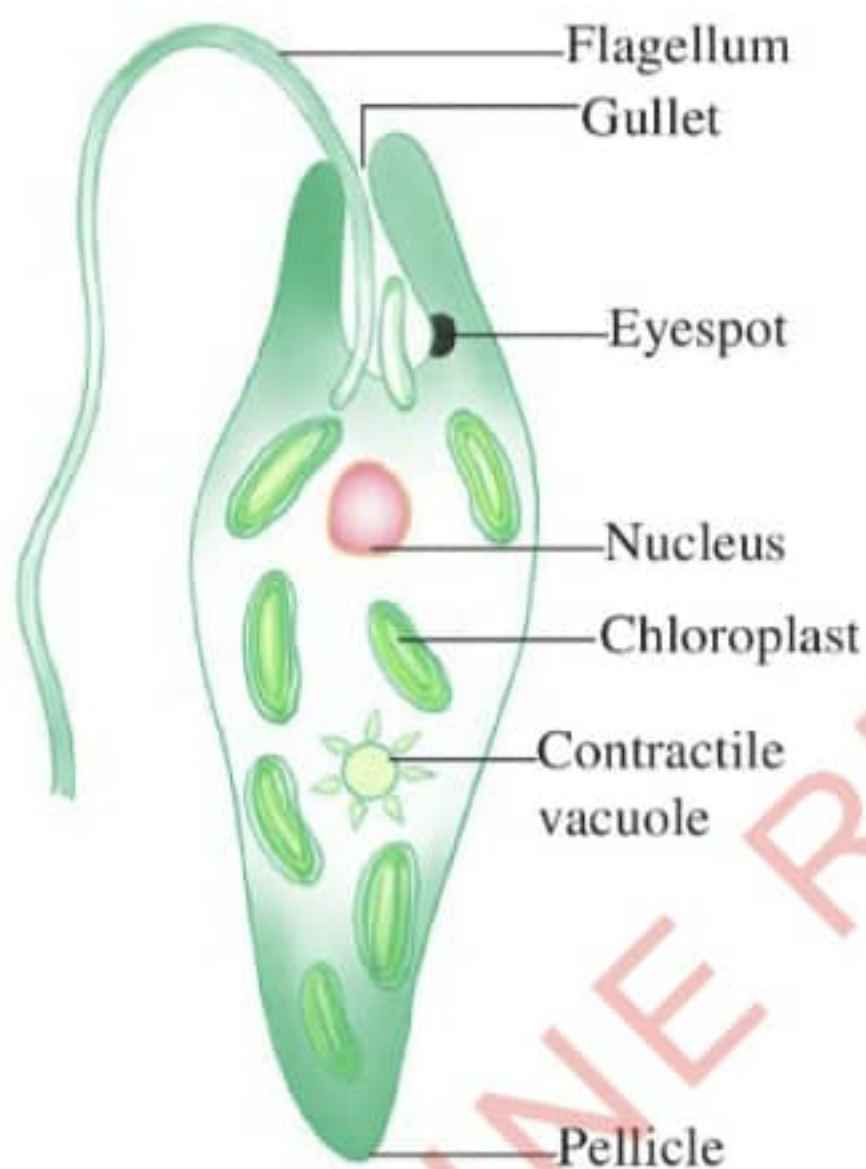


Figure 5.10: Structure of the *Euglena* sp.

Characteristics of *Euglena* sp.

- They are unicellular.
- They have chloroplasts which contain chlorophyll for photosynthesis.
- Most of them are found in freshwater; only a few are found in salt water.
- They are protected by pellicle, which surrounds the cytoplasm. The pellicle allows the euglena to change shape.
- They have flagella for movement.

- They have eye spot for detection of light intensity.
- They are eukaryotic cells.
- They possess both plant and animal characteristics. The plant characteristic is the possession of chloroplasts which contain chlorophyll for photosynthesis. The animal characteristics include ability to move using a flagellum, presence of an eyespot for detection of light intensity, and presence of gullet for ingesting food.

Advantages of *Euglena* sp.

- They are used for sewage treatment due to their heterotrophic mode of nutrition.
- Euglena* sp. is an important source of food to aquatic organisms, such as fish.
- During photosynthesis, *Euglena* sp. produce oxygen gas. This oxygen is useful to aquatic organisms. It also helps to balance oxygen gas levels in sewage treatment plants.

Disadvantages of *Euglena* sp.

Euglena sp. blooms can be harmful to fish. The toxins produced by some *Euglena* species can kill fish and other aquatic organisms.

Phylum Apicomplexa

Organisms found in this phylum are unicellular and parasitic, *Plasmodium* sp. which cause malaria in humans is an example of members of phylum Apicomplexa.

Task 5.6

Search for videos that show characteristics of *Plasmodium* sp. from different reliable internet sources.

Characteristics of *Plasmodium* sp.

- They are unicellular.
- They are eukaryotes.
- They are parasites with a complex life cycle involving the host and the vector.
- They reproduce sexually in the human host and asexually by multiple fission in the mosquito vector and human host.

- When a plasmodium enters the human body, it attacks the red blood cells and liver cells.

Advantage of *Plasmodium* sp.

Plasmodium sp. is used in medical and other researches.

Disadvantage of *Plasmodium* sp.

Plasmodium parasites cause malaria. Malaria can cause inflammation and rupture of the spleen, and miscarriages. It can also cause anaemia due to the destruction of red blood cells. Severe malaria can cause death.

Phylum Ciliophora

Organisms found in this phylum are unicellular and are found in aquatic habitats. *Paramecium* sp. is an example of members of phylum Ciliophora. Figure 5.11 shows a structure of a *Paramecium* sp.

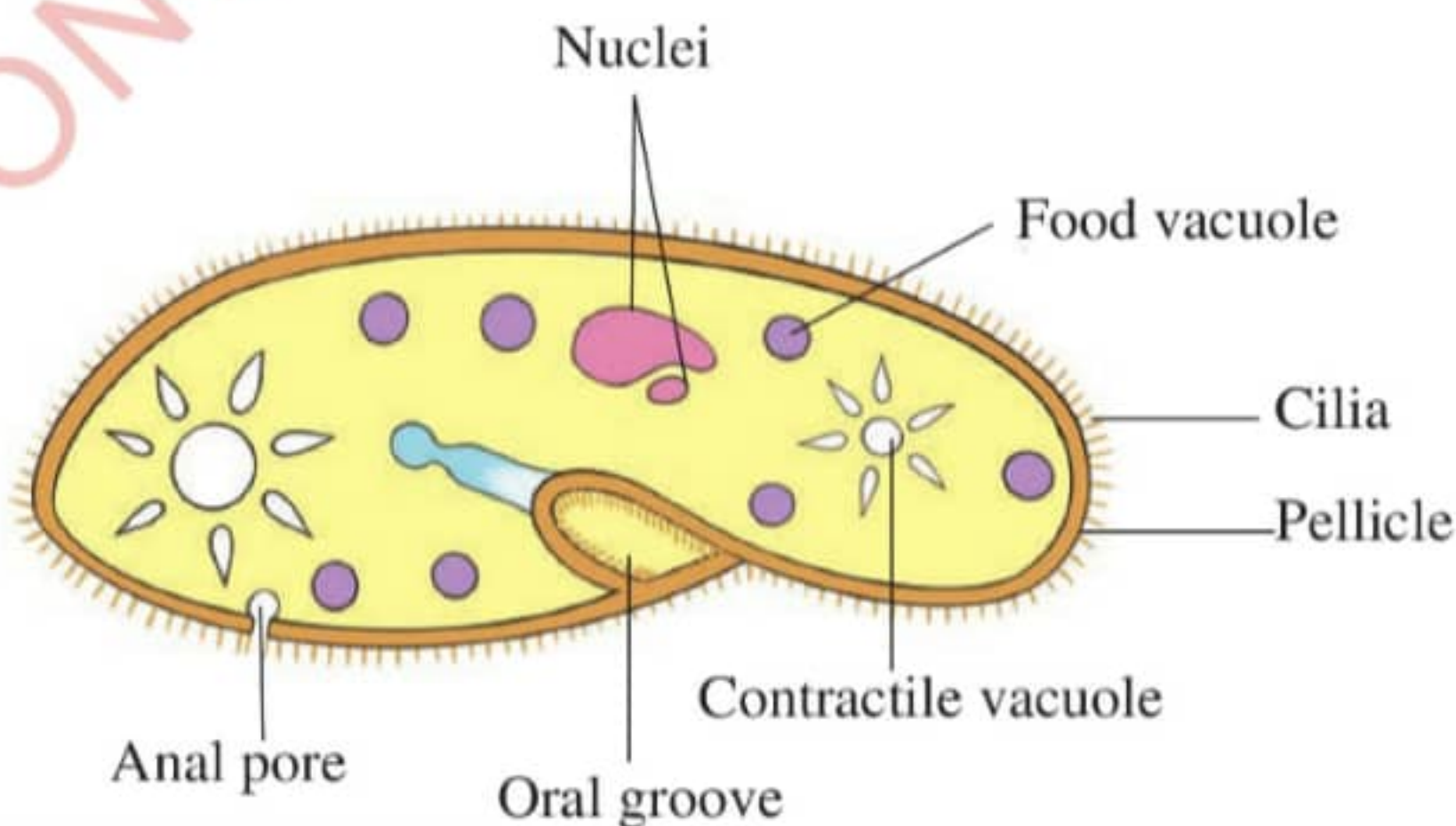


Figure 5.11: Structure of *Paramecium* sp.

Characteristics of Paramecium sp.

- (a) They are unicellular and slipper-shaped.
- (b) They are eukaryotes.
- (c) They are heterotrophic.
- (d) They live in freshwater.
- (e) The body is covered with short hairy structures called cilia.
- (f) They use cilia to move and to create a feeding current.
- (g) The body of the paramecium cell is enclosed by a stiff but elastic membrane, called the pellicle.
- (h) Food enters the organism through an opening called the oral groove.
- (i) They have food vacuoles which are formed to hold and digest food particles.
- (j) They have anal pores which are used for elimination of undigested food.
- (k) **Paramecium** sp. can reproduce either asexually or sexually, depending on their environmental conditions. Asexual reproduction takes place when enough nutrients are available, while sexual reproduction takes place under conditions of starvation.
- (l) Contractile vacuoles regulate the amount of water in the cytoplasm.

Advantages of Paramecium sp.

Paramecium sp. play important ecological role in the environment as they help to clean up small particles of debris in the water as well as feeding on small animals.

Disadvantages of Paramecium sp.

- (a) Some species of **Paramecium** invade and destroy the lining of the intestine, causing a disease called balantidiasis.
- (b) **Paramecium** sp. feed on bacteria which decompose sewage. This may delay the decomposition of sewage.

Phylum Zoomastigina

Organisms found in this phylum are unicellular flagellate protoctists. Generally, they are found in the intestine, but can also be found in blood stream or in the heart. **Trypanosoma** sp. is an example of members of phylum Zoomastigina. Figure 5.12 shows the structure of a **Trypanosoma** sp.

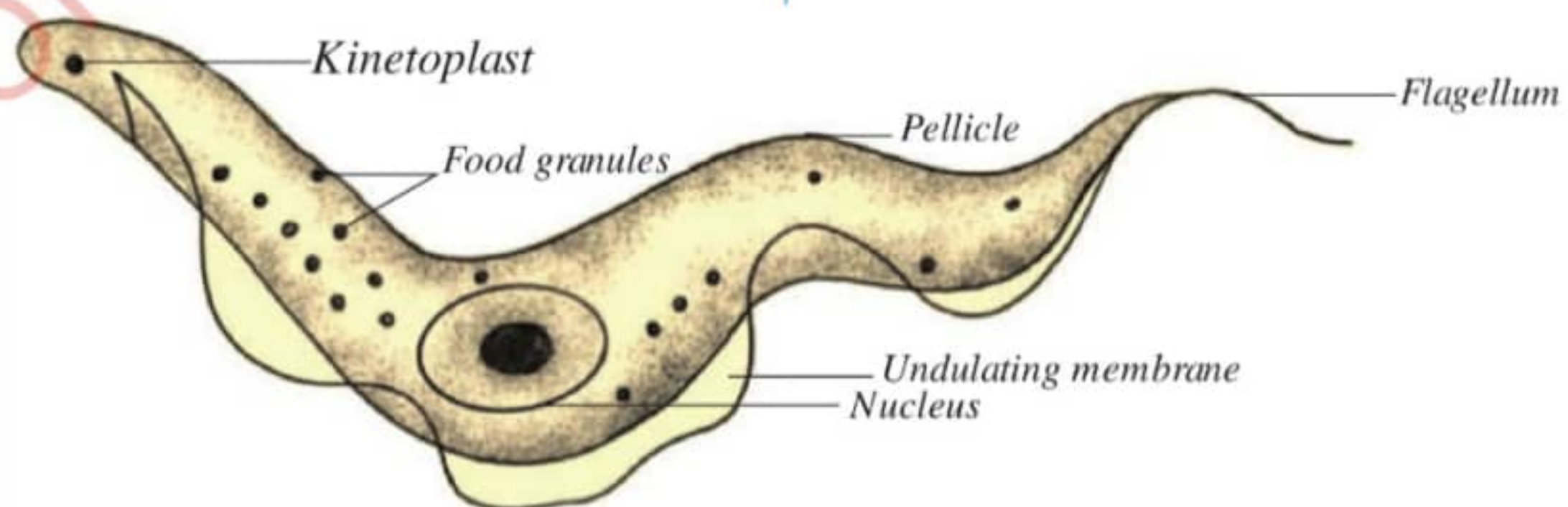


Figure 5.12: Structure of a **Trypanosoma** sp.

Characteristics of *Trypanosoma* sp.

- (a) They are unicellular, slender, elongated, and dorsal ventrally flattened in shape.
- (b) They are parasites in wild animals, domesticated animals, and human beings.
- (c) Have a mass of mitochondrial DNA located near the mitochondrion in a structure called kinetoplast.
- (d) The body is covered with a thin, elastic, and firm pellicle.
- (e) Most of them reproduce asexually by binary fission while inside the host body.
- (f) Some have an undulating membrane for locomotion.

Advantages of *Trypanosoma* sp.

Trypanosoma brucei has been extensively studied by scientists, as it is a model organism for understanding the biology of parasites and the diseases they cause. Research on *Trypanosoma brucei* has led to the development of new drugs and treatments for sleeping sickness and related diseases.

Disadvantages of *Trypanosoma* sp.

- (a) *Trypanosoma brucei* is the causative agent of the disease called human African trypanosomiasis or sleeping sickness. The disease affects mostly African countries. It can be transmitted through bites of a vector called tsetse fly or

through blood contact from an infected individual.

- (b) *Trypanosoma* sp. also cause animal trypanosomiasis or Nagana to cattle, buffalos and other large herbivores.

Exercise 5.4

1. Imagine you are on the field trip to a stagnant water body for your Biology class. Your teacher asks you to collect water samples for microscopic investigation from two different locations in the pond, one near the surface and the other from the bottom. During microscopic examinations you observed presence of *Euglena* sp. from the surface sample and *Paramecium* sp. from the bottom sample. Describe the characteristics of the micro-organisms observed from the samples.
2. You are given a microscope and a slide with a sample of *Trypanosoma* sp. to observe in your Biology class. While looking at the slide under the microscope, your teacher asks you to describe the key characteristics of *Trypanosoma* sp. that you can observe. What features would you look for to explain the characteristics of the sample?

Activity 5.2: Investigating different protoctists

Materials: Specimen bottles, microscope, microscope slides and coverslips, pond water, a notebook, and a pencil

Procedure

1. Go to a pond or any other place with stagnant water.

Caution

Care must be taken when collecting water samples from the pond.

2. Fetch water from the pond using specimen bottles.
3. In the Biology laboratory, put a drop of the pond water on a microscope slide, add stain and cover it with a coverslip.
4. Observe the slide under the microscope. Do you identify any organisms? If Yes, name them.

Kingdom Fungi**Task 5.7**

Read from reliable internet sources, library and other biology books information about fungi.

The kingdom Fungi contains a variety of eukaryotic organisms that consist of bounded cell organelles, such as mitochondria and nucleus. Most of

them are multicellular (made up of many cells) and some are unicellular (made up of a single cell). Example of unicellular fungi is yeast (*Saccharomyces* sp.). Examples of multicellular fungi include bread mould (*Rhizopus* sp.), pin mould (*Mucor* sp.), and mushroom (*Agaricus* sp.).

General characteristics of Fungi

Members of the kingdom Fungi have the following general characteristics.

- (a) They are found in various places including air, water, soil, food, and in the bodies of animals and plants.
- (b) They are eukaryotic organisms with true nucleus which is enclosed in a nuclear membrane.
- (c) They can be either unicellular, example yeast or multicellular, examples mould and mushroom.
- (d) Their body is made up of a mycelium consisting of a network of fine, tube-like filaments called hyphae (except yeast).
- (e) They feed saprophytically, for example mushroom; but some of them are parasitic for example *Candida albicans*.
- (f) Some fungi form symbiotic associations with other species.
- (g) They reproduce both sexually by spores and asexually by budding.
- (h) They store carbohydrates in the form of glycogen.

Distinctive features of Fungi

- They have cell walls made up of chitin, which is a substance containing protein and complex sugars.
- Their bodies are made up of a mycelium consisting of a network of fine, tube-like filaments called hyphae (except yeast).
- They feed saprophytically, for example mushroom; but some of them are parasitic, for example *Candida albicans*.

Phyla of the kingdom Fungi

The members of kingdom Fungi are classified into different phyla. There are three main phyla in kingdom Fungi. These are Ascomycota, Zygomycota, and Basidiomycota.

Task 5.8

Search information from the library and reliable internet sources on yeast, bread mould, and mushroom. Note down their characteristics, advantages and disadvantages.

Phylum Ascomycota

Members of this phylum are commonly called ascomycetes or sac fungi. This is because their sexual spores are enclosed in sac like structures known as asci (singular is ascus). Some ascomycetes such as yeast are single-celled (unicellular) organisms. (See Figure 5.13).

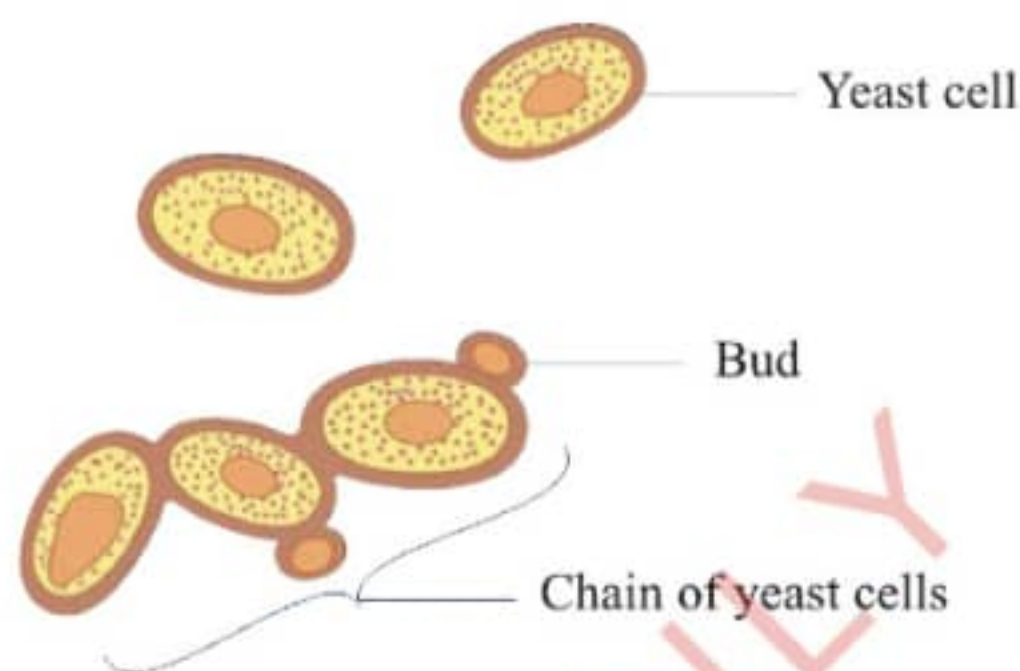


Figure 5.13: Yeast cell

Characteristics of ascomycetes

- Some ascomycetes such as yeast are unicellular and others such as *Penicillium* sp. are multicellular.
- Most ascomycetes have a saprophytic mode of feeding and can grow on the surfaces of dead organic materials such as rotting fruit and other foods.
- They have long tube-like filaments called hyphae with cross walls.
- They reproduce sexually through spores known as ascospores, and asexually through fission, fragmentation, or budding (See Figure 5.14).

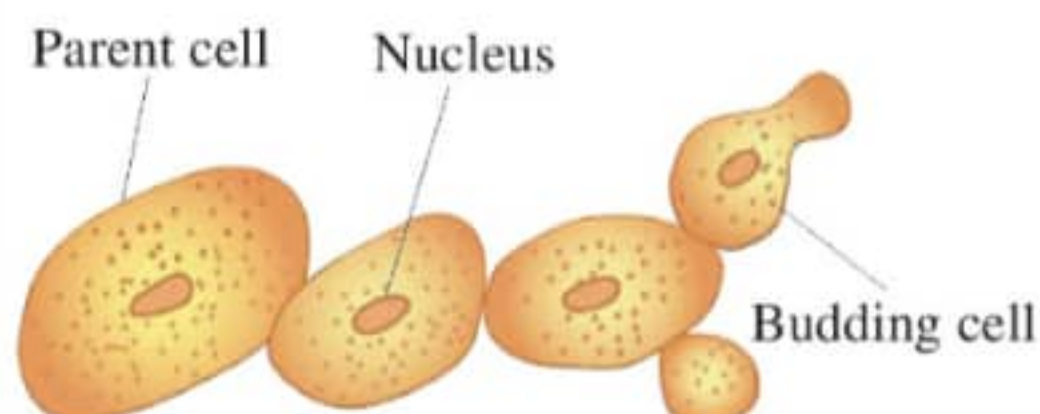


Figure 5.14: Yeast budding

Activity 5.3: Investigating the structure of yeast cells

Materials: Dry yeast, water, methylene blue reagent, microscope, coverslips, microscope slides, dropper, pencil, and notebook

Procedure

1. Dissolve some dry yeast cells in water.
2. Using a dropper, place a drop of the yeast solution on a microscope slide and cover it with a coverslip.
3. Examine the specimen using the low-power objective lens of the microscope.
4. Identify some yeast cells.
5. Describe the structures you have observed in yeast cells when using the high-power objective lens of the microscope.
6. Compare your results with Figure 5.14.

Phylum Zygomycota

The members of this group are known as zygomycetes because they have sexual spores known as zygospores. Zygospores are produced in the structures called zygosporangia. They also have asexual spores known as sporangiospores, which are produced in the structures called sporangia. Zygomycetes grow as a mass of white or dark tiny threads and feed on rotting or decaying food materials like

bread, cassava, pawpaw, and tomato. Examples of organisms in this group include mucor and black bread mould. See Figure 5.15.

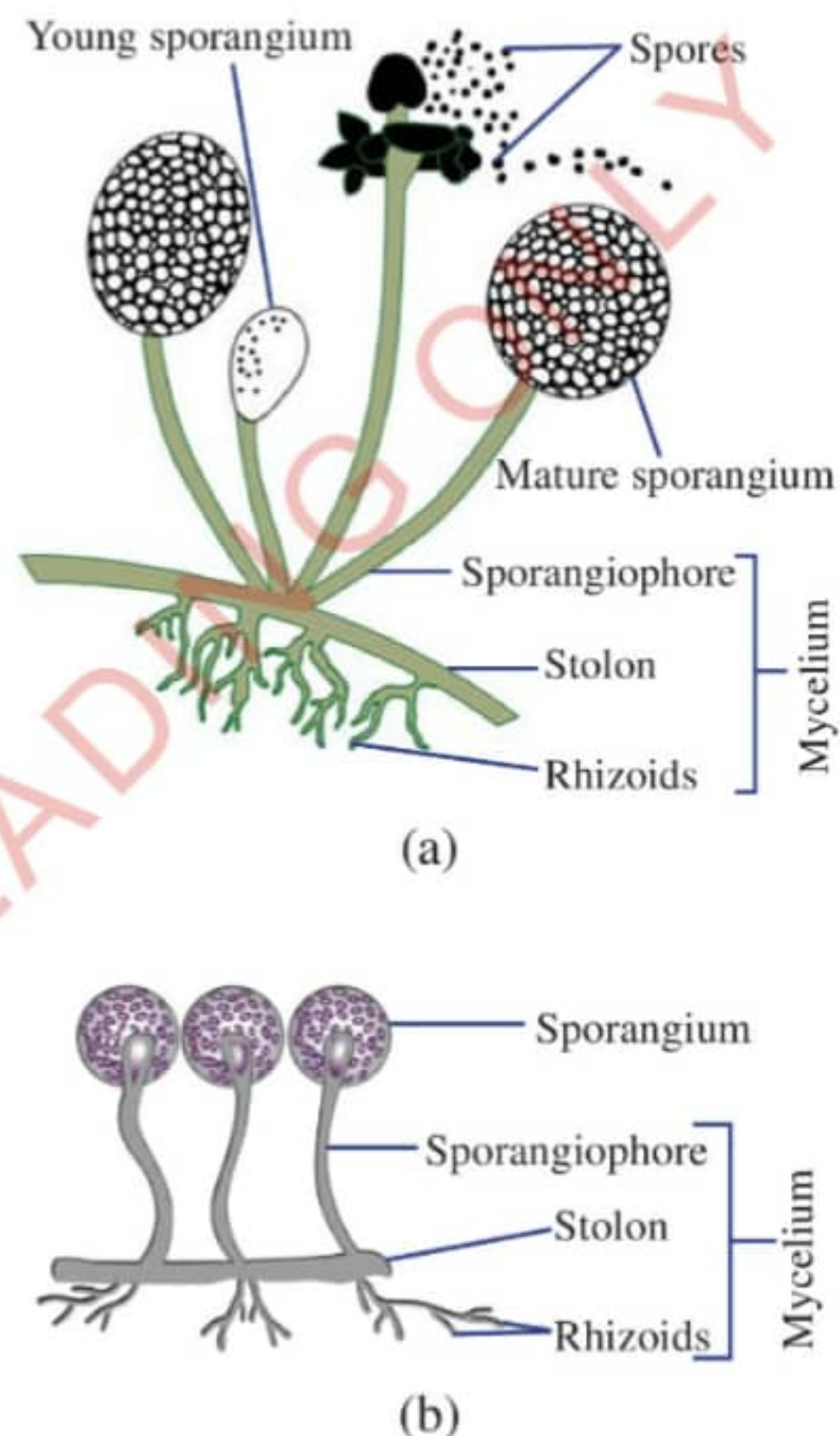


Figure 5.15: Examples of zygomycetes
(a) black bread mould (b) mucor

Characteristics of zygomycetes

- (a) They are multicellular.
- (b) They are saprophytic, growing on decaying organic materials.
- (c) They reproduce sexually through zygospores or asexually through sporangiospores.
- (d) Have hyphae without cross walls.

Activity 5.4: Investigating the structure of bread mould

Materials: A piece of bread, petri dish or watch glass, hand lens, forceps, microscope, microscope slides, pencil, and notebook

Procedure

1. Take a slice of bread, put it on a petri dish and moisten it.
2. Leave the bread on the moist place for 2-3 days until a greyish or dark spots appear on it.
3. Observe the changes that might have occurred on the surface of the bread
 - (a) Use a hand lens and observe the dark spots that have developed on the surface of the bread.
 - (b) Take a small sample of dark spots using forceps and place it on a microscope slide, add a drop of water, and cover with a coverslip.
 - (c) Put the sample on the microscope stage and observe using the low, medium, and high objective lenses.
 - (d) Based on your observations of the structures on the bread using the microscope, describe the features you observed. Compare your observations to the structures shown in Figure 5.15.

Phylum Basidiomycota

Members of the phylum Basidiomycota are called basidiomycetes. They have structures called basidia (singular is basidium) that produce sexual spores. The basidia are usually club-shaped structures. Hence, members of this group are also known as club fungi because they all possess basidia. Their sexual spores are called basidiospores. The common examples of basidiomycetes are mushrooms and toadstools, such as *Agaricus bisporus* (edible mushroom). Other examples of basidiomycetes include brackets, puffballs, and rust fungi. They are well recognisable in various habitats, especially in areas with soil rich in nutrients. Figure 5.16 shows the structure of a mushroom.

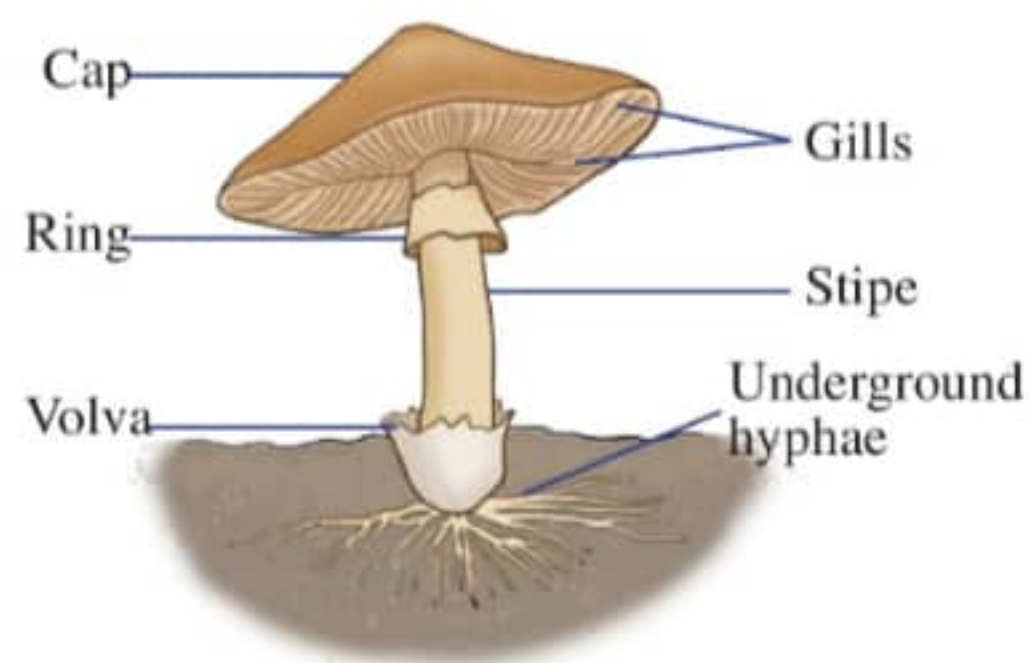


Figure 5.16: Structure of a mushroom

Characteristics of basidiomycetes

- (a) They are multicellular organisms.
- (b) They are saprophytic, growing on decaying organic matter.
- (c) They reproduce both sexually and asexually.

- (d) They produce sexual spores called basidiospores contained in club-like structures known as basidia.
- (e) They have hyphae with cross walls (septate), volva, stipe (stem-like structure), ring, cap, and gills.

Activity 5.5: Investigating the structure of a mushroom

Materials: Mushroom, hand lens, scalpel, petri dish, scissors or sharp knife, pencil, and notebook

Procedure

1. Use a hand lens to examine the mushroom structure.
 2. Describe the key features and structures you observe when you:
 - (a) view from the side; and
 - (b) view from the lower surface of the cap.
 3. Use a scalpel or knife to cut the mushroom vertically into two equal parts.
- Caution**
Care should be taken when using sharp objects, such as knife.
4. Describe the features and structures you see in:
 - (a) the whole mushroom, as seen from the side; and
 - (b) cut the surface of the mushroom.
 5. Compare your results with Figure 5.16.

Question

Based on your individual observations and comparisons with Figure 5.16, what conclusion can you draw regarding the structure of the mushroom?

Advantages of fungi

- (a) Most strains of mushrooms and toadstools, such as **Agaricus** sp. are edible. They are used as sources of food and they have a high content of proteins and vitamins.
- (b) Fungi, such as yeasts (**Saccharomyces** sp.) are used in food manufacturing industries, such as in bakeries (they make dough to rise). Yeast is also used in brewing beer by fermenting sugars to produce alcohol.
- (c) They are used in making lactic acid, citric acid, cheese, and commercial enzymes.
- (d) Some fungi such as **Penicillium** spp. are important in production of drugs such as penicillin.
- (e) Most species of fungi are used in genetic engineering and biological researches.
- (f) Saprophytic fungi are very important in the decomposition of dead organisms and waste materials. Through decomposition by fungi, important nutrients are released into the soil. The nutrients are then absorbed and used by plants.
- (g) Some types of ascomycetes and basidiomycetes develop symbiotic relationship with roots of higher

plant to form mycorrhiza. The hyphae of fungi spread apart and increase the surface area of the vascular plant roots. This helps the plant roots to absorb more nutrients from the soil.

Disadvantages of fungi

- (a) Some fungi cause diseases to human beings and other organisms. Some of the diseases caused by fungi in humans include; athlete's foot, ringworm, meningitis, and skin infections. In some people, the fungal spores can cause allergic reactions.
- (b) In plants, some fungi, such as ascomycetes cause various diseases like potato blight, maize rust, rice blast, and powdery mildew in beans and other legumes. Some fungi attack timber used in building houses and making furniture.
- (c) Some fungi, such as *Candida albicans* live in the bodies of human beings. They grow naturally in the human body, particularly in the mouth, throat, gut, and female reproductive tracts without causing any harm. These fungi can cause infection if they grow in excess and enter deep into the body organs and systems, such as in the blood stream, kidney, heart or brain.

- (d) Some parasitic fungi, such as certain types of ascomycetes produce poisonous substances called mycotoxins. The most common mycotoxin is known as aflatoxin, which is commonly found in harvested maize and groundnuts that have been left in moist places. Aflatoxin is carcinogen (cancer causing agent).
- (e) Some mushrooms, such as *Amanita* sp. are poisonous to humans when eaten.
- (f) Some fungi, such as moulds and other parasitic fungi cause food spoilage.

Exercise 5.5

1. Suppose you are in a class and your Biology teacher brings an unknown specimen. Which characteristics will you look for to confirm that it belongs to the kingdom Fungi?
2. Name the phylum to which each of the following organisms belongs.
 - (a) Bread mould
 - (b) Mushroom
 - (c) Yeast
3. Due to the occurrence of ringworm caused by fungi, many people view fungi as a threat. How would you educate them on the beneficial aspects of fungi in our daily lives?

Kingdom Plantae

Kingdom Plantae is composed of a wide variety of plants. Members of this kingdom vary greatly in size, forms, habitat, means of reproduction, and morphology. Kingdom Plantae includes; moss, ferns, cone bearing plants, and flowering plants. Plants are found in various habitats, such as on land, in oceans, and in freshwater.

Activity 5.6: Investigating plant parts

Materials: A variety of plants, hand lens, notebook and pen or pencil

Procedure

1. Collect a variety of plants from the school or home environment.

Caution

Care must be taken when collecting plants, some plants such as milkweed and oleander are poisonous.

2. Examine each of the plant parts using a hand lens. The parts to be examined include roots, stems, leaves, flowers, and fruits. Do all plants have these parts?
3. Group the plants according to their similarities and differences.

General characteristics of members of the kingdom Plantae

- (a) They are multicellular and eukaryotic organisms.

- (b) They have cell organelles called chloroplasts that contain chlorophyll.
- (c) Green plants are photoautotrophs. This means that they manufacture their own food using sunlight through the process of photosynthesis.
- (d) Some plants reproduce sexually through flowers or cones while others reproduce asexually by means of spores or vegetative propagation.
- (e) Their cells are organised into tissues, organs, and organ systems.
- (f) They show limited movement, for example opening and closing of flower petals and growth movements towards stimuli such as light and water.

Distinctive features of members of the kingdom Plantae

- (a) They have a cell wall made up of cellulose.
- (b) Higher plants have vascular tissues for conduction of water, mineral salts, and manufactured food.
- (c) They store food in the form of starch.

Divisions of kingdom Plantae

The kingdom Plantae is divided into four main divisions. These are: division Bryophyta; division Filicinophyta or Pteridophyta; division Coniferophyta; and division Angiospermophyta.

Division Bryophyta

Members of the division Bryophyta are called bryophytes, which are the most primitive plants. They are found in wet and shaded environments, such as on forest floors, rock surfaces, bare soil, cracks of paved surfaces, bricks, as well as on trunks, and branches of

trees. The sexual reproduction process in bryophytes depends on availability of water, that is why their distribution is restricted to shady and moist places. Examples of bryophytes are mosses and hornworts, as shown in Figure 5.17.



(a)



(b)

Figure 5.17: Examples of bryophytes (a) mosses (b) hornworts

Activity 5.7: Investigating the structure of bryophytes

Materials: Mosses, hand lens, notebook, and pen or pencil

Procedure

1. Collect moss plants from the school or home environment.
2. Use a hand lens to observe the structure of a moss plant.
3. Describe the observed external features of the moss plant.

Question

What is the significance of bryophytes in ecosystems?

General characteristics of bryophytes

- (a) Bryophytes live in moist, damp, and shaded areas.
- (b) They do not produce flowers, fruits or seeds.
- (c) They have root-like structures called rhizoids.
- (d) They reproduce sexually and asexually by spores.

Distinctive features of bryophytes

- (a) They are simple plants which lack true roots, stems and leaves (See Figure 5.18).
- (b) They have an erect plant body, which is leafy in nature.
- (c) They have no vascular tissue, meaning that they have no xylem and phloem.
- (d) They usually depend on water for sexual reproduction.

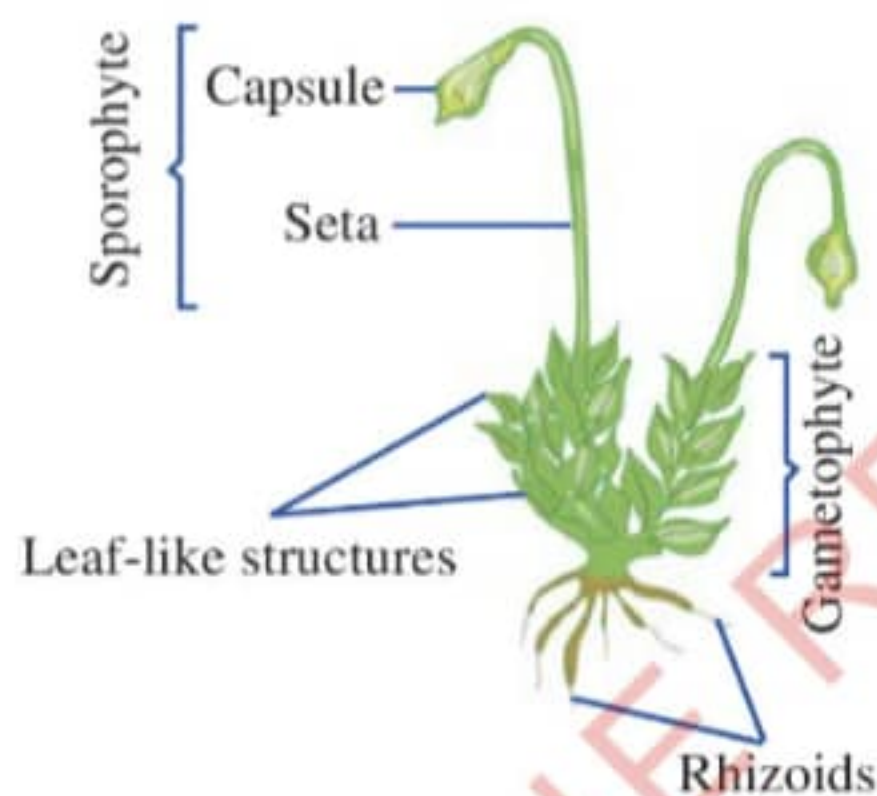


Figure 5.18: Structure of a moss plant

Advantages of mosses

- (a) They help to decompose dead logs and enhance nutrient circulation.
- (b) They are early colonisers of open woodland; hence they help in creating a favourable environment for the development of higher plants.
- (c) They provide food to heterotrophs, such as insects, fungi, and bacteria.
- (d) They play a great role in preventing soil erosion by holding the soil particles together.
- (e) They contribute in the production

of oxygen to the atmosphere, which is used by animals and other organisms.

Disadvantages of mosses

- (a) Moss plants are common weeds in gardens and can be difficult to eliminate.
- (b) Sometimes, they grow on the surface and walls of buildings, making them look old and unattractive.

Division Filicinophyta (Pteridophyta)

Activity 5.8: Investigating the structure of a fern plant

Materials: Fern plants, petri dish, hand lens, notebook, and pen or pencil

Procedure

1. Collect some fern plants from the school or home environment.

Caution

Care must be taken when collecting fern plants. Get only what is required instead of destroying the whole plant.

2. Use a hand lens to examine different types of ferns.
3. Describe their external features.
4. Examine the underside of one of the fronds. Describe the structure you have observed.

Question

What is the significance of filicinophytes in ecosystems?

Members of division Filicinophyta are called filicinophytes or pteridophytes. Examples of filicinophytes are ferns, club mosses, and horsetails, as shown in Figure 5.19. Filicinophytes are much more advanced than bryophytes because they have true roots, stems and leaves.

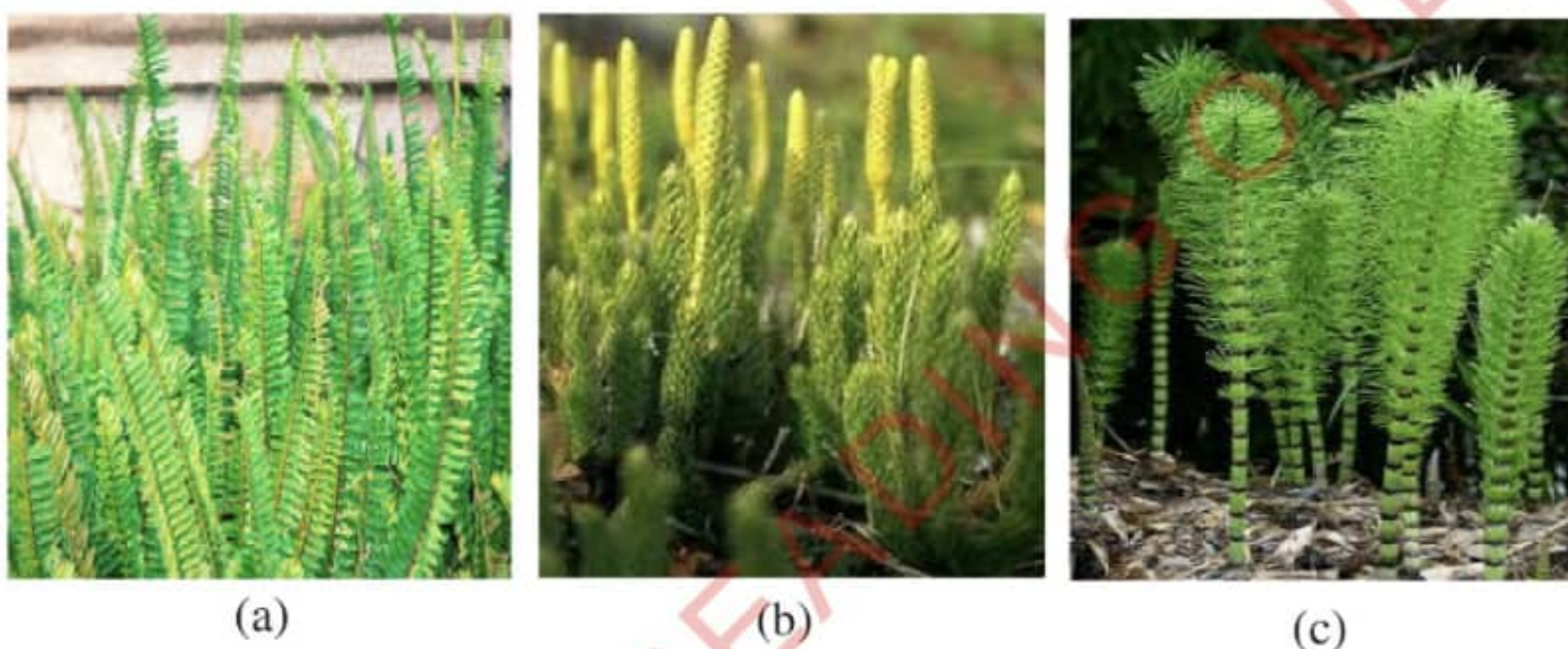


Figure 5.19: Examples of filicinophytes (a) ferns (b) club mosses (c) Horsetail

Leaves of filicinophytes are called fronds. Fronds are usually composed of a leafy blade and petiole (leaf stalk), as shown in Figure 5.20.

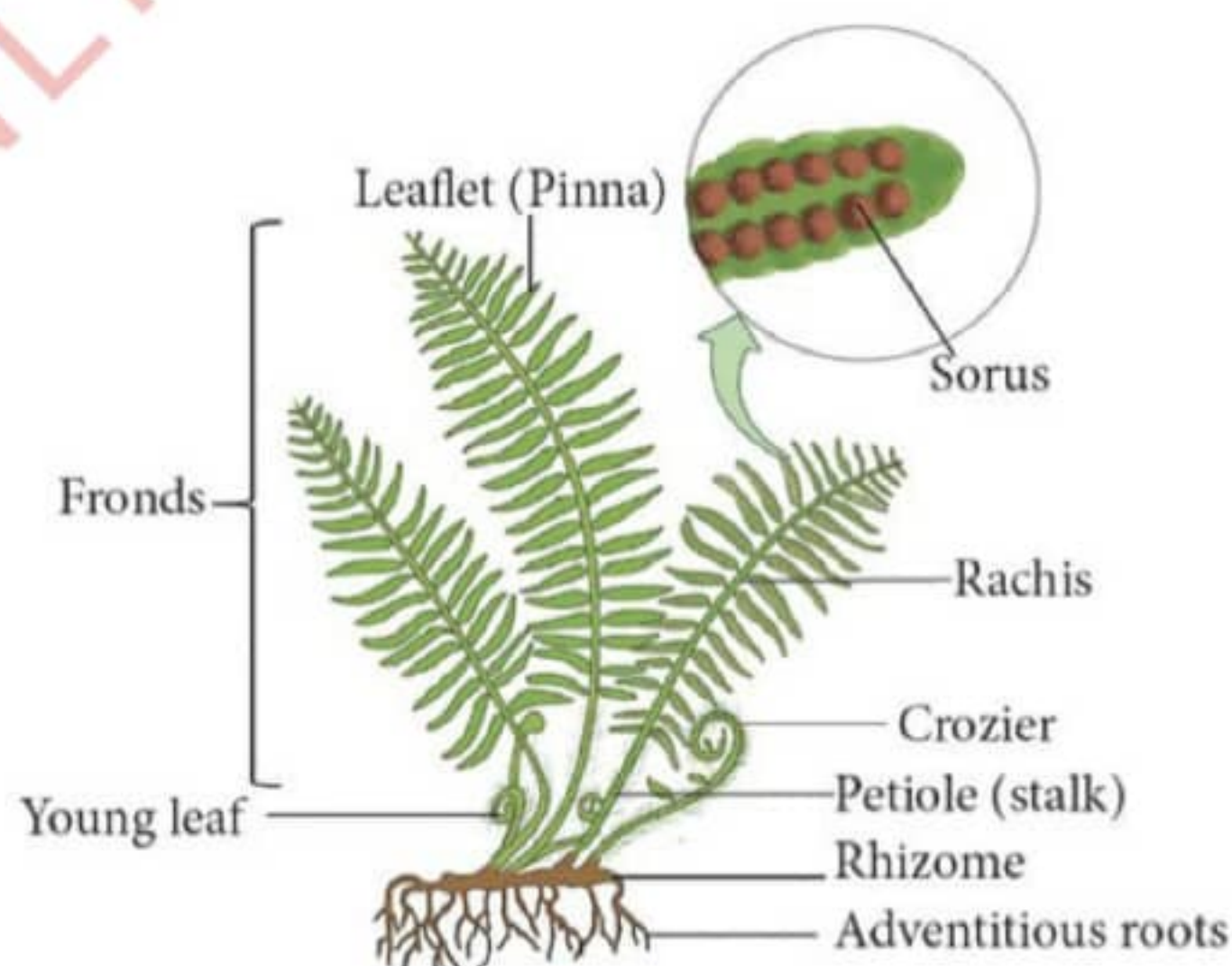


Figure 5.20: Fern plant

General characteristics of filicinophytes

- (a) They live in moist, damp and shady areas.
- (b) They are less dependent on water for survival and reproduction than bryophytes.
- (c) They have true roots, stems and leaves.
- (d) They have vascular tissues.
- (e) They reproduce by means of spores.
- (f) They do not produce flowers and seeds.

Distinctive features of ferns

- (a) Their leaves bear spore-producing structures called sori (singular is sorus). A sorus is a cluster of spore-producing structures called sporangia (singular is sporangium).
- (b) Their leaves are arranged in a clump and are called fronds.
- (c) They have a simple vascular tissue.

Advantages of ferns

- (a) They constitute ground-cover in moist areas.
- (b) They produce food for themselves and for heterotrophic organisms in ecosystem.
- (c) They are used for decoration in homes and offices.
- (d) They are the major components of coal, a fossil fuel which is made up

- of the remains of primitive plants.
- (e) Ferns, such as *Azolla* sp. are used as biological organic fertiliser. They are able to fix nitrogen from the air into compounds that can be absorbed by plants.
- (f) Some ferns, such as *Dryopteris* sp. have a medicinal value.

Disadvantage of ferns

Ferns are regarded as weeds in many places. For example, the giant fern plants are the worst aquatic weeds that threaten the life of other aquatic organisms. In terrestrial environment, some species of fern plants, such as *Pteridium* sp. are common weeds.

Division Coniferophyta

Activity 5.9: Investigating the distinctive features of conifers

Materials: A variety of conifers, such as a whole plant or branches of pine, cedar, cypress, fresh or preserved cones, notebook, and pen or pencil

Procedure

1. Collect the branches of pine, cypress, and cedar plants from the school or home environment.

Caution

Care must be taken when collecting a variety of conifer branches or cones. Get only what is required instead of destroying the whole plant.

2. Observe the structural features that are possessed by these plants which are different from those of other plants.
3. Note down the distinctive features of conifers.

Questions

1. What features have you observed?
2. What features distinguish conifers from other plants?

Coniferophyta is one of the divisions of the kingdom Plantae. Plants under the division Coniferophyta are sometimes called coniferous trees or simply conifers. These conifers bear cones and they are referred to as coniferophytes. The conifers are non-flowering plants whose seeds are not enclosed in ovaries, but are arranged spirally giving the shape of the cone. Examples of conifers include casuarinas, spruces, cedars, pines, and cypress plants. Figure 5.21 (a) and (b) shows the tree and branch of conifer.



(a)



(b)



(c)



(d)

Figure 5.21: Conifers (a) a tree (b) a branch (c) male cones (d) female cone

General characteristics of conifers

Conifers possess the following features:

- (a) They are non-flowering seed-bearing plants.
- (b) They produce naked seeds.
- (c) They have reproductive structures called cones. There are male and female cones. The male cones are smaller and usually occur in clusters. The female cones of most conifers have compound structures. See Figure 5.21 (c) and (d).
- (d) They form soft wood.
- (e) They have needle-like or scaled leaves with thick cuticle for protection and reduction of water loss.
- (f) Conifers are widely distributed but commonly found in areas with cold climate where they form evergreen forests.

Distinctive features of conifers

- (a) They bear cones.
- (b) The pollen grain develops air bladder-like structure which facilitate their movement. Hence, they are wind pollinated plants.
- (c) They have naked seeds, which are born in cones.
- (d) The mature seeds are typically winged. This is an adaptation for seed dispersal by wind.

Advantages of conifers

- (a) Conifers are the evergreen plants, which are grown in large forests and produce useful timber called softwood.
- (b) Conifer forests are home to many birds and other animals.
- (c) They are also used to produce wood pulp for paper making.
- (d) Turpentine which is used as cleaning agent for painting as well as production of disinfectants and insecticides is also a product of conifers.
- (e) Conifer trees are used as firewood; hence they are a source of heat.
- (f) They are a source of food to organisms, such as insects and squirrel.
- (g) Many species of **Pinus** are used as ornaments when planted in parks and gardens. Cypress also is commercially grown and harvested for decoration during festivals, such as Christmas.
- (h) Conifers remove carbon dioxide from the atmosphere through photosynthesis.

Disadvantages of conifers

- (a) Coniferous forest completely shades the ground and prevents the growth of other plants.
- (b) Wood products from conifers are soft and can be easily attacked by termites if not treated.

- (c) They have needle like thorny leaves that can prick and cause injury.
- (d) Resins which are produced by pines catch fire easily. Thus, in case of fire outbreak in a pine forest, fire is likely to spread rapidly because of the resins.

Division Angiospermophyta

Activity 5.10: Investigating the structure of flowering plants

Materials: Hibiscus flower, common bean plant, maize plant, elephant grass plant, hand lens, notebook, and pen or pencil

Procedure

1. Collect hibiscus plant, common bean plants, maize plants and elephant grass plants from the school or home environment. Make sure that you carefully uproot the entire plant in order to display their root system clearly, except for hibiscus.

Caution

- (a) Get only what is required instead of destroying the whole plant.
- (b) Do not collect dangerous plants, such as thorny plant, itching and poisonous plants.
- (c) Be careful of stinging animals, such as bees, wasps, and spider found on plants.

2. Using a hand lens, observe carefully

the leaves, flower arrangement and roots of the collected plants and state the classes to which each plant belongs.

Question

Summarize the morphological differences between the classes of the collected plants in tabular form, including the characteristics, such as leaf shape, flower arrangement, and root structure

Angiospermophyta is the division in kingdom Plantae which comprises plants commonly known as flowering plants or angiosperms. Angiosperms are the most diverse and a successful group of all the plants. They are found in most habitats, such as aquatic and terrestrial habitat including deserts. They range in size from small plants, such as herbs and grasses to big trees like baobab.

General characteristics of angiosperms

The following are the general features of angiosperms.

- (a) Angiosperms are found in terrestrial, freshwater and marine habitats.
- (b) Angiosperms have a well-developed root system, stem and leaves.
- (c) Their seeds are enclosed and protected in the ovary.
- (d) They produce seed bearing fruits.
- (e) They have well developed conducting tissues which contain xylem and phloem.

Classes of the division Angiospermophyta

The division Angiospermophyta is divided into two classes. These classes are Monocotyledoneae and Dicotyledoneae. The word “Mono” means one and “Di” means two. Monocots have been named from having one cotyledon and dicots from having two cotyledons. Cotyledon is the part of seed that develops into the leaves after germination.

Class Monocotyledoneae

This class contains plants with one cotyledon. Examples of plants in this class include maize, sugarcane, wheat, millet, sisal, coconuts and banana plants.

Distinctive features of the class Monocotyledoneae

- (a) Flower parts normally occur in three or multiples of three. See Figure 5.22 (a).

- (b) They have parallel leaf venation, as shown in Figure 5.22 (b).
- (c) The embryo of monocot seeds bears one cotyledon.
- (d) Monocot leaves are composed of an open or closed sheath which encloses the stem.
- (e) Their vascular bundles in stems are scattered, as shown in Figure 5.23 (a).
- (f) They have fibrous root system (Figure 5.23 (b)).
- (g) They have long and thin leaves.

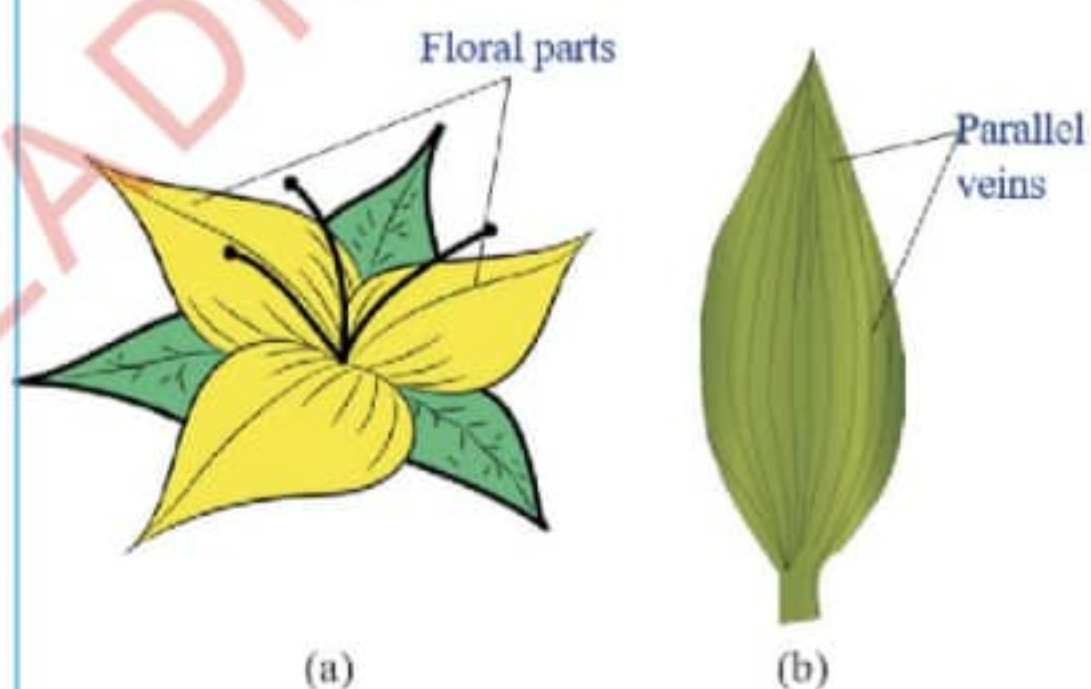


Figure 5.22: Monocot plant (a) floral arrangement (b) parallel leaf venation

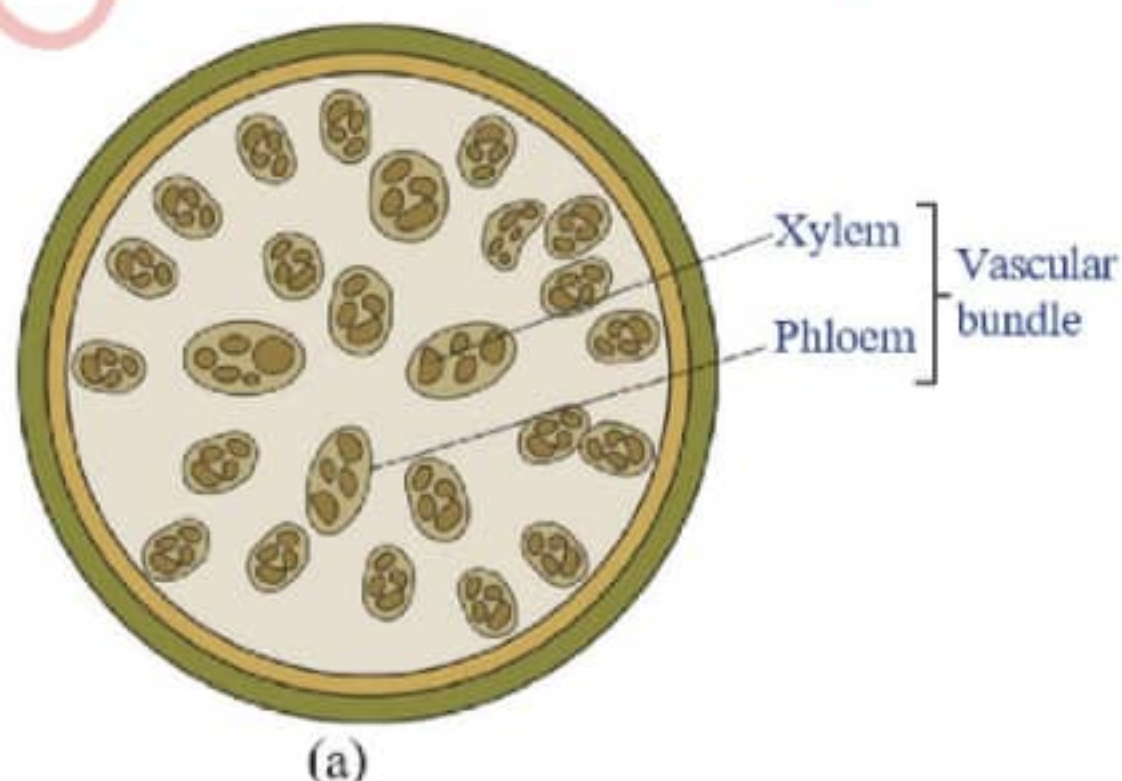


Figure 5.23: Monocot stem and root (a) vascular bundle scattered in stem (b) fibrous root system

Class Dicotyledoneae

This class contains plants with two cotyledons. Examples are beans, mangoes, coffee, groundnuts and sunflower plants.

Distinctive features of the class Dicotyledoneae

- (a) Floral parts are normally in four or five, or multiples of four or five. See Figure 5.24 (a).
- (b) Dicot leaves have net-like venation. See Figure 5.24 (b).
- (c) Their stems have vascular bundles which appear in a ring form, as shown in Figure 5.25 (a).
- (d) They have tap root system, as shown in Figure 5.25 (b).
- (e) Their seed embryo has two cotyledons.
- (f) Dicot plants have petioles that support the leaf.
- (g) They have short and broad leaves.

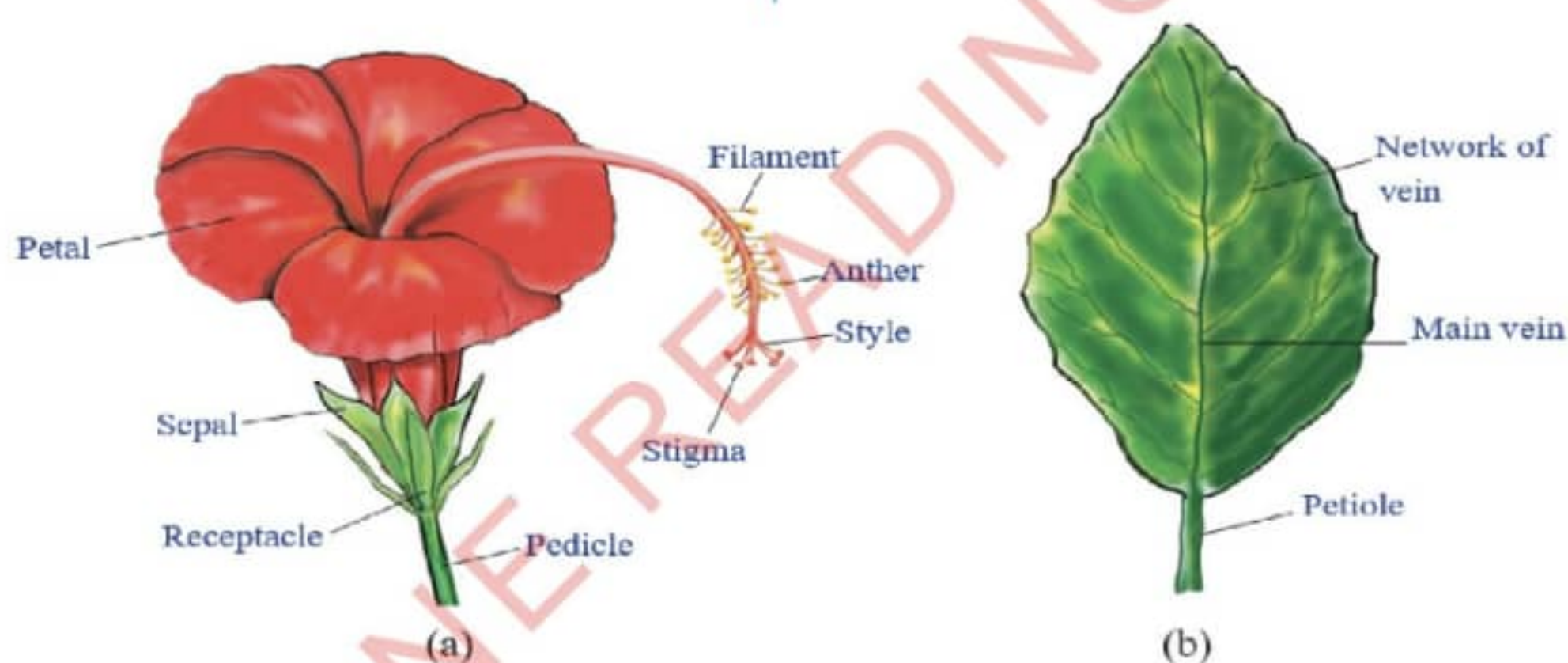


Figure 5.24: Dicot plant (a) floral arrangement (b) network leaf venation

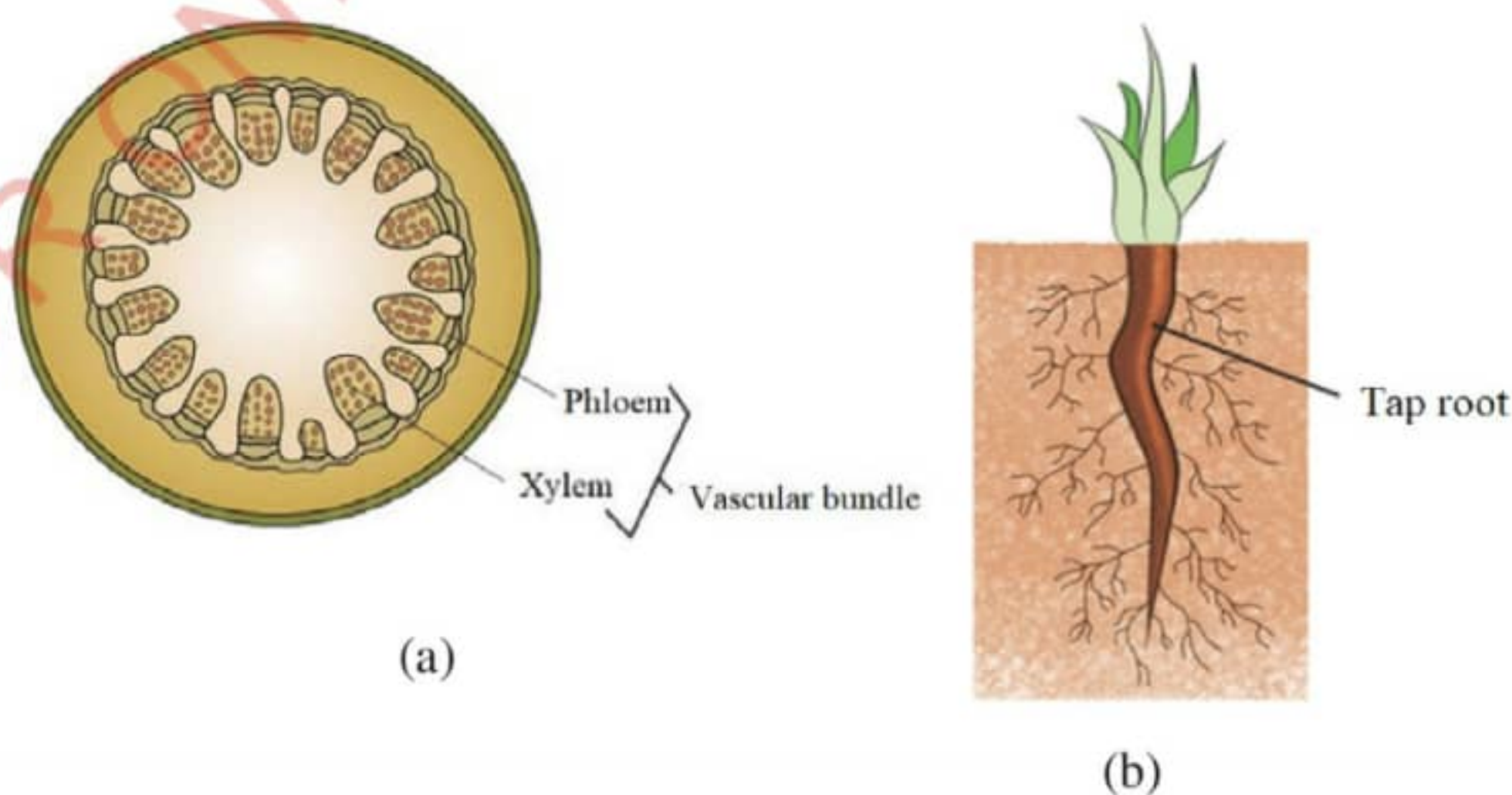


Figure 5.25: (a) vascular bundle in ring form in stem
(b) tap root system

Advantages of angiosperms

- (a) Angiosperms are important for the survival of humans and other animals. For humans, angiosperms provide foods, such as beans, rice, maize, sugarcane, and oranges. Some members of the grass family are used as food for livestock.
- (b) They are habitat to many insects, birds and other animals.
- (c) Different species of Angiospermatophyta are used as medicine for treating various diseases.
- (d) Some angiosperms, such as cotton, flax, and hemp provide fibres that are raw materials in textile industries.
- (e) Angiosperms form ground cover which prevents soil erosion.
- (f) Some plants such as roses, lilies, and hibiscus make the environment attractive.
- (g) They provide varieties of wood for furniture, paper, and building materials.

Disadvantages of angiosperms

- (a) Some angiosperms such as thorn apple, oleander, and milkweed are poisonous. When eaten by humans and other animals as they may cause death.
- (b) Drugs derived from some angiosperms can be dangerous. For example, tobacco and caffeine can be addictive.

- (c) Some angiosperms, such as star grass are weed plants, which compete with food crops leading to reduction of crop yield. Others are parasites to other plant species. For example, parasitic weeds in crops, such as mango, orange and cashew nut trees, and they cause serious loss of crop yield.
- (d) Some aquatic angiosperms, such as water hyacinth can colonise water bodies and affect the ecosystem as well as hindering fishing activities.

Kingdom Animalia

Task 5.9

Read from reliable internet sources and biology books the information about kingdom Animalia.

Kingdom Animalia comprises wide varieties of animals. Animals vary greatly in their structure, morphology and the way their bodies function. Kingdom Animalia includes worms, insects, and higher animals. They are found in various habitats, such as on land, in oceans, and in freshwater.

General characteristics of animals

- (a) They are eukaryotes; therefore, they possess a clearly defined nucleus and their cells have membrane bound organelles.
- (b) They are multicellular, that means they are made up of more than one cell.

- (c) They depend on other organisms as a source of their food.
- (d) Most animals are mobile and hence they can move from one place to another searching for shelter, food, mates, and safety.
- (e) Most of the animals have bilateral symmetrical bodies. This means that their bodies can be divided longitudinally into two equal parts.
- (f) Most of the animals have high level of tissue differentiation and specialised body organs.
- (g) Most of the animals have a well-developed nervous system.
- (h) Most of the animals have anterior and posterior ends, with oral and anal openings.
- (i) The majority of the animals digest food internally in the gut and store carbohydrates in the form of glycogen.

Distinctive features of animals

- (a) They depend on other organisms as source of food.
- (b) Most animals digest food internally in the gut.
- (c) Most animals are capable of locomotion; hence they can move from one place to another. Few animals, such as oysters are sessile. They are found attached on rocks and other hard surfaces. They are not capable of self-locomotion.
- (d) Most animals have a well-

developed nervous system that helps them in detecting changes in their surroundings. This allows them to be aware of the changes occurring in their environment.

Activity 5.11: Differentiating animals from other organisms

Materials: A variety of plants and animals, charts showing a variety of plants and animals, notebook, and pen or pencil

Procedure

- 1 Walk around the school compound and observe a variety of plants, insects, and other animals or observe charts that show a variety of plants and animals.
- 2 Observe the organisms and note as many observable features as possible.
- 3 Group the observed organisms according to their similarities and differences.
- 4 Outline the similarities and differences among the grouped organisms.

Question

Give reasons why animals and plants are placed in different kingdoms.

Phyla of the kingdom Animalia

Kingdom Animalia comprises several phyla. However, only five phyla will be

covered in this section. These are:

- (a) Phylum Platyhelminthes includes pork tapeworm (*Taenia solium*), beef tapeworm (*Taenia saginata*), liver flukes (*Fasciola* species), blood flukes (*Schistosoma* species) and planaria.
- (b) Phylum Nematoda, formerly known as Aschelminthes, includes ascaris, hookworms and filarial worms.
- (c) Phylum Annelida includes earthworms, leeches and their marine relatives, such as ragworms and lugworms.
- (d) Phylum Arthropoda includes prawns, crabs, ticks, spiders, butterflies and cockroaches.

- (e) Phylum Chordata includes advanced organisms, such as human beings, cattle, fish, chicken, frogs and snakes.

Phylum Platyhelminthes

Task 5.10

From reliable internet sources, library and biology books read the examples of flatworms and their characteristics.

Members of the phylum Platyhelminthes are the flatworms and are commonly called platyhelminths. They can be free living or parasites. Figure 5.26 shows examples of platyhelminths.

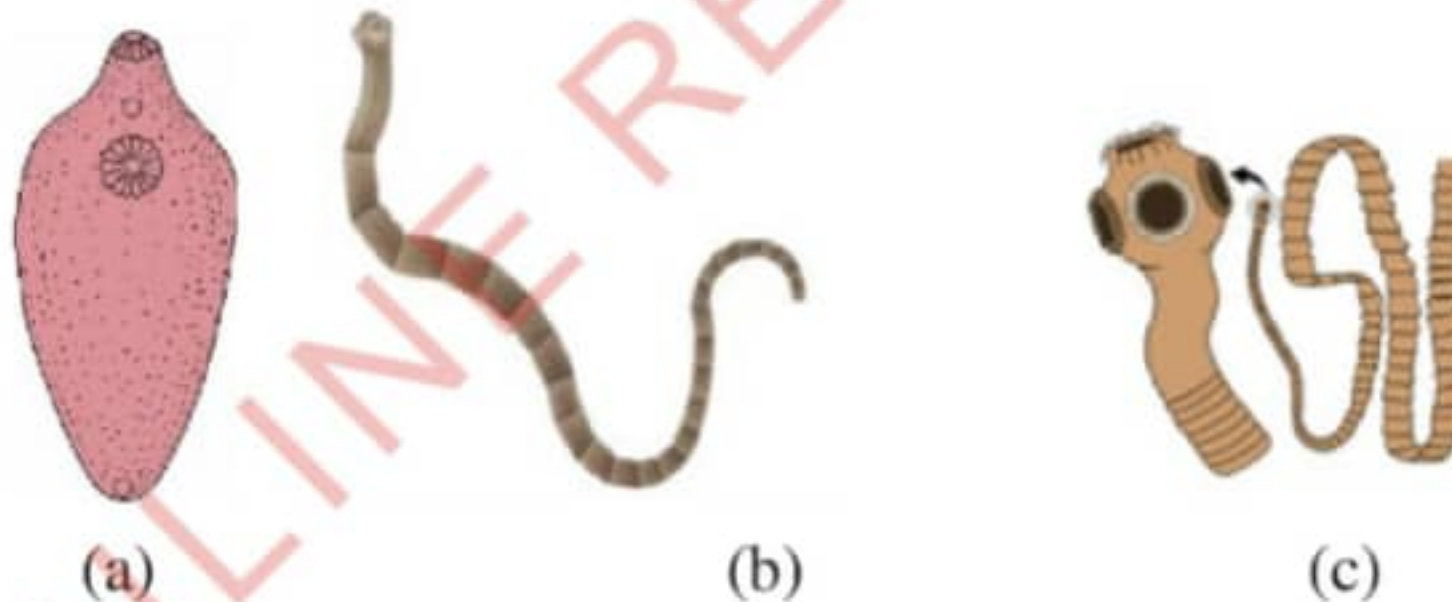


Figure 5.26: Examples of platyhelminths (a) liver fluke (b) beef tapeworm (c) pork tapeworm

Activity 5.12: Observing the external features of the flatworms

Materials: Preserved specimens of tapeworm, liver fluke, petri dishes, forceps, hand lens, notebook and pencil

Procedure

1. Using forceps pick a liver fluke, and tapeworm from preserving containers and put each in different petri dishes.
2. Using a hand lens, carefully observe and identify the following external structures (if any): the mouth, suckers, body shape and the posterior region of the organism.
3. Describe each specimen.

Questions

1. How does the body structure of a tapeworm differ from that of a liver fluke?
2. What is the function of suckers in platyhelminths?

General characteristics of platyhelminths

- (a) They have dorso-ventrally flattened bodies. This is the reason why they are called flatworms.
- (b) Most flatworms have only one opening that is the mouth. However, tapeworms lack both mouth and anus.
- (c) They have no body cavity. A body cavity is a space in an organism which holds and protects internal organs.
- (d) Some members have hooks and suckers for attachment to the host. Such members include tapeworms and the liver fluke. See Figure 5.27.
- (e) Some flatworms, such as planaria have hair-like structure called cilia on their outer body surface for locomotion.
- (f) Few members, such as planaria are free living and inhabit freshwater, marine water and land. Most of its members, such as liver flukes and tapeworms are parasitic.
- (g) They have specialised cells for excretion and osmoregulation called flame cells.
- (h) They lack respiratory and circulatory systems. Transportation of oxygen for respiration and other materials

takes place by simple diffusion.

- (i) Their bodies are bilaterally symmetrical. This means their bodies can be divided longitudinally into two equal parts.
- (j) They are hermaphrodites, meaning that they have both male and female reproductive organs. Hence, their reproduction can either be through self-fertilisation or cross-fertilisation.
- (k) They are heterotrophic, meaning that they get food from other organisms. Flatworms may be parasites, predators or scavengers.

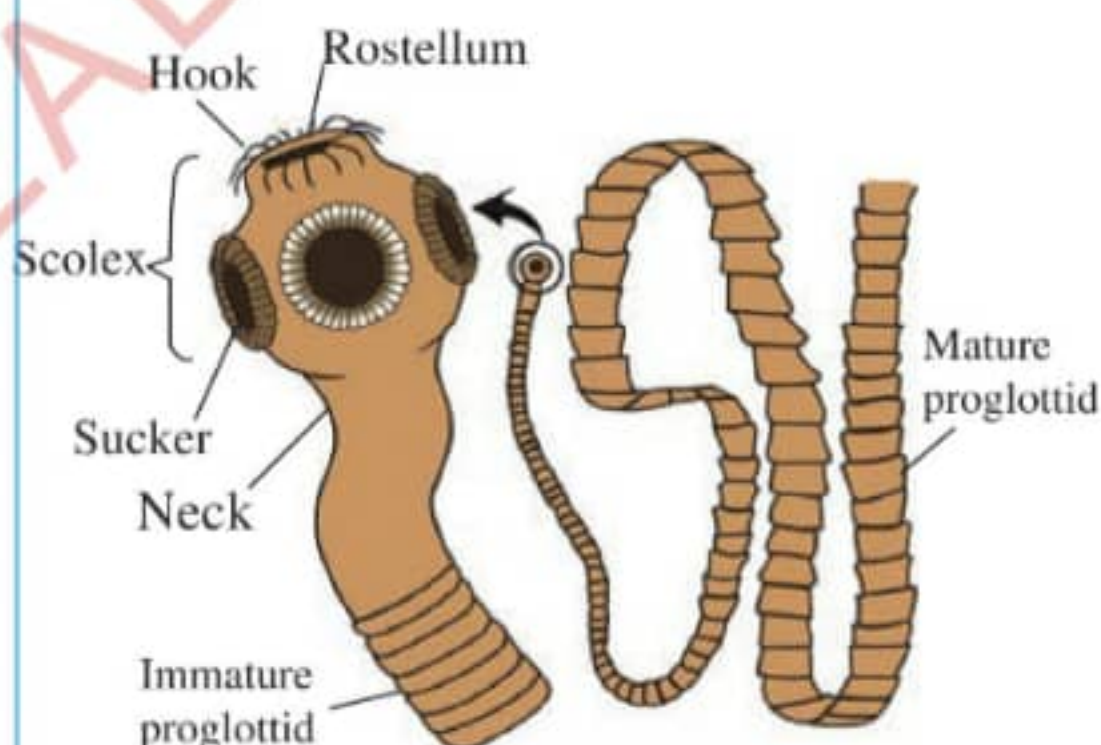


Figure 5.27: External structure of tapeworm

Distinctive features of platyhelminths

- (a) They are dorso-ventrally flattened with ribbon or leaf-like body.
- (b) If present, their digestive system is incomplete. The digestive tract has only a single opening which is the mouth.
- (c) They have specialised cells for excretion and osmoregulation called flame cells.

- (d) They have suckers, hooks, or both for attachment to the host.

Advantages of platyhelminths

- (a) Some flatworms, such as planaria are scavengers, feeding on organic particles from dead bodies of other organisms.
- (b) Some are used as biological specimens in the field of education and medicine. Examples of such flatworms include tapeworms and liver flukes.

Disadvantages of platyhelminths

Task 5.11

Search from reliable internet sources the videos or images showing beef or pork infected with tapeworms. Note down what you have observed.

- (a) Many species are parasites that cause diseases to humans, livestock and wild animals. For example:

- (i) Tapeworm causes taeniasis, which is an intestinal tapeworm infection transmitted through consuming raw or undercooked beef or pork infected with tapeworms. The infection caused by larval stages of tapeworms can lead to heart attack or epilepsy. If the

infection is located in the heart muscles it can cause heart attack and if it is located in the brain, it can cause epilepsy.

- (ii) Liver fluke causes fascioliasis. This disease causes liver rot in sheep, cattle and humans.
- (iii) Some species of blood flukes, such as *Schistosoma* spp. cause bilharzia (schistosomiasis). If untreated it may cause urinary bladder cancer or intestinal cancer.

- (b) Parasitic flatworms can cause reduction in quality and amount of meat and milk produced by livestock due to infections. They can also cause stunted growth and even death in young animals.
- (c) They cause general body weakness and frequent hunger in the hosts as they feed on digested nutrients from the hosts. Working animals, such as oxen cannot perform work to their full capacity if infected with a large number of parasitic flatworms.
- (d) Parasitic flatworms can also cause loss of income due to costs incurred for treatment of diseases.

Exercise 5.6

1. Describe the distinctive features of members of phylum Platyhelminthes.
2. Imagine you are conducting a laboratory investigation. In one of your experiments, you come across a specimen of a tapeworm. What features will enable you to tell whether the specimen is a tapeworm?
3. Why is it not advisable to eat raw or undercooked beef?

Phylum Nematoda**Task 5.12**

Search from reliable internet sources the videos or images that show round worms.

Phylum Nematoda comprises round worms which are among the most numerous multicellular animals on the earth. They inhabit a wide range of environments, including soil, marine and freshwater. Examples of nematodes include the animal parasites, such as ascaris and plant parasites, such as eelworms. See Figure 5.28.



Figure 5.28: Examples of nematodes (a) ascaris (b) eelworm

General characteristics of nematodes

- (a) They have round, cylindrical, elongated and unsegmented bodies with pointed ends.
- (b) They depend on other organisms as source of food.
- (c) They have no blood circulatory system; therefore, transport is by simple diffusion through the thin moist cuticle or through the fluid-filled body cavity.
- (d) Many species of nematodes are parasites of both animals and plants, but there are also many free-living species found in marine and freshwater sediments and in the soil.
- (e) They are dioecious with separate sexes. Males are usually smaller than females.

Distinctive features of nematodes

The following features differentiate nematodes from other animals.

- They have a cylindrical, elongated and unsegmented body with pointed ends.
- They have special sensory structures known as amphids and phasmids.

Activity 5.13: Observing external features of roundworms

Materials: Preserved specimen of ascaris (female and male), hand lens, forceps, notebook, pen and two petri dishes

Procedure

- Use forceps to pick up male and female specimens of ascaris from a preserving container and put them into separate petri dishes.
- Carefully observe the male and female ascaris specimens with a hand lens.
- Describe their external features. Compare the observed features with those depicted in Figure 5.29.

Question

What features distinguish roundworms from flatworms?

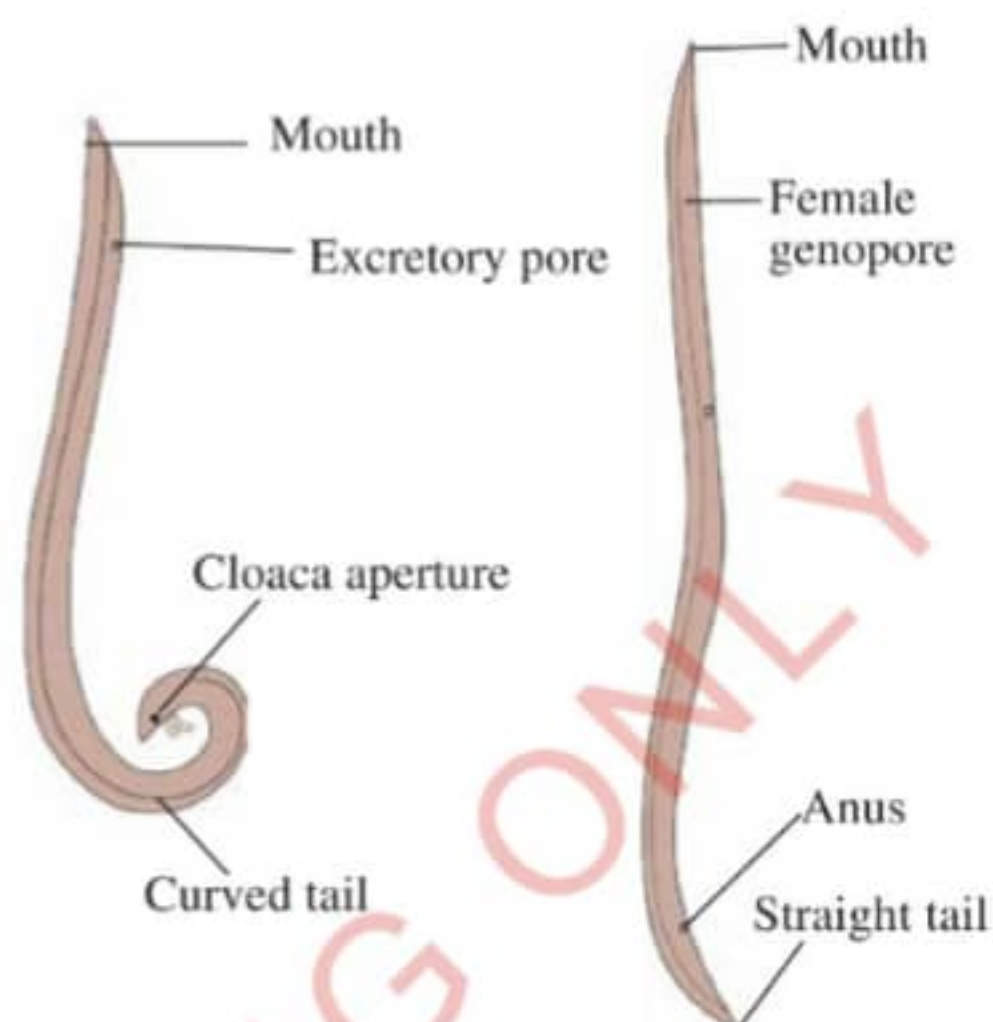


Figure 5.29: External structure of ascaris (a) male (b) female

Advantages of nematodes

- Nematodes have been used in various biological studies. For example, some discoveries in the field of genetics involved the use of nematodes.
- Some nematodes are used as biological control agents to control pests, such as locusts and vectors, such as mosquitoes. Parasitic roundworms attack and eat varieties of caterpillars.
- Soil nematodes are important in recycling of nutrients, hence contributing to soil fertility.

Disadvantages of nematodes

Task 5.13

Search from reliable internet sources the videos or images that show elephantiasis. Note down what you have observed.

- (a) They cause diseases, such as elephantiasis or bancroftian filariasis which affects the human lymphatic system. This results in accumulation of the lymph in legs, hands, mammary glands or scrotal sacs.
- (b) They cause ascariasis in humans and pigs. This may lead to obstruction of the gut and may also cause anaemia.
- (c) Some species of roundworms cause root knot galls in plants,

hence drains photosynthetic products.

Phylum Annelida

The word Annelid is derived from a latin word “annelus” which means little ring. Thus, members of this phylum are known as ringed or segmented worms. Annelida is a large phylum comprising ragworms, peacock worms, leeches, and earthworms. Examples of annelids are shown in Figure 5.30.



Figure 5.30: Examples of annelids (a) marine ragworm (b) peacock worm (c) blood-sucking leech (d) earthworm

General characteristics of annelids

- (a) They have elongated, cylindrical bodies that are divided into distinct segments.
- (b) They have bilateral symmetrical bodies.
- (c) They reproduce both asexually and sexually.
- (d) They have a moist outer covering called cuticle.
- (e) They have closed circulatory system whereby the blood flows within the blood vessels.
- (f) They have a lip-like extension on the

first segment above the mouth called prostomium.

- (g) They are commonly found in soil, freshwater, and marine environment while some species are parasitic.

Distinctive features of annelids

- (a) Most annelids have hair-like structures called chaetae made up of chitin. These are used for locomotion.
- (b) They have a segmented body.

Activity 5.14: Observing external features of the earthworm

Materials: Preserved or live specimen of earthworm, hand lens, petri dish, notebook, pencil, and forceps

Procedure

1. Using forceps, pick an earthworm from a preserving container and put it in a petri dish.
2. Using a hand lens, observe the specimen carefully to identify its external features.

3. Identify its external features, including its mouth, anus, clitellum, and chaeta. Compare the observed features with those depicted in Figure 5.31.

Questions

- (a) How are the earthworms adapted to their environment?
- (b) Explain two features that distinguish earthworms from round worms.

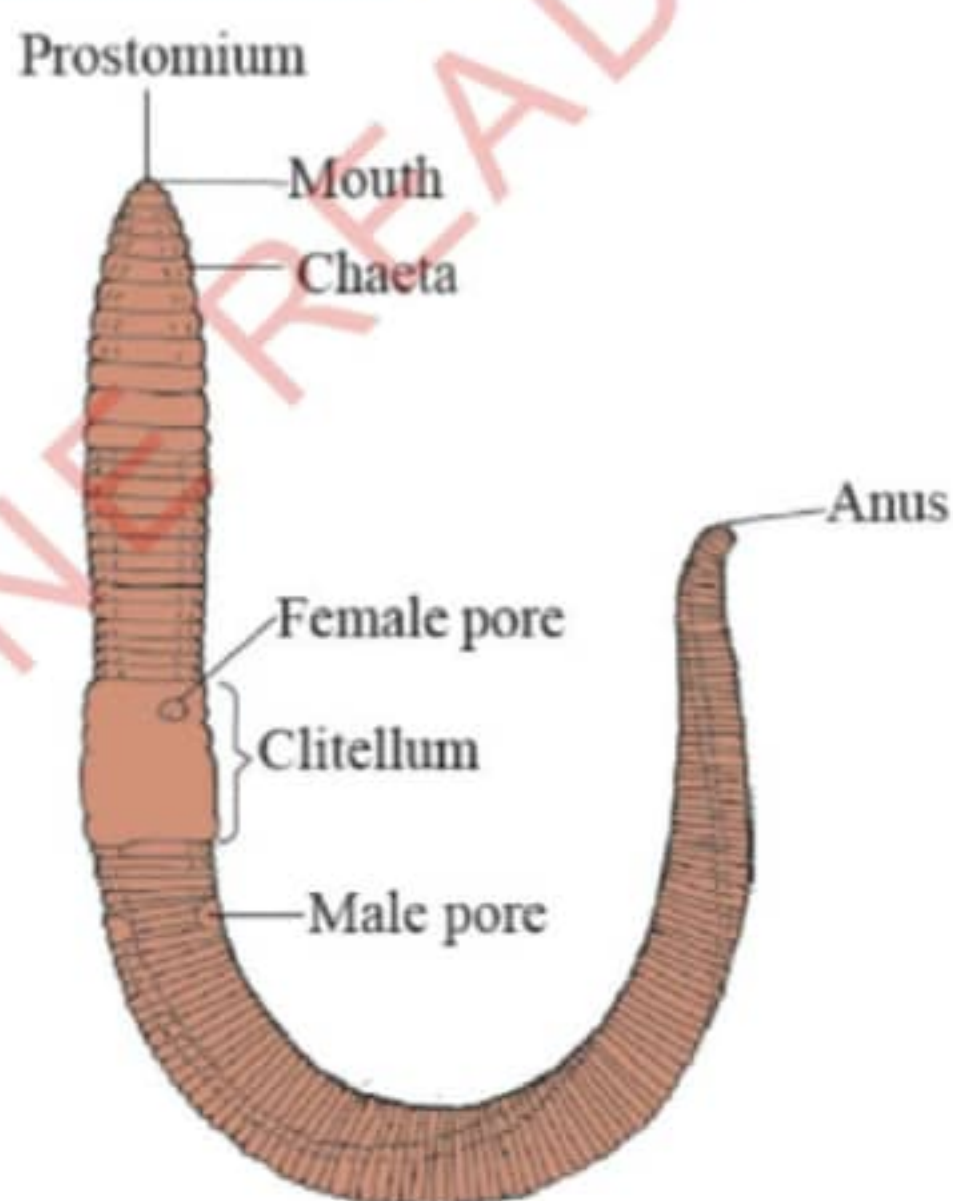


Figure 5.31: External structure of the earthworm

Advantages of annelids

- (a) Some annelids improve soil aeration through burrowing and mixing up of soil layers. Earthworms play important role in improving soil aeration, hence maintaining soil fertility.
- (b) They are used for commercial production of compost manure in the process known as vermicomposting.

- (c) Some annelids are used as bait in the fishing industry. Examples of annelids used as bait include earthworms and lugworms.
- (d) They are used as source of nutritious food for some fish and birds. They are rich in proteins and vitamins.
- (e) Some annelids are ecological decomposers; hence, they contribute to nutrient circulation in the ecosystems.
- (f) They remove pollutants from the soil and clean the environment by transforming organic wastes during feeding.
- (g) Faeces and urine of some annelids have considerable value as manure and they increase soil fertility.
- (h) They are used as specimens in biological studies.
- (i) The excretory waste of some annelids which are in the form of cast increases sticking together of soil particles. This in turn increases water holding capacity of the soil.

Disadvantages of annelids

- (a) Some annelids damage young roots of the growing plants.
- (b) Leeches are harmful to mammals, reptiles and fishes, because they suck blood from them.

- (c) Earthworms can increase soil porosity and aeration which may cause water loss by seepage from the farms.

Exercise 5.7

1. What morphological features differentiate earthworms from tapeworms?
2. What is the function of chaeta found in most annelids?
3. Imagine you are walking around school farm and you come across someone who is killing earthworms. Educate that person the importance of earthworms.

Phylum Arthropoda

The term Arthropoda originates from two Greek words, “*arthron*” meaning joint and “*podos*” meaning foot. Literally, arthropods are organisms with jointed appendages. Arthropods are the most successful animals on the earth. They are found almost in every habitat, including marine, freshwater, and terrestrial habitats. They vary widely in their habitats and feeding habits. Examples of arthropods include prawn, scorpion, tick, beetle, butterfly, crab, spider, centipede, millipede and grasshopper, as shown in Figure 5.32.

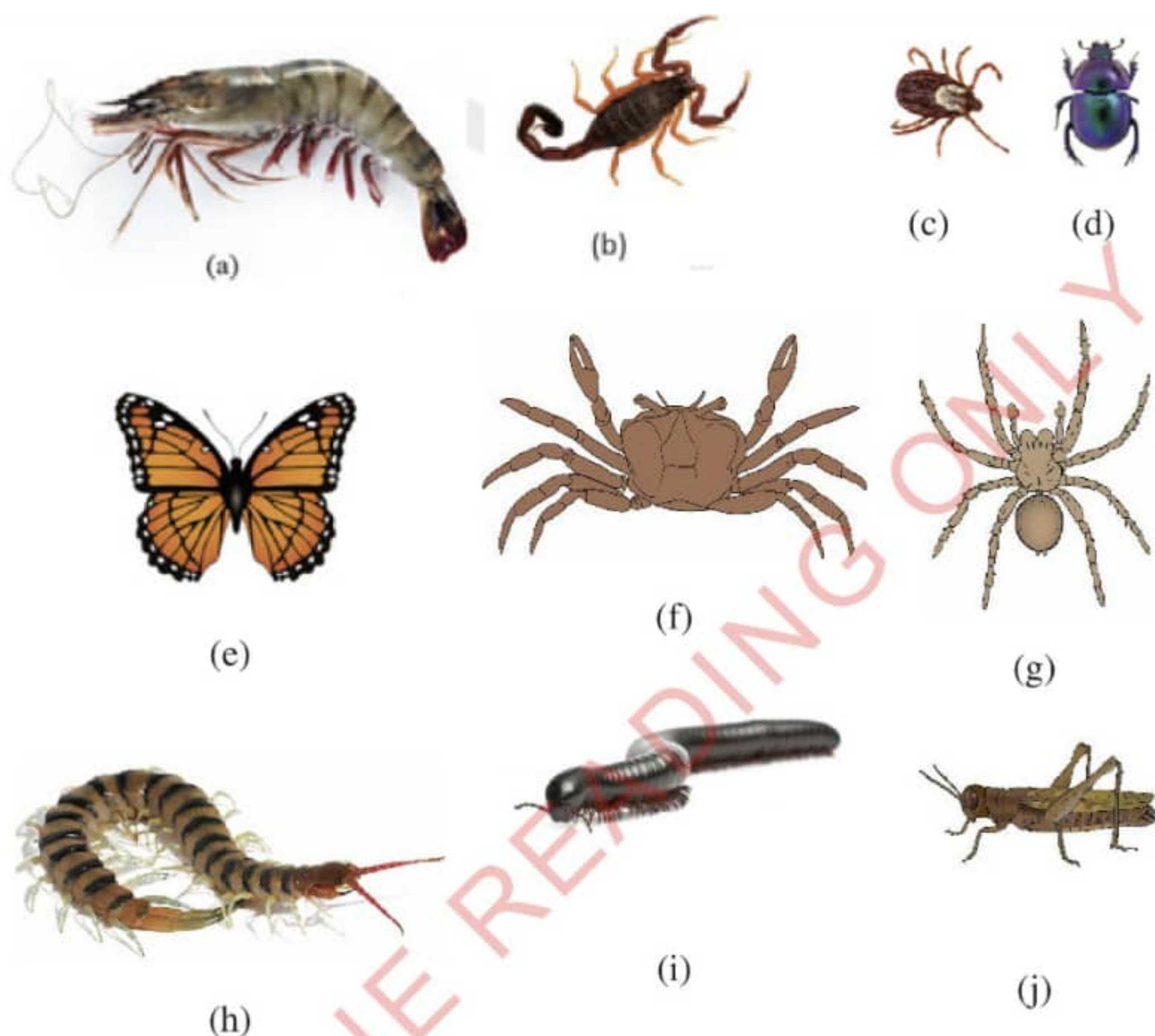


Figure 5.32: Examples of Arthropods (a) prawn (b) scorpion (c) tick (d) beetle (e) butterfly (f) crab (g) spider (h) centipede (i) millipede (j) grasshopper

General characteristics of arthropods

- (a) They have exoskeleton made up mostly of chitin. The exoskeleton serves different purposes, such as muscles attachment, support and protection against physical damage.
- (b) Some arthropods grow by periodically shedding off their exoskeleton in a process called moulting or ecdysis.

- (c) They have segmented bodies. In most arthropods, the body is organised into three major regions. These regions are the head, thorax and abdomen. In some arthropods, such as arachnids and crustaceans, the thorax and head are fused together to form the cephalothorax. Therefore, these arthropods have two major body parts, namely cephalothorax and abdomen.

- (d) They have jointed and paired appendages that may be modified in various ways to form structures, such as antennae, mouthparts, legs and reproductive organs. The modified appendages are used for various functions, such as feeding, gaseous exchange, locomotion and sensation.
- (e) Their body plan is bilaterally symmetrical. This means, their bodies can be divided longitudinally into two equal parts through a central axis.
- (f) They have compound or simple eyes.
- (g) They have open circulatory system where blood is not confined to the blood vessels and heart only. Blood flows into the body cavity to reach various tissues and organs.

Distinctive features of arthropods

- (a) They have jointed appendages which serve different purposes, such as feeding, locomotion and sensation.
- (b) They have segmented bodies organised into regions called tagmata.
- (c) They have an exoskeleton made up of mostly of chitin.

Classes of phylum Arthropoda

Task 5.14

Search from the library and reliable internet sources the information on classes of phylum Arthropoda, then make short notes.

Based on the presence of antennae, position and numbers of appendages, habitats, life histories and habits, arthropods are classified into five classes. These are class Crustacea which includes crabs and lobsters; class Arachnida which includes spiders and scorpions; class Chilopoda which includes centipedes; class Diplopoda which includes millipedes; and class Insecta which includes cockroaches, termites, and grasshoppers.

Activity 5.15: Observing the external structure of crab, spider, grasshopper, and millipede

Materials: Preserved specimens or pictures of crab, spider, grasshopper, and millipede, notebook, pen and hand lens

Procedure

1. Using forceps, pick specimens from a preserving container and put them in a petri dish.
2. Using a hand lens, observe the specimens carefully to identify their external features.

Questions

- (a) List down two observable features which show that these specimens are arthropods.
- (b) What two external features that distinguish a spider from a grasshopper?

Advantage of arthropods

- (a) They are used as food for human beings and other animals. They are rich in protein and calcium minerals. Example of edible arthropods are lobsters, prawns, shrimps, edible grasshopper, and larvae of beetles and butterflies.
- (b) They are used for production of various substances. For example, honey bees produce honey, wax and bee venom; silk worms produce silk that can be used for commercial purpose; and silk from spiders is used to produce surgical threads.
- (c) They are used in biological control of pests. Some arthropods have been used effectively as predators to kill or control pests. For example, lady beetles are used to kill aphids. Aphids are the insect pests that transmit viruses to plants.
- (d) Insects, such as bees and butterflies are useful in agriculture as they act as pollinators. Many plants depend on insects for pollination.
- (e) They are used in scientific

- researches particularly in the fields of physiology, genetics and evolution. Fruit flies (*Drosophila melanogaster*) are among the insects used for this purpose.
- (f) Some insects are used in the production of medicine. Such insects include blister beetles that produce cantharidin which is used for treatment of urinogenital diseases.
- (g) Some arthropods are scavengers and feed on rotten or decayed materials, hence helping in reducing wastes and ensuring nutrient circulation in the ecosystem. Examples of such arthropods are millipede and cockroaches.
- (h) Some arthropods improve soil aeration and soil fertility. For example, termites break down plant tissues and diplopods feed on decaying plants and animals, thus improving soil fertility.
- (i) Some crustaceans are used for decoration purpose especially crabs and crayfish.
- (j) Some arachnids, such as mites play an important role in the formation of humus from decomposed leaf litter and wood.

Disadvantages of arthropods

- (a) Chilopods, such as centipedes have venomous claws. They can bite humans and other animals

causing pain, small wounds and blisters.

- (b) Some arthropods destroy crops and other properties. For example, burrowing diplopods can cause some damage to the roots or germinating seeds, mites attack food crops (on farm and in storage) and timber.
- (c) Some arthropods, such as scorpion, wasps and bees can cause painful bites and stings. Stinging insects that live in colonies can cause significant injury and even death.
- (d) Some insects are vectors of animal and plant diseases. For example, mosquitoes carry *Plasmodium* spp. which cause malaria and tsetse flies carry *Trypanosoma* spp.

which cause sleeping sickness to human beings. In addition, white flies carry tomato yellow leaf curl virus which causes chlorosis in plants.

- (e) Some arthropods are parasites to mammals and other vertebrates. For examples ticks, jiggers, and bedbugs.

Phylum Chordata

This phylum comprises animals with a high degree of body organisation. The majority of animals under this phylum have a supporting column of bones known as backbone or vertebral column.

Examples of chordates are shown in Figure 5.33.

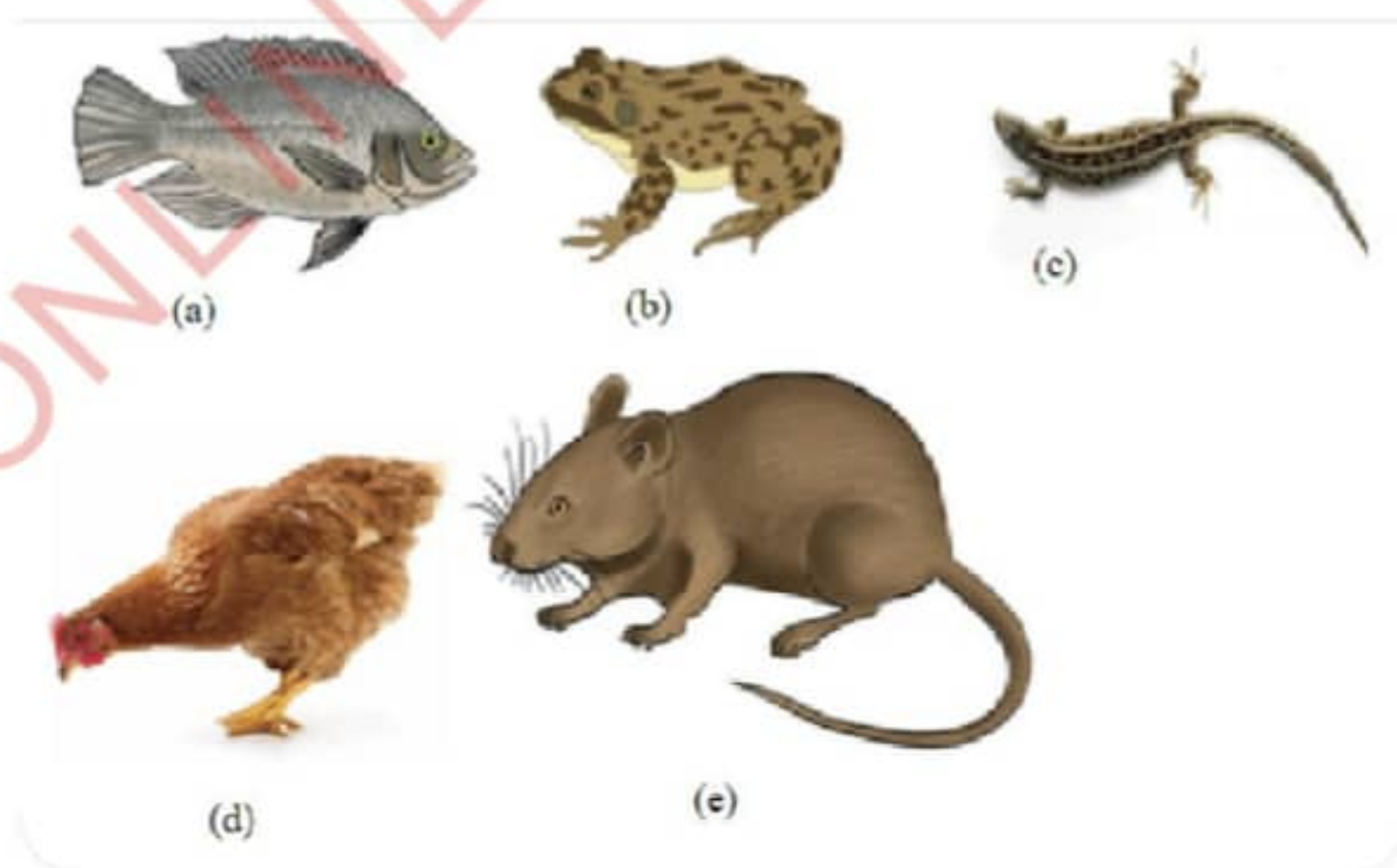


Figure 5.33: Examples of chordates (a) fish (b) frog (c) lizard (d) chicken (e) mouse

Activity 5.16: Observing the external features of fish, chicken, frog, lizard and rat or mouse

Materials: Preserved or fresh specimens of fish, frog, chicken, catfish, lizard, rat, charts showing pictures or drawing of fish, frog, chicken, lizard, rat, hand lens, scalpels, notebook, pen and a tray

Procedure

1. Place each specimen on a tray.
2. Observe each specimen carefully.
3. Describe the external features you observed, including the arrangement of scales, presence of gills, type of fins, and the overall shape of each specimen.

Question

What features distinguish a fish from a frog?

General characteristics of chordates

- (a) They occupy both terrestrial and aquatic habitats.
- (b) Most chordates have a complete digestive system with two openings, the mouth and anus.
- (c) Most chordates have a ventral heart.
- (d) Most chordates have a closed circulatory system.
- (e) Most have endoskeletons made up of bones and cartilages.
- (f) Some chordates have central

nervous system made up of the well-developed brain and a spinal cord.

Distinctive features of chordates

Chordates have unique features that differentiate them from other animals. Chordates have the following features at some stages of embryological development.

- (a) A supporting skeletal structure called notochord. In many vertebrates, the notochord is replaced by the vertebral column during development.
- (b) A dorsal hollow tubular nervous tissue known as nerve cord that is found inside vertebral column.
- (c) Gill slits at least during early developmental stages and may later develop into various structures, such as gills in fish or component of the ear and throat in mammals.
- (d) Post anal tail. In humans it is only present during embryonic stages.

Advantages of chordates

- (a) Chordates, such as fish, chicken, goats, sheep, cattle and buffalo are used as source of food to human beings and other animals.
- (b) Their excreta in the form of faeces or dung are used as organic manure which adds nutrients to the soil, thus improving soil fertility.
- (c) Products from mammals have

commercial value as they are used in making different products used by human beings. For example, cattle, sheep and goats provide skin which is used for making leather in tannery industries. Leather is used to make belts, bags, shoes and clothes. Many mammals provide horns and hooves which are used for making buttons and ornaments. The bones of mammals are ingredients used in making chicken feed. Sheep provide wool that is used for production of various products, including clothes.

- (d) Some chordates, such as rats (rodents) are used in scientific research. They are trained to detect narcotics at the airports and landmines. They are also used to detect explosives buried underground during the war. They are also used to diagnose diseases such as, tuberculosis. Police dogs are used to track down criminals.
- (e) Some domestic mammals, such as cattle and donkeys are trained to perform different duties, such as cultivation of crops and transportation of goods.
- (f) Some chordates are source of income. Live animals and animal product such as meat and milk are sold in domestic and foreign markets; hence, they are an important source of local and foreign currency. Some animals are kept in zoos, national parks and game reserves to attract tourists.

- (g) Their excreta are used for production of biogas which is used as a source of energy.
- (h) Chordate derived products such as fish oil and shark collagen are used to treat diseases in human beings and livestock. Snake skin, tortoise shell and shark fins are used in traditional medicine.

Disadvantages of chordates

- (a) Some chordates are dangerous because they can attack humans, causing injury or death. Examples of such chordates include crocodiles, sharks, lions, leopards, buffalo, elephants and hyenas.
- (b) Some chordates are vectors of diseases. For example, birds can transmit viruses that cause bird flu. Cats and rats can spread fleas which transmit bacteria and other pathogens that cause diseases, such as plague. Infected dogs can also transmit rabies, which is a viral disease of dogs and can also infect humans.
- (c) Some chordates are crop pests, destroying crops in storage or in the field. Examples of such chordates include birds, rats, monkeys, and pigs.
- (d) Some snakes, frogs, toads and newts produce toxins which cause skin irritation and even death to human beings.

Chapter summary

1. A virus is an extremely small microscopic agent. It is not a cell.
2. Kingdom Monera consists of bacteria and blue-green algae. Bacteria can be pathogenic or non-pathogenic.
3. Kingdom Protocista includes *Amoeba* sp., *Plasmodium* sp., *Euglena* sp., *Paramecium* sp., and *Trypanosoma* sp.
4. The main phyla of kingdom Fungi are: Ascomycota, for example baker's yeast; Zygomycota, for example black bread mould and mucor; and Basidiomycota, for example mushrooms, and toadstools.
5. The main divisions of kingdom Plantae are: division Bryophyta, division Filicinophyta or Pteridophyta, division Coniferophyta, and division Angiospermophyta.
6. Vertebrates are animals that possess vertebral column. Examples of such animals include fish, frog, and human being. Invertebrates are animals that do not possess the vertebral column. Examples of such animals include worms and grasshoppers.
7. Features that differentiate members of kingdom Animalia from other organisms include ability to move from one place to another and possession of a well-developed nervous system that helps them in detecting changes in their surroundings.
8. Phyla of the kingdom Animalia include Platyhelminthes (flatworms), Nematoda (round worm), Annelida (segmented worms), Arthropoda (animals with jointed appendages), and Chordata (animals possessing supporting skeletal structure called vertebral column).

Revision exercise 5

Choose the correct answer in questions 1- 7.

1. The following are diseases caused by fungi in plants and animals EXCEPT _____.
 - (a) wheat rust and maize rust
 - (b) candidiasis
 - (c) influenza
 - (d) powdery mildew

2. Which of the following is the advantage of bryophytes?
 - (a) They cause fungal diseases in human beings
 - (b) They help to retain water in the soil
 - (c) They reproduced resins
 - (d) They are good sources of timber

3. One of the following plants is an example of conifers.
 - (a) Mango tree
 - (b) Orange tree
 - (c) Coconut tree
 - (d) Pine tree

4. Which of the following groups of organisms belong to the same Phylum?
 - (a) Liver fluke, hook worm, earthworm, and spider
 - (b) Tapeworm, earth worm, grasshopper and rat
 - (c) Butterfly, grasshopper, frog, and elephant
 - (d) Crabs, grasshopper, spider, and centipede

5. Which of the following is among the distinctive features of annelids?
 - (a) Their bodies have similar body segments
 - (b) They have a cylindrical, elongated and unsegmented body
 - (c) They are dorso-ventrally flattened
 - (d) They have suckers, hooks or both

6. Which of the following is not a distinctive feature of animals?
 - (a) They depend on other organisms as source of food
 - (b) Most of animals are capable of locomotion
 - (c) They have well developed nervous system
 - (d) They have exoskeleton

7. Which of the following is correctly matched?
 - (a) Platyhelminthes - segmented worms
 - (b) Nematoda – round worms
 - (c) Annelida – jointed appendages
 - (d) Arthropoda –flatworms

Write **TRUE** for a correct statements and **FALSE** for an incorrect statements in the spaces provided.

8. A virus is an extremely small organism. _____
9. Kingdom Monera includes *Amoeba* sp. _____
10. *Plasmodium* sp. is a parasite that causes sleeping sickness. _____
11. Rhizoids are found in moulds and mosses. _____
12. Coniferophyta is one of the divisions of the Kingdom Plantae. _____
13. Explain the effects of the following organisms to humans.
 - (a) *Entamoeba* sp.
 - (b) *Plasmodium* sp.
 - (c) *Trypanosoma* sp.
14. List characteristics of phylum to which *Euglena* sp. belongs.
15. A bread was put in a moist cupboard. After a few days black thread-like structures which ended up in club-like structure appeared on the bread.
 - (a) Write the common name of the organism that grew on the bread surface.
 - (b) Name the kingdom to which the observed organism belongs.
 - (c) Name the phylum or division in which the observed organism belongs.
 - (d) Outline the advantages of the members of the kingdom you mentioned in (b) above.
16. What features make cockroach and human being to belong in the same kingdom?
17. Classify each of the following organisms to phylum level; (a) tape worm, (b) housefly and (c) monkey.
18. Mtakuja villagers believe that all insects are dangerous. They decided to kill them by spraying insecticide. As a biologist, convince them that insects are beneficial to their lives.



Chapter Six

Nutrition in plants

Introduction

Plants are organisms that can manufacture their own food, and therefore are called autotrophs. Other organisms that can manufacture their own food include some bacteria and protoctists. Although plants can make their own food, they require some chemical elements from the environment. Such chemical elements are important for normal growth and development. In this chapter, you will learn about photosynthesis process and its importance, as well as the structure of a leaf in relation to photosynthesis. You will also learn roles of essential mineral elements to plants. The competencies developed will enable you to explore the importance of plant nutrition to life.



Think

The world without nutrition in plants

Concept of nutrition

Nutrition is the process of feeding and utilising of food for energy provision, growth, development, repair and maintenance of the overall body health. Nutrition also refers to the study of the relationship between diet, health, and

diseases. There are two major types of nutrition based on how organisms obtain their food. These are autotrophic nutrition and heterotrophic nutrition. In this chapter, you will learn about autotrophic nutrition.

Autotrophic nutrition

Autotrophic nutrition is a type of nutrition in which organisms manufacture their own food using available energy sources. An organism that manufactures its own

food is called an autotroph. The term 'autotroph' originates from two Greek words 'autos' which means 'self' and 'trophy' which means 'feed'; hence 'autotrophy' means 'self-feeding'. This means that autotrophs are organisms that are capable of making their own food for their own use and for other organisms. Examples of autotrophic organisms include green plants, algae, and some bacteria such as cyanobacteria. The autotrophs can be divided into two groups based on how they obtain their energy. These are chemoautotrophs and photoautotrophs.

Chemoautotrophs are organisms that obtain their energy from chemical substances like hydrogen sulphide, iron, methane, and ammonia. They utilise these chemicals to make their own food in the form of carbohydrates through a process called chemosynthesis. Examples of chemoautotrophs are some bacteria (archaebacteria) that live in harsh environments. Such environments include the deep sea and around volcanic sites where there is no sunlight and where many other organisms cannot survive. Photoautotrophs are organisms that obtain their energy from sunlight in order to make their own food. They make their food using water and carbon dioxide in the presence of chlorophyll and sunlight through a process called photosynthesis.

Photosynthesis

Task 6.1

Search from the library and reliable internet sources the information on the meaning of photosynthesis.

Photosynthesis is the process by which green plants, some bacteria and protoctists make their own food using water and carbon dioxide in the presence of sunlight energy. Organisms that make their own food are called primary producers or autotrophs. They are the primary source of food for all other organisms. During the process of photosynthesis, six molecules of carbon dioxide and twelve molecules of water combine to form one molecule of glucose (a simple sugar), six molecules of water, and six molecules of oxygen. The leaf is the main site for photosynthesis in plants. However, photosynthesis can take place in other green parts of the plant, such as stem. The following chemical equation represents the photosynthesis process in plants.



Photosynthesis produces a six-carbon sugar molecule (a hexose) called glucose. Plants convert these hexose sugars into other carbohydrates, such as complex sugars, starch and cellulose. Plants are also capable of converting hexose sugars into other organic compounds, such as proteins and fats. The food formed by plants during photosynthesis is stored in the form of starch.

Biologists often use the presence of starch in some parts of plants as an indication that photosynthesis has taken place.

Structure of the leaf in relation to photosynthesis

The internal and external structures of a leaf make it well adapted for photosynthesis. In most plants, the leaf is the main site for photosynthesis, although some plants, such as cactus (plural is cacti) use their stems for photosynthesis. The leaf structure provides a very efficient means for absorbing carbon dioxide and sunlight.

External structure of a leaf

The external features of a leaf can be viewed using either a hand lens or with the naked eye. Figure 6.1 shows the external structure of a leaf.

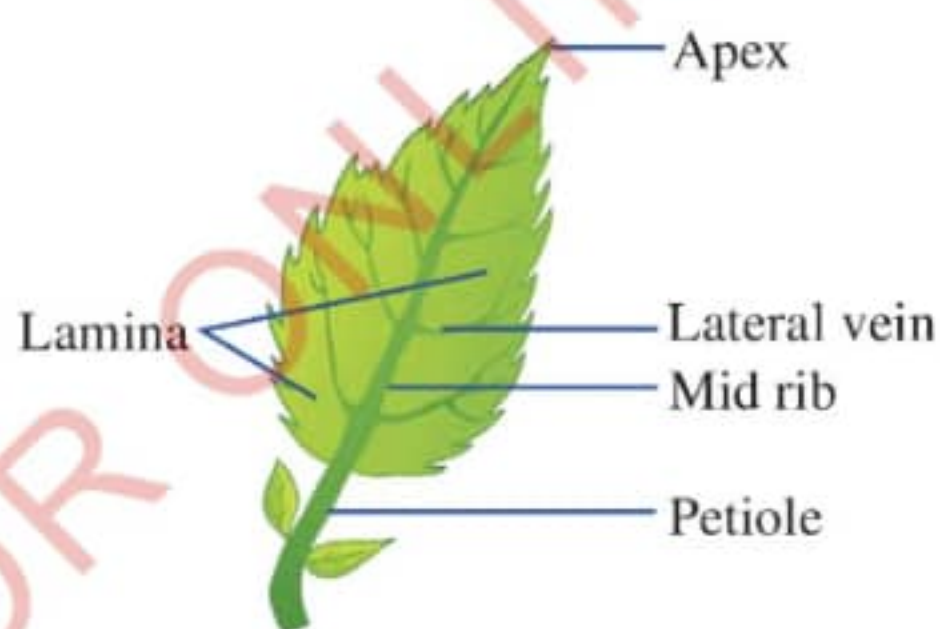


Figure 6.1: External structure of a leaf

The petiole or leaf stalk attaches the leaf to the branch or stem. It keeps the lamina in a position that will enable it to get maximum amount of sunlight. The lamina or leaf blade is the expanded portion of a leaf. It has a large surface

area. This maximizes the absorption of sunlight energy and carbon dioxide. The lamina is also thin so that carbon dioxide gas can diffuse and sunlight energy can penetrate over a short distance to reach cells. The mid-rib and veins contain vascular tissues, namely xylem and phloem. Xylem transports water and dissolved minerals while phloem transports manufactured food in the plants.

Activity 6.1: Observing the external features of a leaf

Materials: Leaves from various plants, hand lens, pen or pencil, notebook and rubber

Caution

Be careful not to collect hairy or thorny leaves.

Procedure

1. Collect different plant leaves from the school or home environment.
2. Observe each of the leaves using your naked eyes.
3. Observe each of the leaf using the hand lens. Note as many details as you can.
4. Describe each of the observed leaves.
5. Explain how each of the plant parts aids photosynthesis.

Question

State the observed differences in structure among the variety of leaves.

Internal structure of a leaf

Task 6.2

Search from library and reliable internet sources the information on internal structures of a leaf that are related to photosynthesis.

The internal features of a leaf can be viewed under a microscope. Figure 6.2 shows the internal structure of a leaf. The leaf has lower and upper parts which are covered by a thin waxy layer called cuticle. On the upper part of the leaf is the upper epidermis while the lower epidermis is located on the lower part of the leaf.

The leaf has small openings called stomata. Depending on the species of a plant and the environment in which it

grows, the leaf can either have stomata on both sides or on one side (lower or upper part). But in most plants, the lower epidermis has more stomata than the upper epidermis. Immediately below the epidermis is another tissue layer called mesophyll, that is photosynthetic because it contains chloroplasts.

The mesophyll tissue is divided into two parts, namely the palisade and spongy mesophyll. The palisade mesophyll cells are usually tightly packed. The spongy mesophyll has loosely packed cells. It also consists of air spaces and vascular tissues which are made up of xylem and phloem cells, as shown in Figure 6.2.

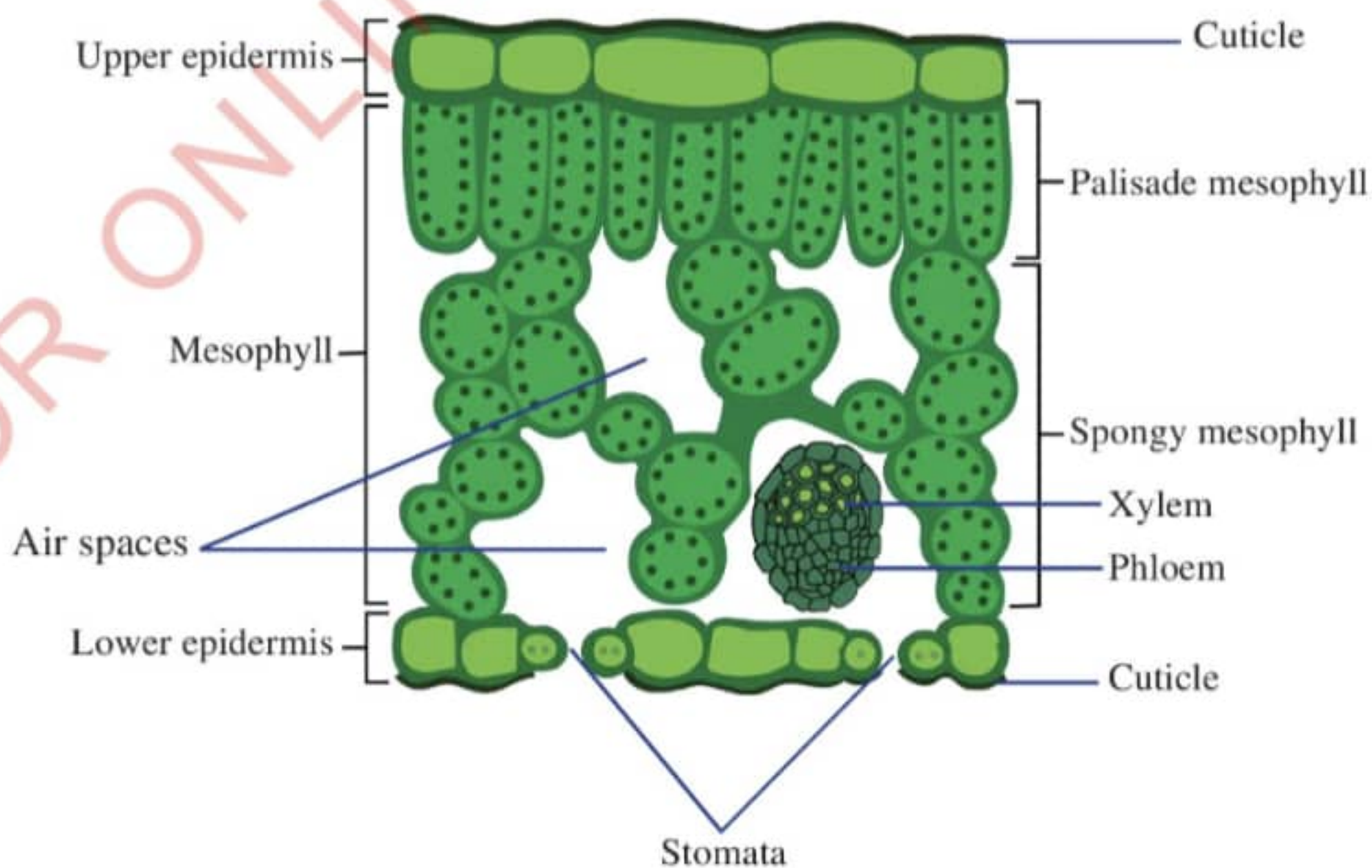


Figure 6.2: Internal structure of a leaf

The cuticle: This is the outermost transparent and waxy layer of the leaf. It allows light to penetrate into the photosynthetic cells. It also protects the leaf from injury, pests, and excessive loss of moisture. In order to allow gaseous exchange, there is no cuticle on the stomata.

The epidermis: This is the outermost layer of cells found on both lower and upper surfaces of a leaf. It is transparent and only one cell is thick, hence it allows sunlight to penetrate the leaf easily. The epidermis has pores called stomata.

Stomata: Stomata (singular is stoma) are small pores in the epidermis. They allow oxygen and carbon dioxide to diffuse in and out of the leaf. Stomata are surrounded by guard cells that close and open the pores.

Mesophyll: This is made up of palisade layer and spongy layer (See Figure 6.2). The palisade mesophyll is made up of cells that are elongated and arranged at right angles to the surface of the leaf. It is found just below the upper epidermis. Being near the upper epidermis, the palisade cells are exposed to maximum sunlight. This enables the cells to absorb maximum amount of sunlight energy. The spongy mesophyll is just above the lower epidermis. It is a loosely constructed layer of irregularly shaped cells separated by large intercellular

air spaces that connect with each other and linked to the atmosphere through the stomata pores. The air spaces in the spongy mesophyll provide an immediate source of carbon dioxide.

Activity 6.2: Investigating the internal structure of a leaf

Materials: Prepared slide of a transverse section of a leaf, microscope, notebook, and pencil or pen

Procedure

1. Mount the prepared slide on the microscope.
2. Observe the slide under the low power objective lens. What do you see?
3. Describe what you have observed.

The process of photosynthesis

Task 6.3

Search from reliable internet sources the simulations that show the process of photosynthesis. Note down what you have observed.

Photosynthesis takes place in cell organelles known as chloroplasts. These are mostly found in the green leaves. Chloroplasts contain chlorophyll that is responsible for trapping sunlight energy that is used during photosynthesis. Photosynthesis takes place in two stages, namely light reaction and dark reaction.

The light reaction or light dependent reaction stage

This stage takes place in specialised structures of the chloroplast called grana. The grana (singular is granum) contain chlorophyll that absorbs light energy from the sun. The light dependent reaction changes light energy into chemical energy. The formed energy is stored in a chemical compound called ATP (Adenosine Triphosphate). This energy is used in the dark reaction stage of photosynthesis.

Light energy causes photolysis, a chemical process whereby water molecules (H_2O) are split into hydrogen ions (H^+) and hydroxyl ions (OH^-). This can be represented using the following equation.



The hydroxyl ions undergo further reactions to produce water and oxygen. Some oxygen is released into the atmosphere. The rest of the oxygen is used for respiration in the plant. Hydrogen ions are used in the dark reaction stage to synthesise food together with carbon dioxide from the air.

The dark reaction or light independent reaction stage

This stage takes place in the stroma in the absence of light. The stroma is a colourless matrix of fine material found in the chloroplast. During

the dark reaction stage, hydrogen ions from light reaction stage and carbon dioxide from the atmosphere combine to form glucose that is the simplest carbohydrate. Glucose is later converted into starch. Starch is insoluble and acts as a temporary storage of excess carbohydrates. It can easily be converted into glucose when a plant needs energy. Glucose may also be converted into cellulose (the structural material of plants). Lipids and proteins are also formed from glucose through complex processes.

Activity 6.3: An experiment to determine the presence of starch in a leaf

Materials: A leaf, absolute ethanol, iodine solution, source of heat, test tube, beaker, dropper, forceps, water, white tile, cotton wool, notebook, and pencil

Procedure

1. Take a leaf from a plant that has been in the sunlight for at least 6 hours.
2. Put the leaf in boiling water for 2-3 minutes, as shown in Figure 6.3. This stops further chemical reactions in the leaf because boiling kills the leaf cells.

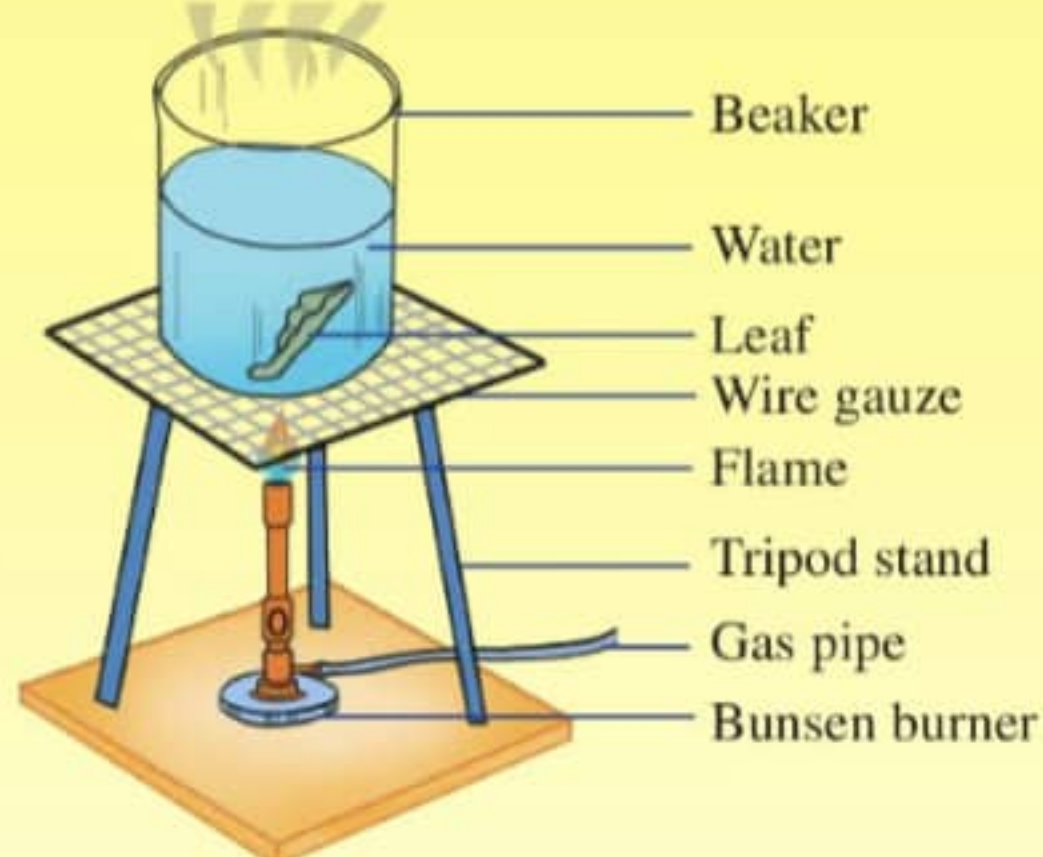


Figure 6.3: Boiling a leaf to kill the cells

- Put the boiled leaf in the test tube containing absolute ethanol. Cover the test tube with cotton wool. Place the test tube in a beaker of boiling water, as shown in Figure 6.4. The absolute ethanol dissolves the chlorophyll.

Caution

Do not heat absolute ethanol directly because it is highly flammable. Cover the test tube with cotton wool.

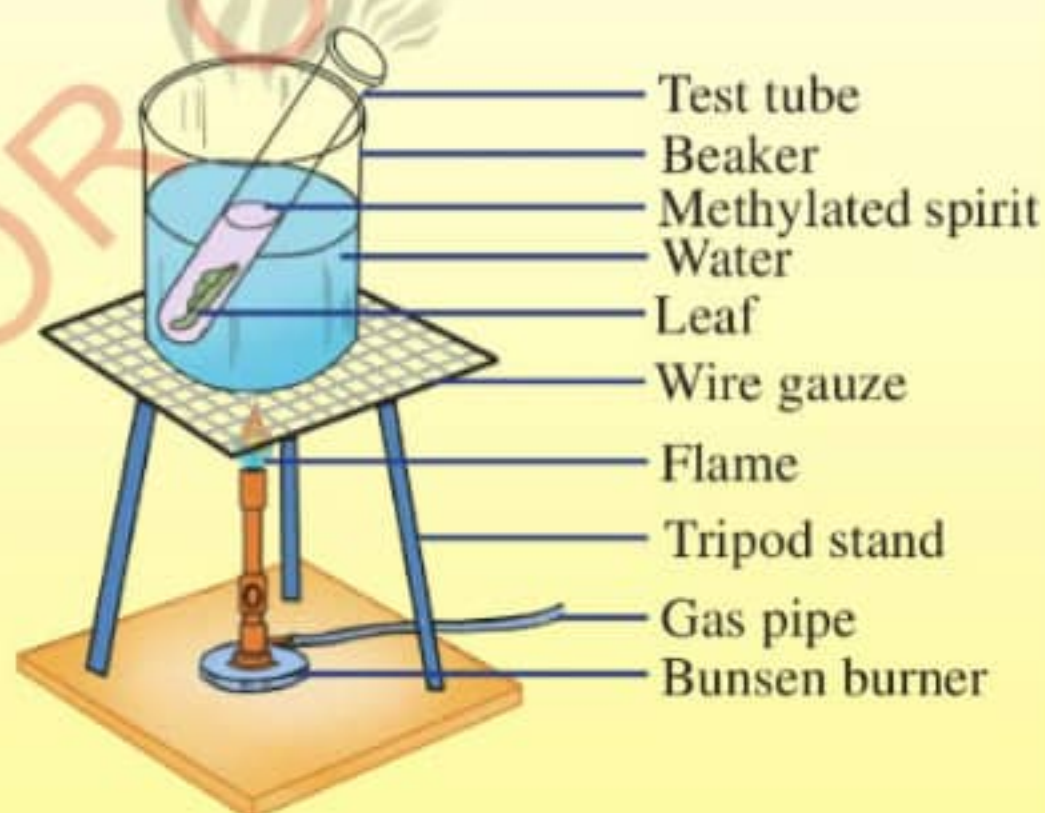


Figure 6.4: Decolourising a leaf by using absolute ethanol

- Remove the leaf from the absolute ethanol and rinse it in warm water. The water softens the leaf.
- Put the leaf on a white tile. Add 2-3 drops of iodine solution on the leaf. What do you observe?
- Record your observations.

Note: The blue-black colour indicates the presence of starch.

Activity 6.4: An experiment to determine the importance of carbon dioxide in photosynthesis

Materials: Conical flasks, thread, corks, soda lime, water, iodine solution, dropper, potted plant, notebook, absolute ethanol, white tile, test tube, source of heat, beaker, pen or pencil

Procedure

- Place the potted plant in dark place for at least 24 hours to destarch its leaves.
- Enclose a leaf (A) in a conical flask containing soda lime, as shown in Figure 6.5. This leaf will not have a supply of carbon dioxide from the air because soda lime absorbs carbon dioxide.
- Enclose a second leaf (B) in a conical flask containing some water.

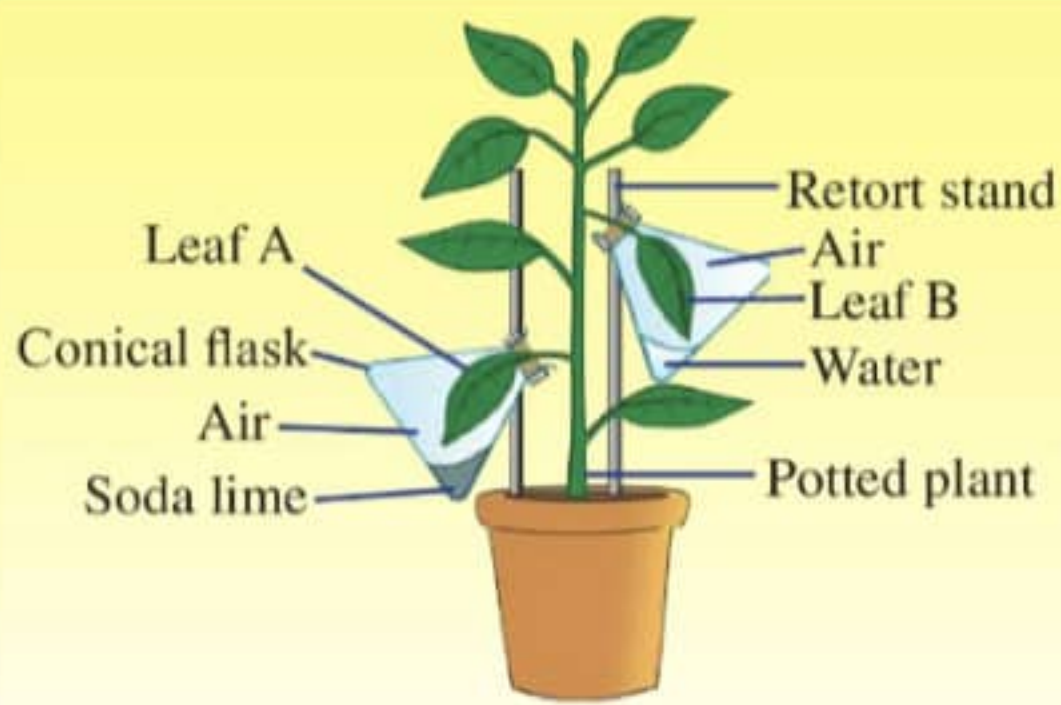


Figure 6.5: Experimental set-up to determine the importance of carbon dioxide in photosynthesis

4. Place the plant in sufficient light for 6 hours at room temperature.
5. Remove the two leaves (leaf A and B) from flasks.
6. Test for starch by using standard procedures as in Activity 6.3.
7. Record your observations.

Activity 6.5: An experiment to determine whether chlorophyll is necessary for photosynthesis

Materials: Variegated leaves, iodine solution, absolute ethanol or spirit, beaker, white tile, water, heat source, pencil, and notebook

Procedure

1. Pick a variegated leaf from a plant that has been exposed to sunlight for at least 6 hours. Variegated leaves are green but mixed with patches of other colours, such as

yellow, red and white. The green areas have chlorophyll but areas with other colours do not have chlorophyll. Figure 6.6 shows variegated leaves with green and yellow patches.



Figure 6.6: Variegated leaves with green and yellow patches

2. Describe a colour map of the leaf.
3. Test the leaf for the presence of starch using standard procedures as in Activity 6.3.
4. Compare the stain patterns with the described colour map. What do you notice?
5. Record your observations and then share with your teacher.

Question

Based on the results you have obtained, what conclusion can you draw about variegated leaves in relation to photosynthesis?

Activity 6.6: An experiment to determine whether sunlight is necessary for photosynthesis

Materials: Potted plant, manila card, paper clips, absolute ethanol or methylated spirit, iodine solution, test tube, beaker, water, white tile, notebook, and pen or pencil

Procedure

1. Place the potted plant in a dark place for 24 hours to destarch its leaves.
2. Use manila card (opaque paper) and paper clips to cover the upper and lower epidermis of one leaf as shown in Figure 6.7.



Figure 6.7: Leaf in a potted plant covered with manila card

3. Put the plant in bright sunlight for 6 hours.
4. Detach the leaf that has been covered by manila card from the plant.
5. Remove the paper clips and manila card.

6. Test the detached leaf for starch using the standard procedure. What do you notice?
7. Discuss your observations with your classmate.

General conclusion

The above experiments have shown that chlorophyll, water, carbon dioxide, and sunlight are necessary for photosynthesis. The process of photosynthesis cannot take place in the absence of any of these. If sunlight or carbon dioxide are in low supply, the rate of photosynthesis will decrease. In addition, magnesium and iron elements are essential for the production of chlorophyll. If they are in low supply, the rate of photosynthesis decreases because chlorophyll pigment is insufficient.

Activity 6.7: An experiment to verify that photosynthesis produces oxygen

Materials: Test tube, beaker, sodium hydrogen carbonate, filter funnel, pond weed, wooden splint, a lighter, notebook, and pen or pencil

Procedure

1. Put the pond weed in a beaker containing water. Put one spatula of sodium hydrogen carbonate in the water. This serves as a source of carbon dioxide.
2. Put a plant under a funnel and collect the bubbles in an inverted test tube, as shown in Figure 6.8.

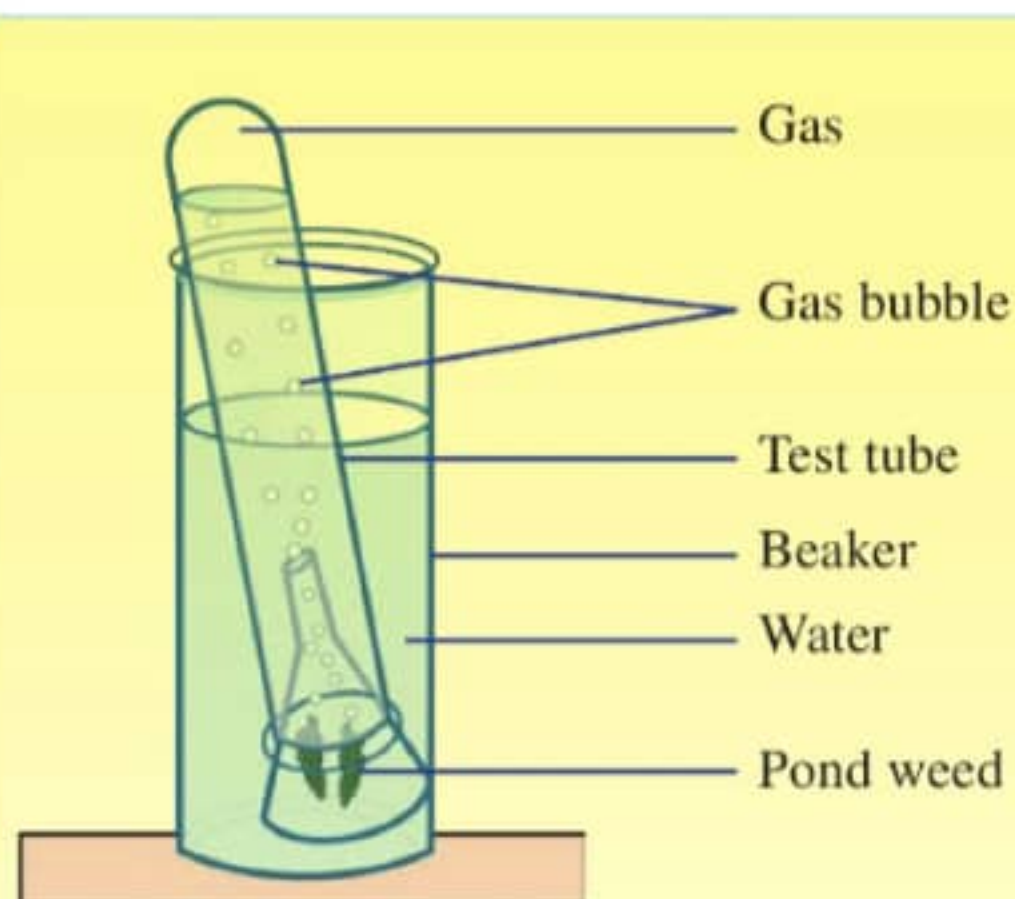


Figure 6.8: Experimental set-up to show that photosynthesis produces oxygen

3. Leave the set-up exposed to the sunlight for 6 hours.
4. Remove the test tube slowly, covering its mouth while it is still under water.
5. Open the test tube slightly but quickly and introduce a glowing splint.
6. Record what you have observed.

Questions

1. What did you observe when the experimental set-up was exposed to sunlight?
2. What did you observe when a glowing splint was introduced into the test tube?
3. Why did gas bubbles evolve from the bottom part of the experimental set-up?
4. Name the gas that evolved from the experimental set-up and give reason to support your answer.

5. What conclusion can you draw from this experiment?

Importance of photosynthesis

Task 6.4

Read from reliable internet sources and biology books information about the importance of photosynthesis

Photosynthesis is a vital process for the continuous survival of plants and other organisms on earth. The importance of photosynthesis includes production of oxygen, reduction of atmospheric carbon dioxide, conversion of solar energy into chemical energy, and production of food.

Production of oxygen

Photosynthesis produces oxygen. This replenishes the atmospheric oxygen that has been used during burning, respiration, rusting, and other processes. All aerobic organisms require oxygen for respiration.

Reduction of atmospheric carbon dioxide

The process of photosynthesis uses carbon dioxide from the atmosphere. This process reduces atmospheric carbon dioxide, since it is converted into carbohydrate in the presence of water, sunlight, and chlorophyll. Being

a by-product of respiration, carbon dioxide is harmful if excessive amounts accumulate in the atmosphere. The accumulated carbon dioxide forms a layer, which traps heat escaping from the earth's surface and re-radiates it back to the earth's surface, thus causing global warming.

Conversion of solar energy into chemical energy

Photosynthesis converts sunlight energy into chemical energy. The sun is the primary source of all forms of energy used in life processes. This energy is used by plants as well as animals. It is made available to living things through photosynthesis. Animals obtain energy by feeding on plants or animals that eat plants. Thus, sunlight energy is converted into chemical energy that is stored in organic food molecules. The energy stored in organic food molecules is then released into the cells during respiration. This energy is used in life processes, such as movement, reproduction, sensitivity, growth, and excretion.

Production of food

Photosynthesis produces food for the plants. Some animals, such as cattle, goats, and zebra depend directly on plants as their source of food. Such animals are called herbivores. Other animals called carnivores, such as lions, hyenas, and leopards feed directly on herbivores by killing and eating them. Animals, such as human beings and monkeys depend on both plants and animals as their source of food. These are called omnivores.

The extra food produced by plants is stored in various plant organs. The stored food is used by the plants during adverse conditions, such as drought when the plant cannot synthesise adequate amounts of food. The underground storage organs of the plants are formed from modified stems, leaves or roots. There are several types of underground storage organs, such as bulbs, tubers, roots, corms, and rhizomes.

Bulb: This is an underground storage organ formed from the modification of the plant stem and leaves. An onion is an example of a bulb, as shown in Figure 6.9.

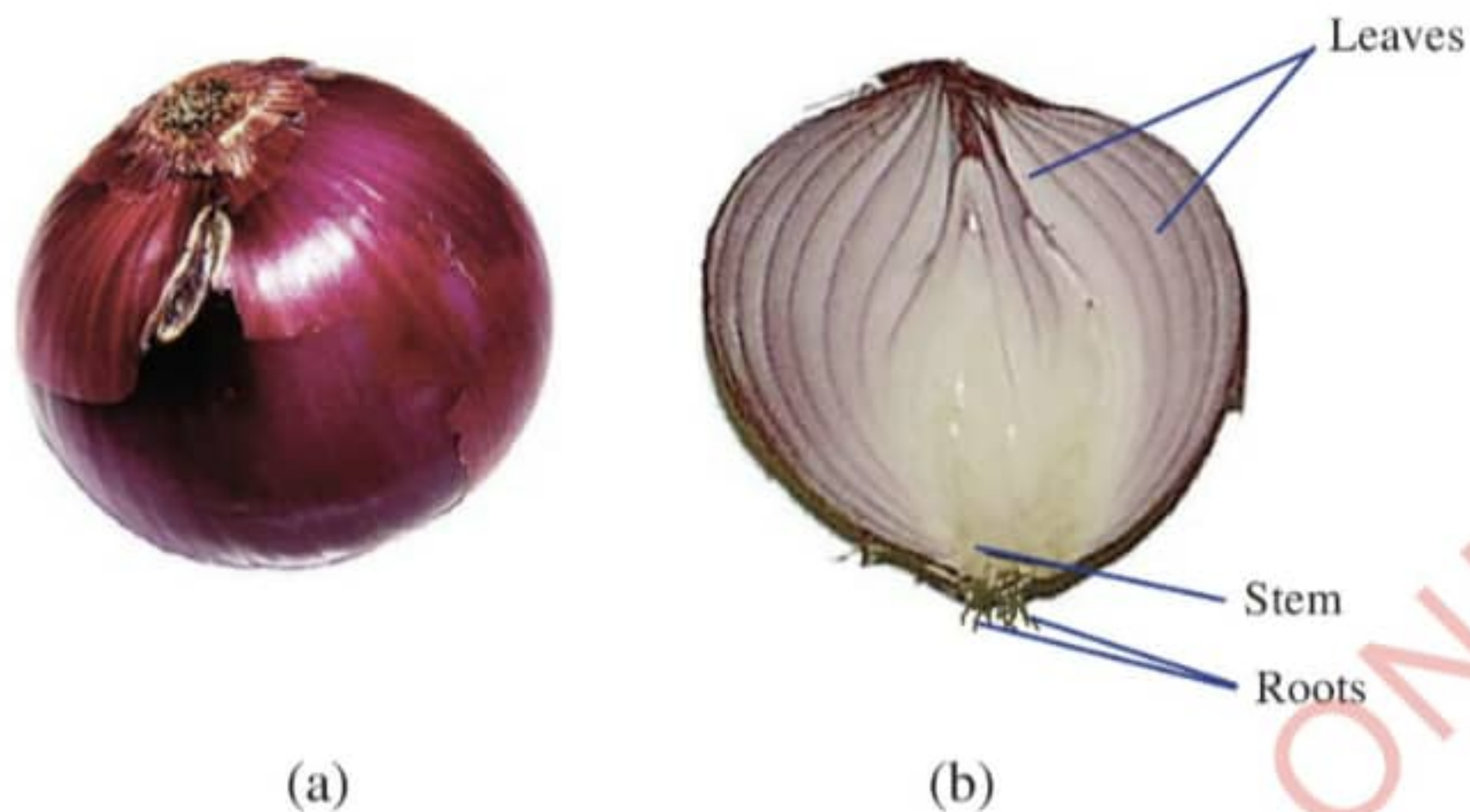


Figure 6.9: Onion bulb (a) whole (b) cut in half

Tuber: This is a fleshy storage organ formed from either a stem or a root. An examples of a stem tuber is Irish potato while an example of a the root tuber is sweet potato (See Figure 6.10).

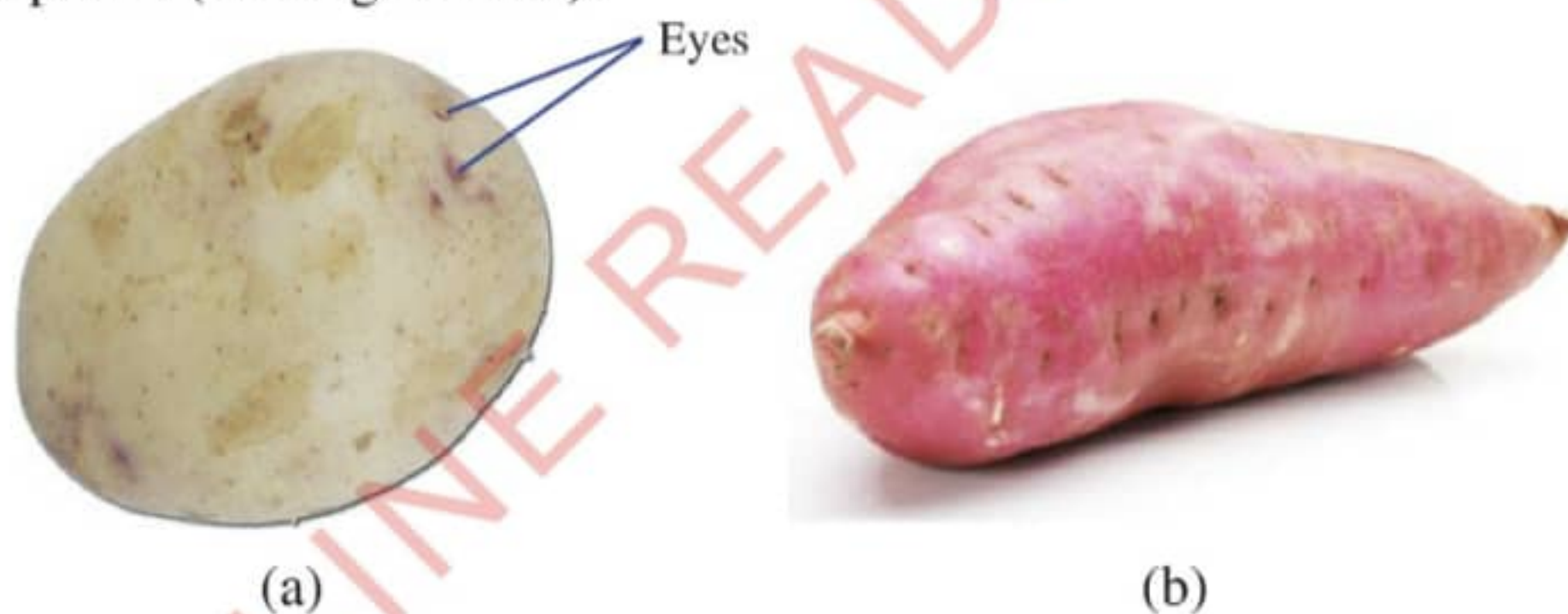


Figure 6.10: Tubers (a) stem tuber (Irish potato) (b) root tuber (sweet potato)

Taproot: Some taproots such as those of carrot, sugar beet, and beetroot are specialised for food storage (See Figures 6.11).

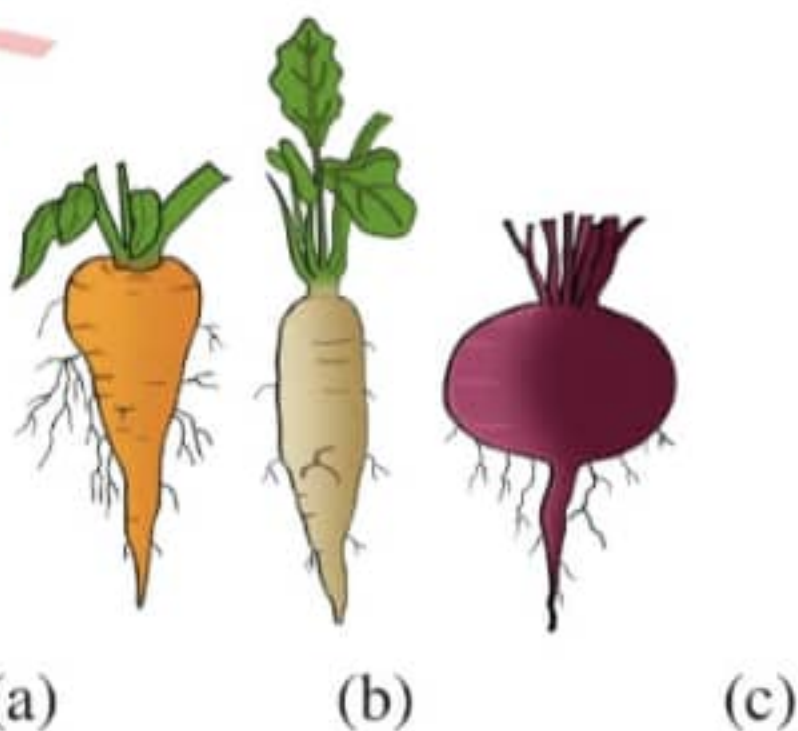


Figure 6.11: Examples of taproots (a) carrot (b) sugar beet (c) beetroot

Corm: This is an underground storage organ formed from the plant stem. It is a mass of solid tissues with a dry papery cover made of modified leaves. Examples of corms are yams, cocoyams and crocuses (See Figure 6.12).

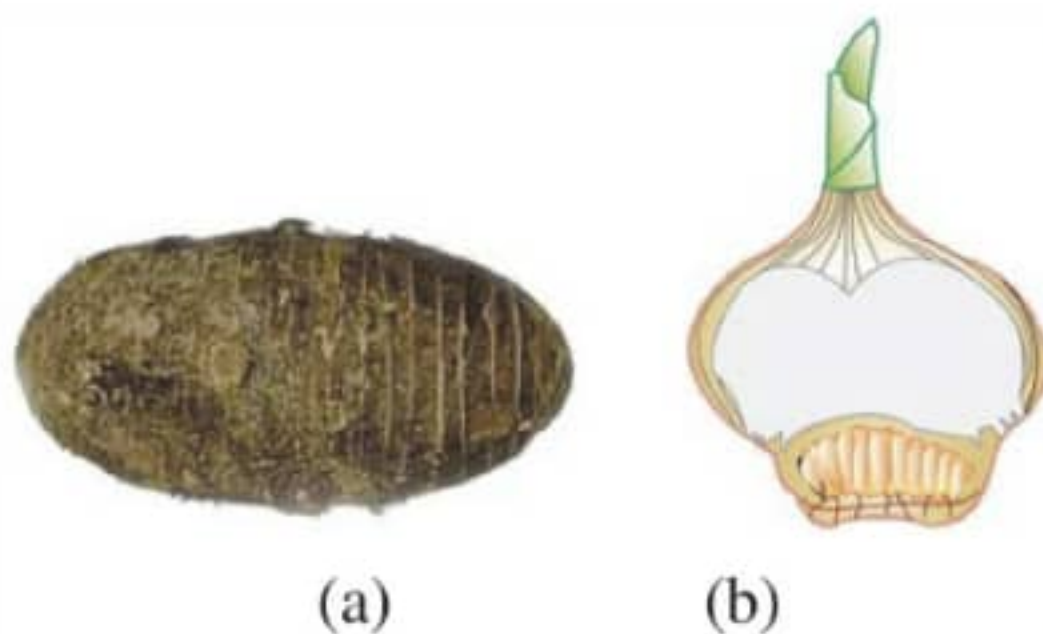


Figure 6.12: Examples of corms (a) cocoyam (b) crocus

Rhizome: A rhizome is a swollen underground stem, which grows horizontally on the surface or just below the surface. An example is ginger, as shown in Figure 6.13.



Figure 6.13: Ginger

Exercise 6.1

1. Explain the importance of chloroplasts in plants.
2. Describe the internal structure of a leaf.
3. What are the roles of underground storage organs in plants?
4. Why are vascular bundles important in plants?
5. Explain why it is not advisable to sleep in a room with a potted plant.

Essential and non-essential elements in plants

Task 6.5

From reliable sources read the information about the essential and non-essential elements in plants.

There are many chemical elements which are known to be important for plant growth and development. These chemical elements are divided into two major groups, namely essential and non-essential elements. The essential elements are vital for plant development and survival.

The plant cannot complete its life cycle without them. These include nickel (Ni), carbon (C), hydrogen (H), oxygen (O), nitrogen (N), phosphorus (P), calcium (Ca), iron (Fe), manganese (Mn), potassium (K), magnesium (Mg), sulphur (S), boron (B), copper (Cu), zinc (Zn), molybdenum (Mo), and chlorine (Cl). The non-essential elements are used to stimulate growth in plants. Examples of non-essential elements include lithium (Li), iodine (I), sodium (Na), mercury (Hg), silver (Ag), tin (Sn), radium (Ra), silicon (Si), bromine (Br), and cobalt (Co).

The essential elements can also be divided into two groups, which are

macroelements and microelements. The macroelements are those elements that are required by plants in relatively large amounts. They are also known as macronutrients. These include phosphorus (P), potassium (K), nitrogen (N), sulphur (S), magnesium (Mg), calcium (Ca), carbon (C), oxygen (O), and hydrogen (H). Among the macroelements, some are non-minerals, such as carbon, oxygen and hydrogen, that are obtained from the air. Other macroelements are minerals, such as magnesium, calcium, phosphorus and potassium. Both non-mineral and

mineral elements are essential for normal plant growth and metabolic activities. For example, carbon is a major constituent of plants, hydrogen is required during photosynthesis and oxygen is required during respiration. The essential mineral elements are mainly absorbed by plants from soil. They usually dissolve in water and are absorbed by plants in form of ions. The ions enter the plant through roots and are transported upward through the vascular system. Table 6.1 shows various sources of macroelements in plants.

Table 6.1: Sources of different macroelements in plants

Macroelement	Sources
Phosphorus	Obtained from commercial fertilisers, crop residues, and manure
Potassium	Obtained from soil minerals, organic materials, and commercial fertilisers
Nitrogen	Obtained from commercial fertilisers and from the air. <i>Rhizobium</i> bacteria in the roots of leguminous plants fix nitrogen from the air for use by the plants
Sulphur	Obtained from rainwater, gypsum, and commercial fertilisers
Magnesium	Obtained from soil minerals, organic materials (such as compost), commercial fertilisers, and lime
Calcium	Obtained from lime, gypsum, and commercial fertilisers

Functions of macroelements in plants

Task 6.6

Read from reliable online sources and books about the functions of macro and microelements in plants

Each mineral element has a specific function in the plant body. Some are used in the formation of building materials for proteins and carbohydrates in plants while others play important roles in other metabolic activities of the plant. The deficiency of such elements will eventually affect plant growth and metabolic activities. Table 6.2 shows the functions, signs of deficiency, and effects of excess macroelements in plants.

Table 6.2: Functions, signs of deficiency, and effects of excess macroelements in plants

Macroelement	Functions	Signs of deficiency	Effects of excess
Nitrogen	(i) Synthesis of proteins and transfer of energy (ii) Formation of chlorophyll (iii) Increases seed and fruit production (iv) Speeds up growth rate	(i) Slow growth (ii) Yellowish or light green leaves, as shown in Figure 6.14(b) (iii) Reduced yield of fruits and seeds (iv) Poorly developed leaves	(i) Very dark green succulent leaves (ii) Break down of vascular tissue and restricted transport of water (iii) Easily damaged stems because of excess sap
Phosphorus	(v) Germination and growth of seeds (vi) Production of flowers and fruits (vii) Growth of roots	(i) Reduced plant growth (ii) Delayed development (iii) Small leaves that drop early	Reduces the plant's ability to take up required micronutrients, particularly iron and zinc

Macroelement	Functions	Signs of deficiency	Effects of excess
	(viii) Ripening of seeds and fruits	(iv) Bluish-green or purplish leaves, as shown in Figure 6.14 (c)	
Potassium	(i) For steady growth (ii) Increases resistance to diseases (iii) Useful in protein formation	(i) Scorched brown leaf edges and tips, as shown in Figure 6.14 (d) (ii) Rolling of leaves (iii) Susceptibility to diseases	Deficiency of calcium and magnesium
Magnesium	(i) Formation of chlorophyll (ii) Activation of plant enzymes necessary for growth	(i) Yellowish leaves, as shown in Figure 6.14 (e) (ii) Leaves fall without withering	(i) Stunted growth (ii) Lack of food needed for growth and development due to impaired photosynthesis
Calcium	(i) Formation of cell walls (ii) Increases mechanical strength of the plant (iii) Growth of roots (iv) Normal transport and retention of other elements in the plant	(i) Poorly developed roots with weak tips (ii) Curling of leaf margins, as shown in Figure 6.14 (f) (iii) Hooked leaf tips (iv) Internal decay	Reduced uptake of essential elements, particularly, phosphorus and magnesium

Macroelement	Functions	Signs of deficiency	Effects of excess
Sulphur	(i) Production of protein (ii) Formation of chlorophyll (iii) Improved root growth and seed production	(i) Slow growth (ii) Small leaves that roll up and are stiff and brittle (iii) Premature shedding of leaves (iv) Tips of buds die	Premature ageing of plants

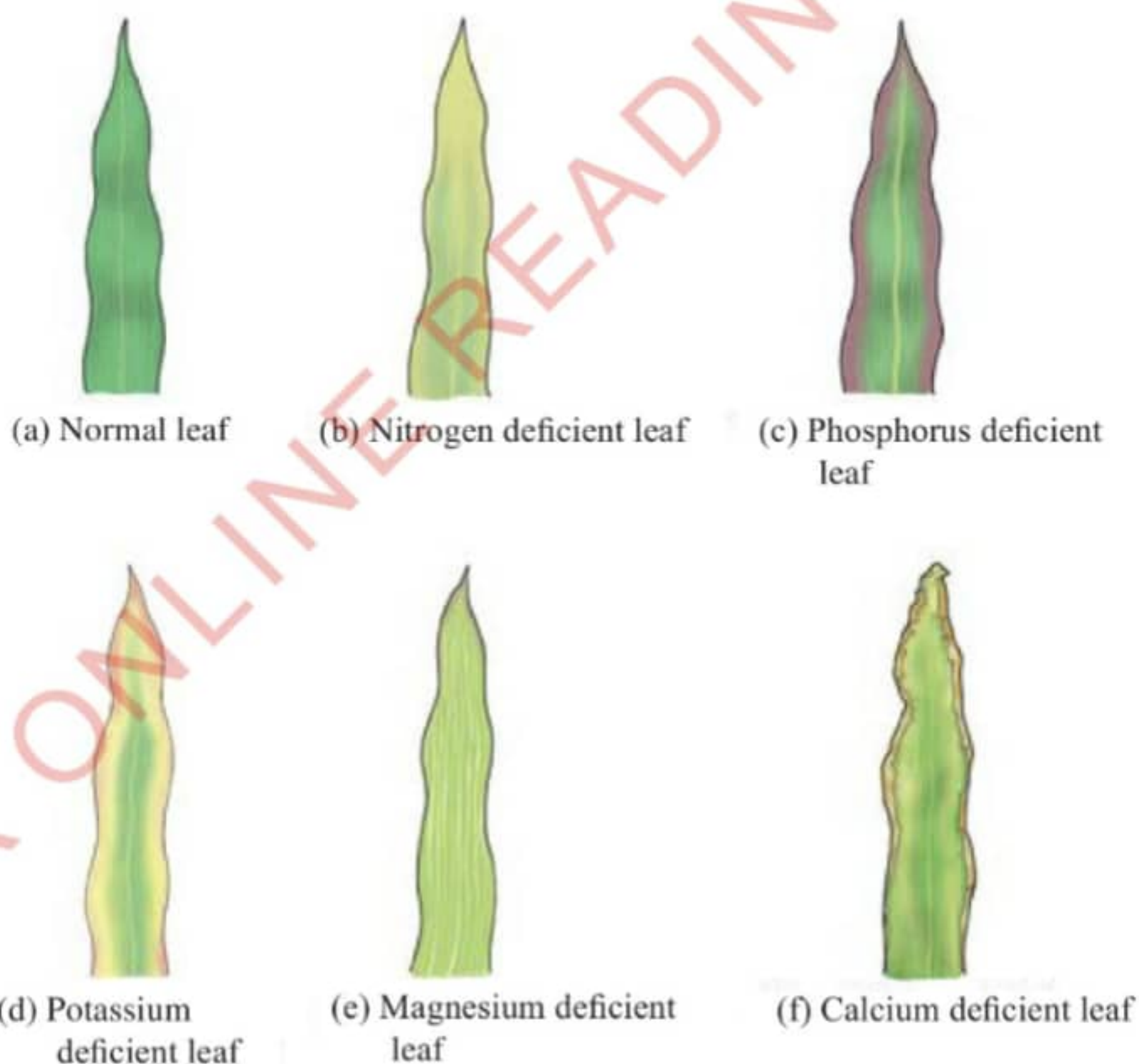


Figure 6.14: Signs of deficiency of some macroelements in a maize leaf

Microelements are essential mineral elements that are needed for plant growth and survival but only in very small quantities. However, deficiency or excess of microelements in plants can have effects on physiological activities of plants.

These elements are also referred to as trace elements. They include boron, copper, iron, chlorine, manganese, zinc, nickel, and molybdenum. Table 6.3 shows functions of microelements in plants, deficiency signs, and effects of their excess levels.

Table 6.3: Functions, deficiency signs, and effects of excess microelements in plants

Macroelement	Functions	Signs of deficiency	Effects of excess
Boron	(i) Aids in the production of sugar and starch (ii) Aids in water intake by cells (iii) Important for seed and fruit development (iv) Keeps Calcium in a soluble form	(i) Distorted or dead growing tips (ii) Hollow stems (iii) Deformed fruits (iv) Discoloured leaves	(i) Scorched leaves (ii) Falling off of leaves
Copper	(i) Important for normal growth and development (ii) Aids in the formation of proteins	(i) Bluish-green leaves (ii) Withering of leaves (iii) Leaves failing to unfold (iv) Distorted growth tips	(i) Iron deficiency (ii) Suppressed growth
Chlorine	Important for plant metabolism	(i) Wilting (ii) Stumpy roots	Scorched leaf edges

Macroelement	Functions	Signs of deficiency	Effects of excess
Iron	(i) Important for the formation of chlorophyll (ii) Important in transportation of oxygen	Yellowing of leaves between the veins	Brown spots on leaves
Manganese	(i) Catalyst for enzyme action (ii) Required for the formation of chlorophyll	(i) Yellowing of leaves between veins (ii) White or grey spots on leaves	(i) Iron deficiency (ii) Brown spots on leaves surrounded by a pale circle
Molybdenum	Helps in the formation of root nodules in legumes	(i) Yellow spots on leaves (ii) Dead spots on leaves (iii) Distorted or dead growing tips	(i) Pale-green leaves (ii) Rolled leaf margins (iii) Stunted growth
Zinc	(i) Important for regulation of plant growth and maturity (ii) Helps in the formation of proteins	(i) Yellowing of leaves between veins (ii) Appearance of purple or dark spots on leaves (iii) Small, deformed leaves (iv) Reduced fruiting	(i) Iron deficiency (ii) Pale green leaves

Activity 6.8: Investigating the effects of mineral elements on plant growth

Materials: Pots or tins, seeds (such as maize and beans), fertilisers containing different minerals, notebooks, and a pen

Procedure

1. Prepare pots or tins for planting seeds.
2. Plant the same types of seeds in all pots or tins.
3. Your teacher will explain what is in each type of fertiliser (for example, 'NPK' has Nitrogen, Phosphorus and Potassium) and the proper amounts of each type of fertiliser required by the plants.
4. Vary the amounts of fertiliser that you apply in the pots or tins as follows:
 - (a) Do not put any fertiliser in the first pot or tin.
 - (b) Put adequate amounts of fertiliser in the second pot or tin.
 - (c) Put excess amounts of fertiliser in the third pot or tin.
5. Water all experimental pots or tins every morning and evening with the same amount of water for a period of 30 days.
6. Make daily observations of the plants in the pots or tins, including height and leaves colouration. Record the observations.

Questions

1. Is there any difference in growth of the plants in the three pots or tins? Give reasons.
2. Why was the fertiliser not applied in the first pot or tin?

Exercise 6.2

1. As a Biology student, you are investigating the role of elements in plant growth. Explain the fundamental difference between essential and non-essential elements in the environment of plant nutrition.
2. You are working on a research project focusing on the importance of individual elements in plant growth. Explain why phosphorus is considered crucial for plant growth and development.
3. You are leading a discussion on the differences between microelements and macroelements in your classroom. Explain the differences between these two categories of nutrients.

Chapter summary

1. Photosynthesis is the process by which green plants, some bacteria and some protoctists make their own food using carbon dioxide and water in the presence of sunlight.
2. A plant leaf comprises features that make it well adapted to photosynthesis.
 - (a) Waxy transparent cuticle
 - (b) Thin epidermis with stomata
 - (c) Palisade cells with chloroplasts
 - (d) Palisade cells positioned just below the upper epidermis
 - (e) Spongy mesophyll with air space
 - (f) Veins that have vascular tissue
 - (g) Broad flat lamina
 - (h) Petiole to attach the leaf to the stem or branch at an appropriate angle to access sunlight
3. Photosynthesis involves two stages, namely the light reaction stage and the dark reaction stage.
4. During the light reaction stage, sunlight causes the production of ATP and splitting of water molecules into hydrogen and hydroxyl ions. This reaction is called photolysis.
5. In the dark reaction stage, carbon dioxide and hydrogen ions combine to form glucose. This reaction is called carbon dioxide fixation. Glucose is later used to make starch, proteins and lipids.
6. Plants require both macroelements and microelements to survive.
7. The macroelements include nitrogen, phosphorus, potassium, sulphur, calcium, and magnesium.
8. The microelements include boron, copper, iron, chlorine, manganese, molybdenum, and zinc.
9. Lack of the necessary elements affects plant growth and crop yields. However, excessive amounts have negative effects. Plants should get optimum amount of minerals.

Revision exercise 6

Section A

Choose the correct answer.

1. Which of the following is NOT a type of an underground storage organ?
 - (a) Bulb
 - (b) Seed
 - (c) Corm
 - (d) Rhizome
2. The process by which plants and some bacteria use the energy from sunlight to produce glucose is called _____.
 - (a) Photolysis
 - (b) Hydrolysis
 - (c) Photosynthesis
 - (d) Plasmolysis
3. Boron, copper, iron, chlorine, manganese, molybdenum and zinc are examples of:
 - (a) microelements
 - (b) macroelements
 - (c) non-mineral elements
 - (d) organic elements
4. The part of a leaf that provides a large surface area for maximum absorption of sunlight and carbon dioxide is:
 - (a) petiole
 - (b) stomata
 - (c) lamina
 - (d) midrib
5. Write **TRUE** for correct statements and **FALSE** for incorrect statements in the spaces provided.
 - (a) The leaf is the main site for photosynthesis in plants. _____
 - (b) Essential elements are necessary for plant growth, development, and reproduction. _____
 - (c) Carbon, oxygen, hydrogen, and nitrogen are non-minerals. _____

- (d) *Rhizobium* bacteria helps to fix nitrogen from the air into the soil.

- (e) Microelements are needed by plants in large quantities. _____
- (f) Copper is an example of trace elements. _____
- (g) Lamina is the external part of a leaf while midrib is the internal part.

- (h) A bulb is an underground storage organ formed from the plant roots.

- (i) Sunlight energy is not needed during photosynthesis. _____
- (j) Unlike carnivores, autotrophs make their own food in the form of carbohydrates. _____

Section B

1. Briefly explain the following terms.
 - (a) Photosynthesis
 - (b) Palisade mesophyll
 - (c) Photolysis
2. Differentiate the light reaction stage from dark reaction stage of photosynthesis.
3. Describe how a plant leaf is adapted to photosynthesis.
4. Why is photosynthesis a vital process for all living organisms?
5. Why is it necessary to destarch a leaf in an experiment investigating the importance of photosynthesis to plants?
6. Answer the following questions.
 - (a) Mention any two sources of sulphur in plants.
 - (b) Give any two signs of sulphur deficiency in plants.
 - (c) List down at least three functions of sulphur in plants.
 - (d) Why should a leaf be boiled in alcohol when testing for starch?
 - (e) List down the necessary conditions for photosynthesis.

7. Study the two plant leaves in Figure 6.15 and then answer questions (a) and (b).

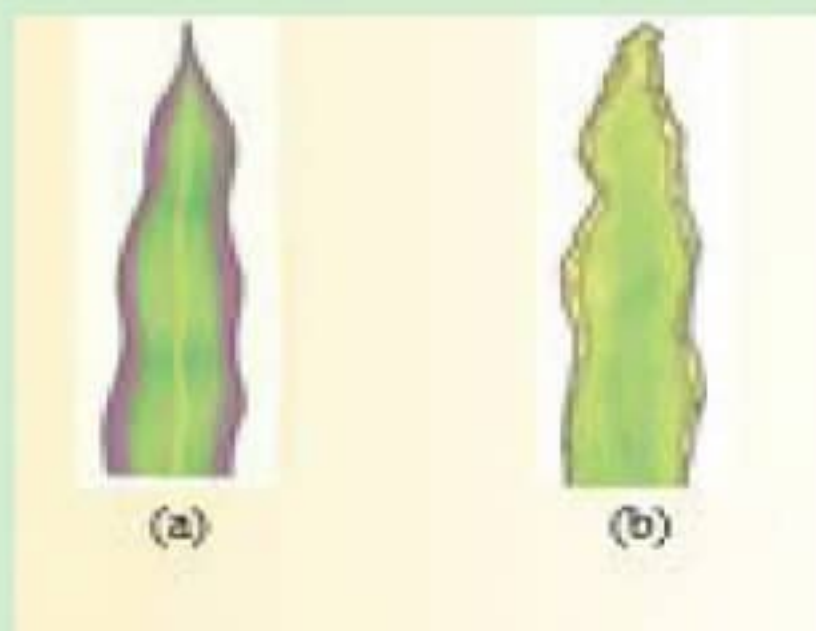


Figure 6.15: Leaves with mineral deficiency

- (a) For each of the signs shown in figures 6.15 (a) and (b), name the macroelement that is in limited supply.
- (b) Outline the effects of excessive amounts of the macroelement stated (a) above.
8. An agricultural officer advised farmers to apply fertiliser that contains nitrogen, phosphorous and potassium in their farms with mineral deficiency. Which signs were shown by plants that made the agricultural officer give such advice to farmers?

Glossary

Amoeba

aquatic or parasitic unicellular organism that uses pseudopodia for locomotion and engulfing food

Aflatoxin

poisonous substances produced by fungi, which are mostly found in maize and groundnuts

Analogous structure

structures with different origins but perform the same function

Analogy

similarity of use

Aquarium

place or container for keeping live marine or freshwater organisms

Basidia

microscopic club-shaped structure in basidiomycetes, such as mushrooms in which sexual spores are produced

Binomial nomenclature

scientific system of naming organisms whereby every name is made up of two parts

Botany

study of plants

Capsid

protein shell that encloses genetic material in a virus

Cell

basic unit of life

Conjugation

sexual reproduction where genetic material is exchanged between two organisms

Cell sap

liquid inside the large central vacuole of a plant cell

Chlorophyll

green pigment found in the leaves of green plants, and stems of some plants

Chloroplast

cell organelle with chlorophyll where photosynthesis occurs

Class

third taxonomic rank of classification

Cones

light-sensitive cells in the retina that are important for colour vision

Conifers

cone bearing plants whose seeds are not enclosed in ovaries but are arranged spirally in cones

Cotyledon

seed leaves found within a plant embryo

Dioecious plant

Organism with separate sex

Division

second taxonomic rank of classification in plants

Ecology

branch of Biology that deals with the study of the relationship between living things and their environment

Embryo

an organism in the very early stages of development from a zygote

Eukaryotic cells

cells whose nuclei are bound by nuclear membranes

Eukaryotic organism

an organism that has a clearly defined nucleus surrounded by a nuclear membrane. It also possesses membrane bound organelles, such as mitochondria, endoplasmic reticulum, and lysosomes

Fronds

leaves of a fern plant. They are made up of leaflets called pinnae

Homologous structures

structures with same origin, but may perform the same or different functions

Hyphae

filaments or thread-like structures making up the fungi

Kingdom

first taxonomic rank of classification

Laboratory

a room specialised to conduct scientific experiments

Locomotion

type of movement whereby the whole body moves

Metabolism

chemical processes that take place inside living cells

Mesophyll

The palisade layer and the spongy layer of the leaf

Microbe

organisms that are too small to be seen by naked eyes

Microscope

an instrument that is used in scientific studies to enlarge very small things, such as microorganisms so that they can be easily seen and examined

Monocotyledon plant

plant that produces seeds with one cotyledon. An example of such plants is a maize plant

Moss

small non-vascular plant with leaf-like structure anchored by root-like structures called rhizoids

Multicellular organisms

many celled organisms

Mycelium

dense network of hyphae in fungi

Nutrition

the process by which living things take in food for growth, metabolism, and body repair

Order

fourth taxonomic rank of classification

Organ

a group of tissues that carry out the same function

Organelles

small structures found inside the cell having a specialised function, for example mitochondria, chloroplasts, and vacuoles

Osmoregulation

active regulation of the osmotic pressure of body fluids so that they do not become excessively diluted or concentrated

Phloem

tissues made up of sieve-tube elements and companion cells that transport manufactured food in vascular plants

Phylum

second rank of classification in animals

Pollen

the male gametes of a flowering plant

Prokaryotic cell

cell that does not have a true nucleus or membrane-bound organelles

Proteins

organic compounds made up of amino acids

Protoctista

kingdom that includes amoeba, plasmodia, euglena, and paramecia

Pseudopodia

cell extensions in amoeba used for either locomotion or engulfing food particles

Radicle

part of plant embryo that develops into a root

Rhizoids

root-like structure of fungi and other non-vascular plants like mosses

Root hair

extension of an epidermal cell of a root that helps to absorb water and minerals from the soil

Scientific methods

a set of steps used in science to prove a hypothesis

Sense organs

groups of tissues that enable us to observe things through seeing, smelling, hearing, tasting, or touching them

Sensitivity

ability to detect and respond to stimuli. It is also referred to as irritability

Skin

the organ forming the outermost covering of the body of vertebrates

Specific epithet

the second part of a scientific name

Sporangia

spore-producing organs of fungi and some plants

Spore

Reproductive cells in fungi and some plants

Specialised cell

a cell that performs a specific function

Specimen

an organism, part of a body of an organism, or any object used for scientific study

Stimulus

anything or condition in the environment that causes an organism to respond (plural is stimuli)

Sting

Piercing, needle-like structure found in bees, hornets, and wasps

Stomata

tiny pores in the epidermis of leaves that allow gaseous exchange and transpiration to occur in plants

Taxon

a group into which organisms are classified based on shared characteristics

Taxonomy

the field of Biology that deals with classifying organisms

Temperature

the degree of hotness or coldness of an object or environment

Tissue

a group of cells working together to perform a specific function

Theory

ideas intended to explain something or a cause of something

Unicellular organisms

single-celled organisms

Vitamin

complex organic micronutrients essential for the body wellbeing and protection against diseases

Xylem

tissue made up of vessels and tracheid that transport water and minerals in vascular plants

Bibliography

- Campbell, N.A., Reece, J.B., Taylor, M.R., & Simon, E.J. (2005). *Biology concepts and connections* (5th ed.). Benjamin Cummings 1301 Sansome St. San Francisco, CA 94111.
- Chand, S., Verma, P.S., & Pandey, B.P. (2010). *Biology for class XI* (2nd ed.). S. Chand & Company Ltd.
- Lawrence, E. (Editor) (2008). *Henderson's dictionary of biology*, 14th ed. Pearson Education Limited.
- Mackean, D. G. (1995). *GCSE Biology* (2nd ed.). John Murray (publishers) Ltd.
- Mackean, D.G. (2005). *IGCSE Biology*. Hodder Murray, an imprint of Hodder Education.
- Ministry of Education and Vocational Training. (2008). *Biology for secondary schools Form 1 & 2* (1st ed.). Oxford University Press.
- Roberts, M.B.V. (1986). *Biology, a functional approach* (4th ed). Thomas Nelson and sons Ltd.
- Stone R.H., & Cozens, A.B. (2006). *New biology for tropical schools* (3rd ed.). Longman Group Ltd.
- Tanzania Institute of Education. (2022). *Biology for secondary school student's book form four*. Tanzania Institute of Education.
- Tanzania Institute of Education. (2021). *Biology for secondary school student's book form three*. Tanzania Institute of Education.
- Tanzania Institute of Education. (2021). *Biology for secondary school student's book form two*. Tanzania Institute of Education.

Tanzania Institute of Education. (2021). *Biology for secondary school student's book form one*. Tanzania Institute of Education.

Tanzania Institute of Education. (2019). *Biology for advanced level secondary school student's book form five*. Tanzania Institute of Education.

Tanzania Institute of Education. (1989). *Biology for secondary schools, book one*. Ibina Publishers Ltd.

Tanzania Institute of Education. (2013). *Diploma in secondary education: Biology pedagogy module*. Tanzania Institute of Education.

United Republic of Tanzania. (2023). *Biology syllabus for ordinary secondary education, Form I-IV*. Tanzania Institute of Education.

Windelspecht, M. (2013). *Biology* (11th ed.). McGraw-Hill companies.

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